Effects of expanding higher education vacancies: effects on marginal and infra-marginal students

Preliminary Results

22 de octubre de 2026

Overview: what we are trying to do

Motivating question. Why don't we increase capacity in high-return programs?

Expanding a program's capacity has two sides, and we study each one:

- **At the margin (RDD).** What happens if we change capacity at the margin? Applicants just above the admission cutoff get the seat, and those just below do not. Comparing them tells us what an additional seat does for the student who receives it.
- **Inframarginal students.** What happens to the students who would have been admitted anyway if we change capacity? A larger program may change class size and who their classmates are, so more seats could affect students already in the program.

Together, the two pieces tell us whether expanding capacity is worthwhile: the gain for the marginal student, and what happens to everyone else in the program.

Overview: the five inframarginal designs

I. Yearly vacancy changes Every year-to-year change in a program's vacancies, used as an instrument for the size of its entering cohort (2SLS).

II. Own sudden expansion A selective program suddenly increases its seats. Difference-in-differences with an event study.

III. Entry of new universities (SUA) Programs more exposed to the 2012 entry of new universities, compared with less exposed ones.

IV. Close competitors' sudden expansions A program is treated once a close competitor (same market, entrant PSU within 50 points) suddenly increases its seats. Difference-in-differences with an event study.

V. Design I at the individual level The same yearly vacancy changes, but with one row per inframarginal student, individual controls and fixed effects for the student's application portfolio (ROL). 2SLS.

In every design we look at three things: **enrollment** (does the program get more or fewer students?), **composition** (does the type of entrant change?) and **graduation of the inframarginal students**.

Main results: changing capacity at the margin (RDD)

An additional seat for applicants at the cutoff (crossing the admission cutoff):

- **Target-program enrollment**: +50,7 pp.
- **Target-program graduation (8 years)**: +18,0 pp.
- Graduation from *any university*: +2,9 pp. From *any higher-education institution*: +2,4 pp.
- PSU retaking: −4,9 pp.

Reading. The seat changes *where* the student enrolls and graduates, much more than *whether* they do: most rejected applicants find another program. The effect is present in every field and is larger when the target program is much more selective than the next-best option.

Main results: changing capacity, effects on inframarginal students

I. The first stage is strong (about 0.74 more entrants per added seat). The effect on inframarginal graduation is zero with year FE and small within university and year: $-0{,}33$ pp per 10 additional entrants (mean 36.9 %). Graduation from any university does not fall, so students switch programs rather than drop out.

II. A sudden expansion adds about 23 entrants and lowers entrant PSU by 5 to 9 points. The negative effect on inframarginal graduation comes from a decline that was already under way before the expansion, and disappears with university $\times$ year FE ($-1{,}5$ pp, not significant).

III. The first stage has the opposite sign to the one predicted, rejects the pre-trend test and is no longer significant once region $\times$ year FE are added. The reduced form on inframarginal graduation is negative, but the event study shows no break in 2012.

IV. When a close competitor expands, the program's enrollment *rises* by about 3 students and its entrants are weaker (NEM -4, PSU -2 to -5), with flat pre-trends. Inframarginal graduation at 8 years does not change. Among selective programs, program graduation falls by 4 pp with program and year FE, but not with field $\times$ year FE or CS.

V. Design I at the student level (432,184 students). The first stage is

Main take-away

1. **At the margin**, an additional seat decides which program the student attends, not whether they study. It raises graduation from the target program, with larger effects when the target program is much more selective than the next-best alternative.
2. **Inframarginal students** are barely affected when capacity changes. The cleanest estimate (Design I) is small and points to switching programs, not dropping out.
3. The larger effects in Designs II–IV **do not survive** event studies or richer fixed effects. Design V shows that the sign of Design I depends on whether program trends are removed (levels vs. first differences).
4. So the evidence does not show a large cost of expanding capacity for the students already in the program. The question of *why capacity is not expanded in high-return programs* is therefore not answered by costs to inframarginal students.

Outline

1. Data and Sample
2. RDD: Threshold crossing effects
   - Main Results
   - Heterogeneity
   - Selective Program-Years
   - Selectivity Gaps and Next-Best Alternatives
   - Same-field vs Different-field Alternatives
3. Inframarginal Program-Year Analysis
   - Inframarginal Design No. 1
   - Inframarginal Design No. 2
   - Inframarginal Design No. 3
   - Inframarginal Design No. 4
   - Inframarginal Design No. 5
4. Appendix

Data and Sample

Data and sample

- We combine DEMRE and MINEDUC administrative records for admission cohorts 2007–2016.
- The resulting database links applications, enrollment outcomes, graduation outcomes, and PSU scores at the applicant level.
- The main analysis focuses on applicants within 25 points of the regular admission cutoff.

Statistic	Sample	N	Mean
Applications	Full sample	4,434,346	–
Unique students	Full sample	786,071	–
Program-years	Full sample	11,789	–
Running variable	BW25	839,108	-2.326
Above cutoff	BW25	839,108	0.433
Enrolls target	BW25	839,108	0.369
Enrolls university	BW25	839,108	0.833
Enrolls HE	BW25	839,108	0.870
Graduates target (8y)	Grad. sample	838,307	0.140
Graduates university (8y)	Grad. sample	838,307	0.402
Graduates HE (8y)	Grad. sample	838,307	0.423

RDD: Threshold crossing effects

RDD around admission cutoffs

- Running variable: application score centered around the regular admission cutoff,

$$r_{ipt} = score_{ipt} - cutoff_{pt}.$$

- Treatment indicator:

$$A_{ipt} = 1(r_{ipt} \geq 0).$$

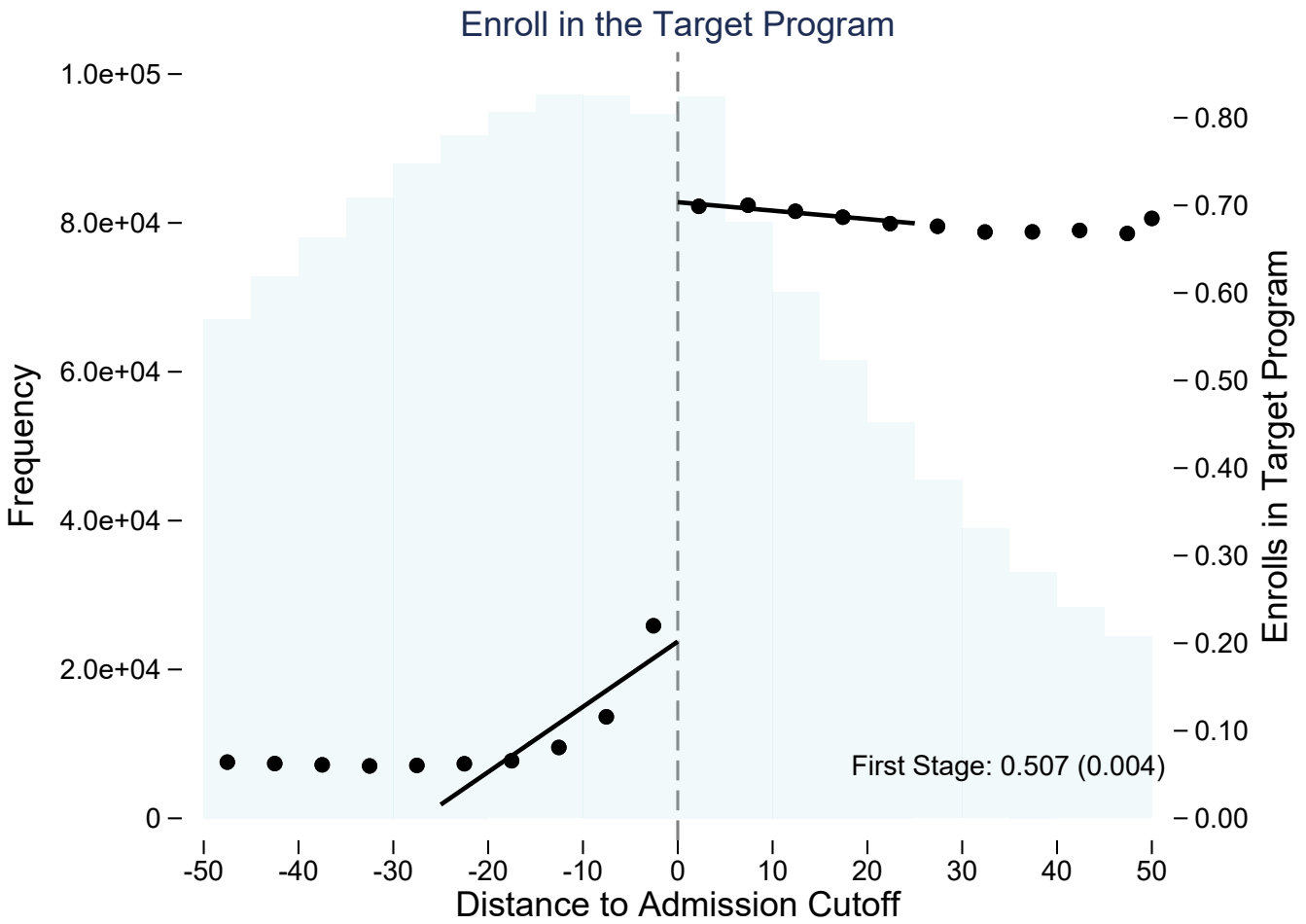
- We compare applicants marginally above and below the cutoff within the same program-year.
- Main specification:

$$Y_{ipt} = \beta A_{ipt} + f(r_{ipt}) + \mu_{pt} + \varepsilon_{ipt}$$

Main bandwidth: $|r_{ipt}| \leq 25$. Standard errors are clustered at the program-year level.

Main Results

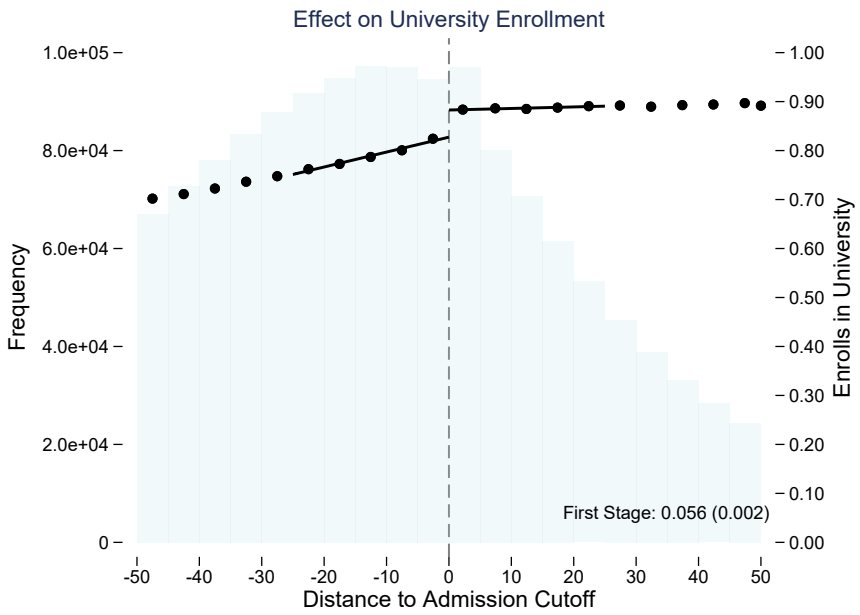
Enrollment in the Target Program

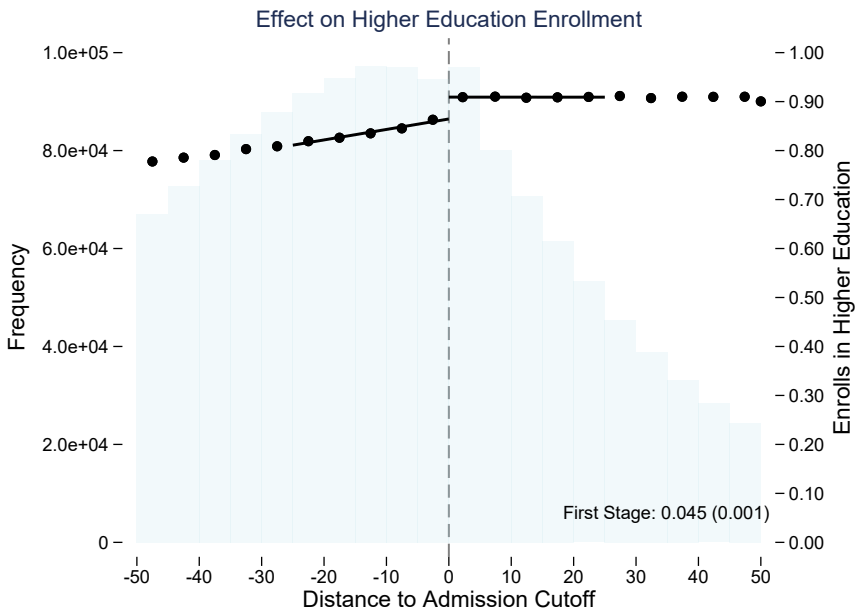


RDD estimate
0.507 (0.004)

Crossing the cutoff increases target-program enrollment by approximately 51 percentage points.

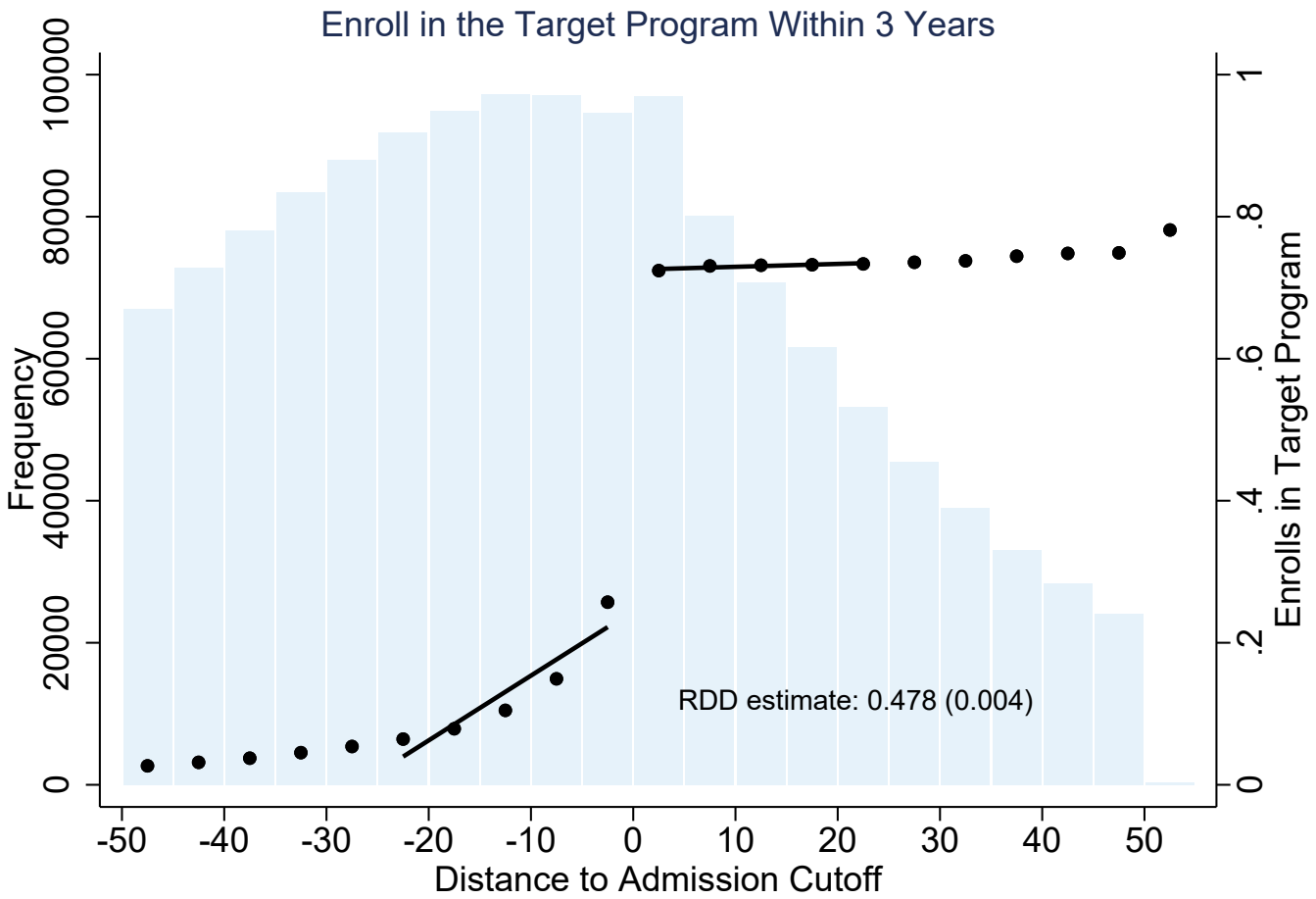
Enrollment on University and HE Program





- Most applicants below the cutoff still access university or higher education through alternative options.
- Crossing the cutoff mainly changes **where students enroll**, rather than **whether they enroll**.

Target Enrollment Within Three Years

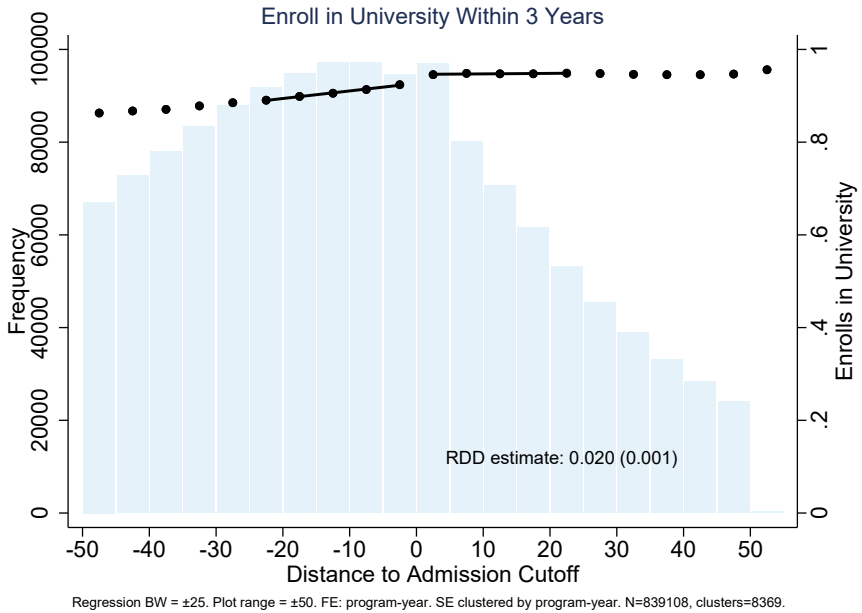


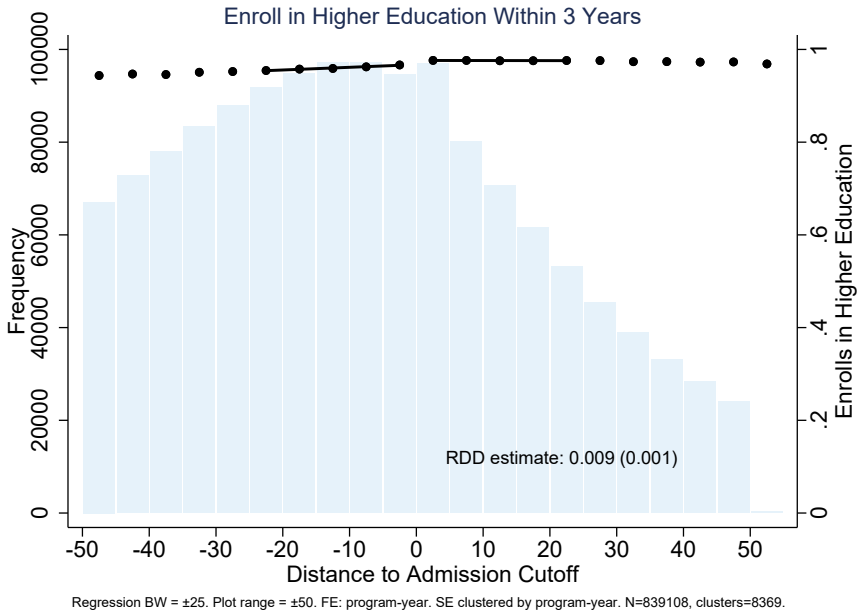
Regression BW = ±25. Plot range = ±50. FE: program-year. SE clustered by program-year. N=839108, clusters=8369.

RDD estimate
0.478
(0.004)

Crossing the cutoff increases target-program enrollment within three years by approximately 48 percentage points.

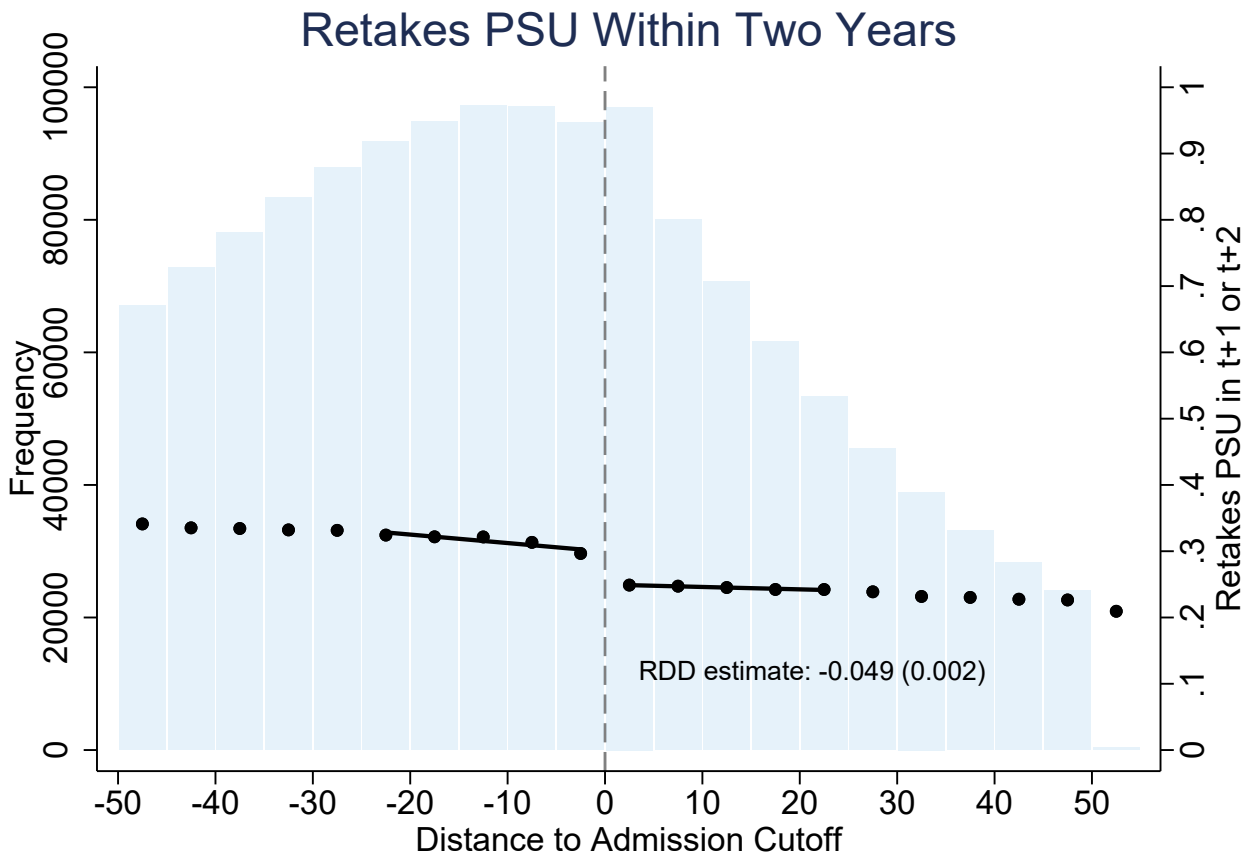
Broader Enrollment Outcomes Within Three Years





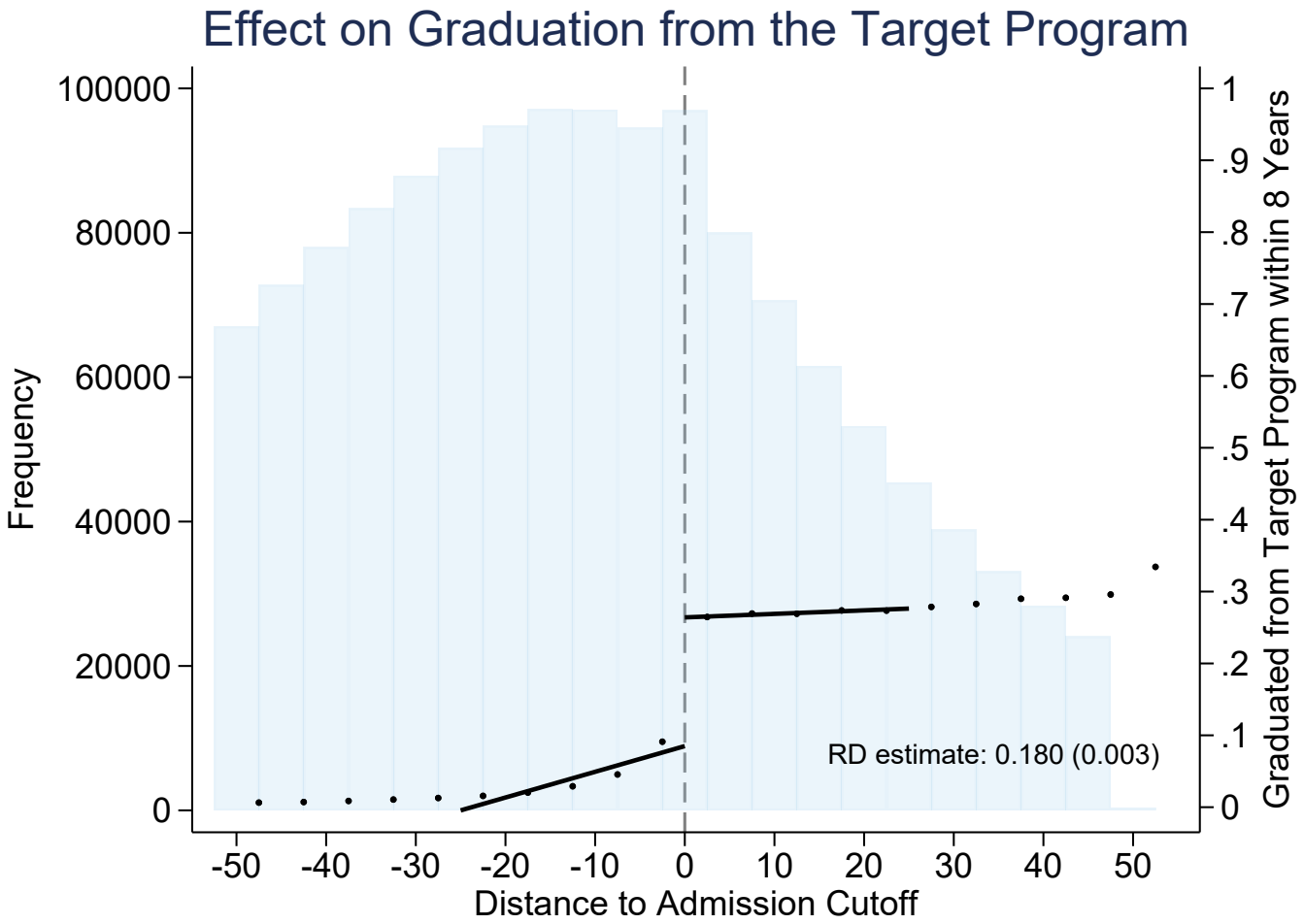
- Most applicants below the cutoff still access university or higher education through alternative options.
- The effect on overall higher-education participation remains small.
- The main effect continues to operate through the specific program attended.

PSU Retaking Within Two Years



- Crossing the cutoff reduces PSU retaking within the next two years.
- Applicants above the cutoff are less likely to re-enter the admission process.
- Admission eligibility lowers incentives to improve future admission outcomes.

Target Graduation Within Eight Years

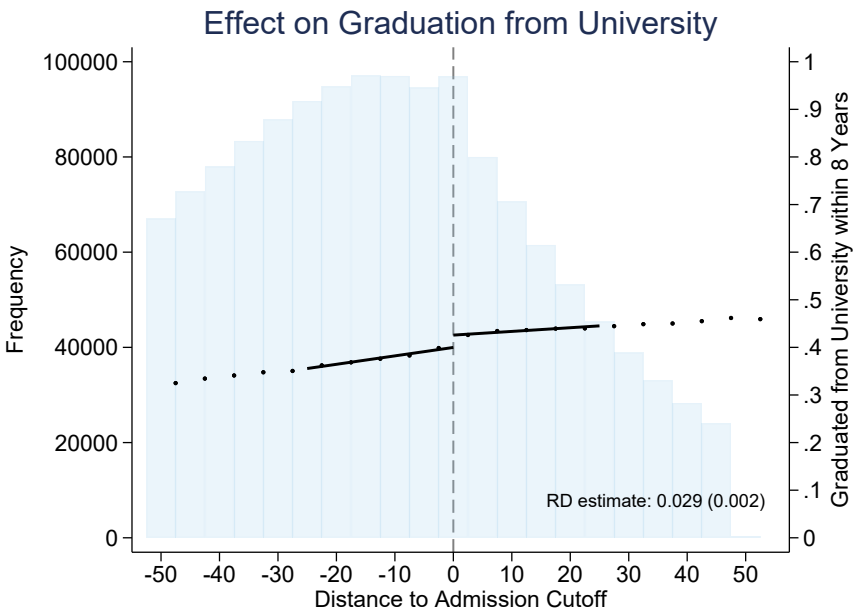


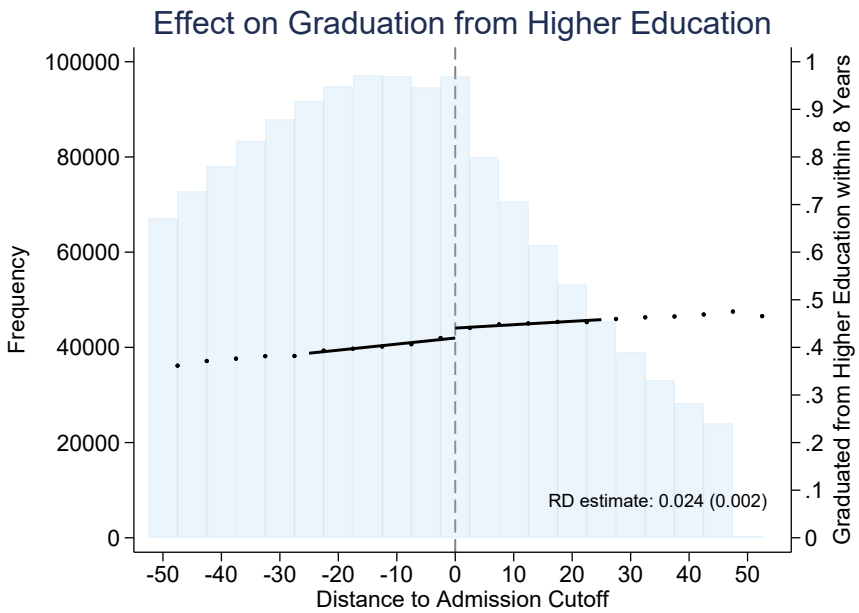
RDD estimate
0.180
(0.003)

Crossing the cutoff increases graduation from the target program by approximately 18 percentage points.

Graduation is measured within eight years of the application year.

Broader Graduation Outcomes Within Eight Years

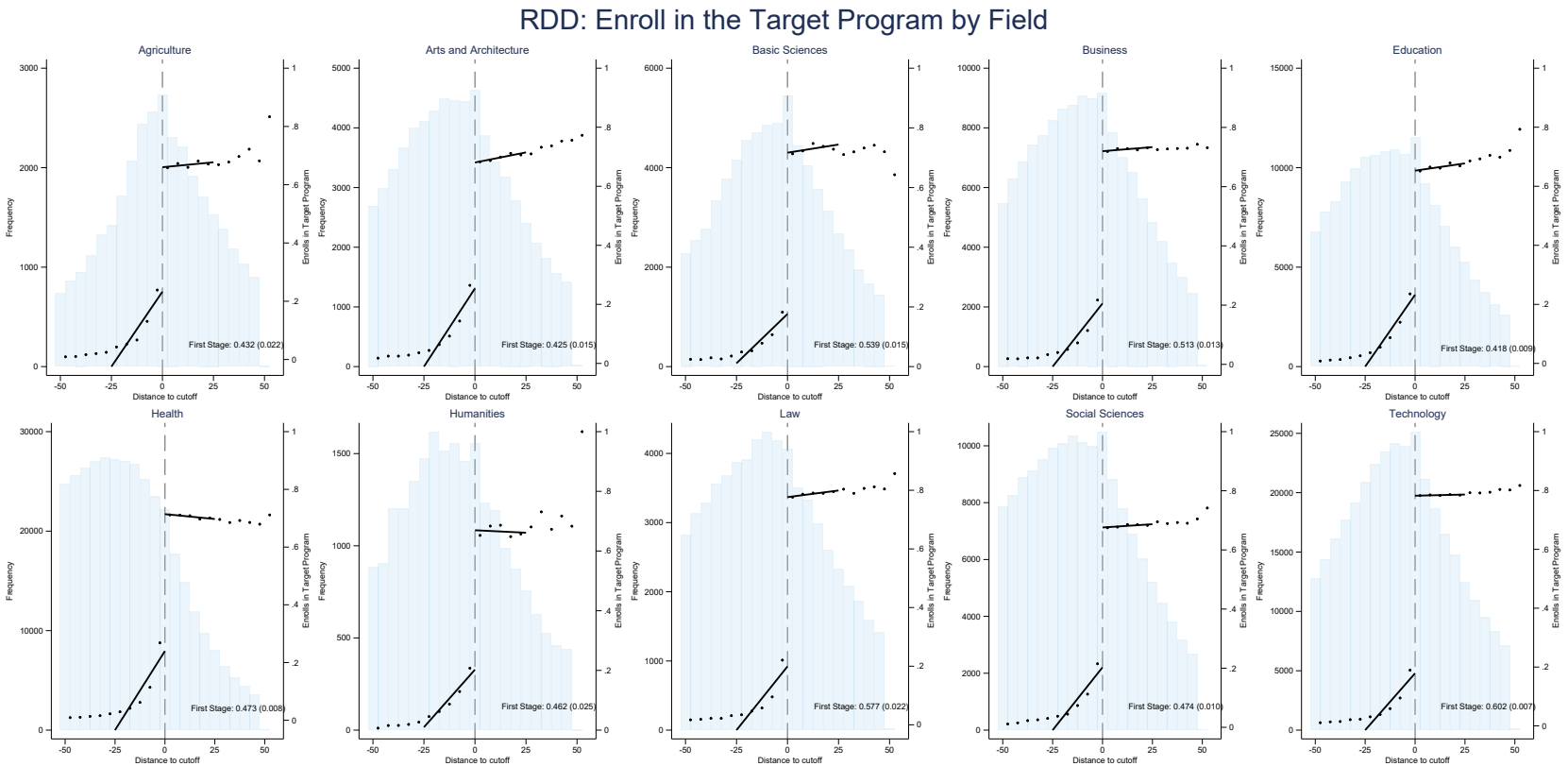




- Broader graduation effects are positive but much smaller than the target-program effect.
- Crossing the cutoff mainly changes where students graduate, rather than whether they complete higher education.

Heterogeneity

Target Enrollment Effects by Field

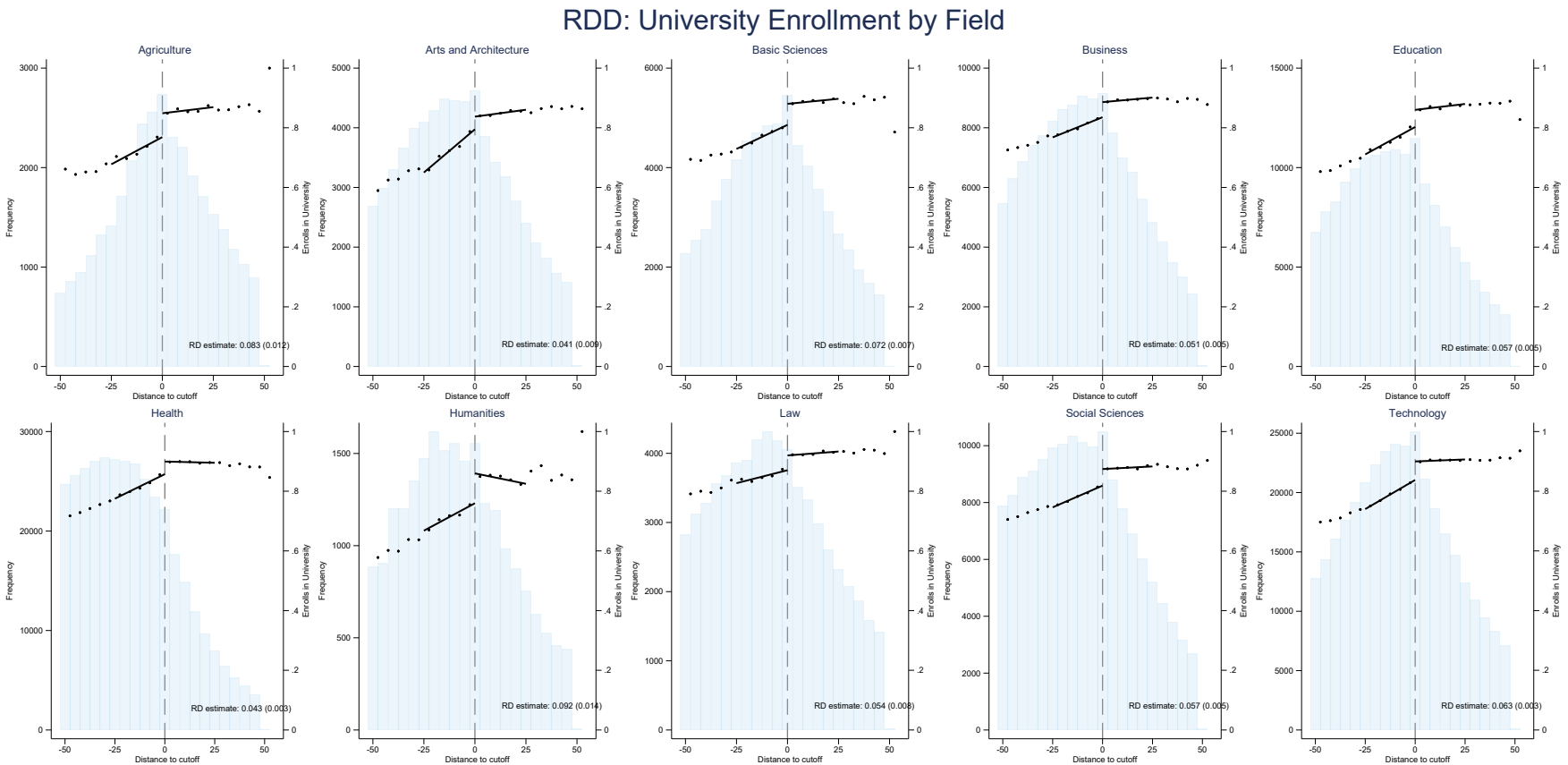


- Crossing the cutoff increases target-program enrollment in every field.
- Effects are largest in Technology, Law, Basic Sciences, and Business.

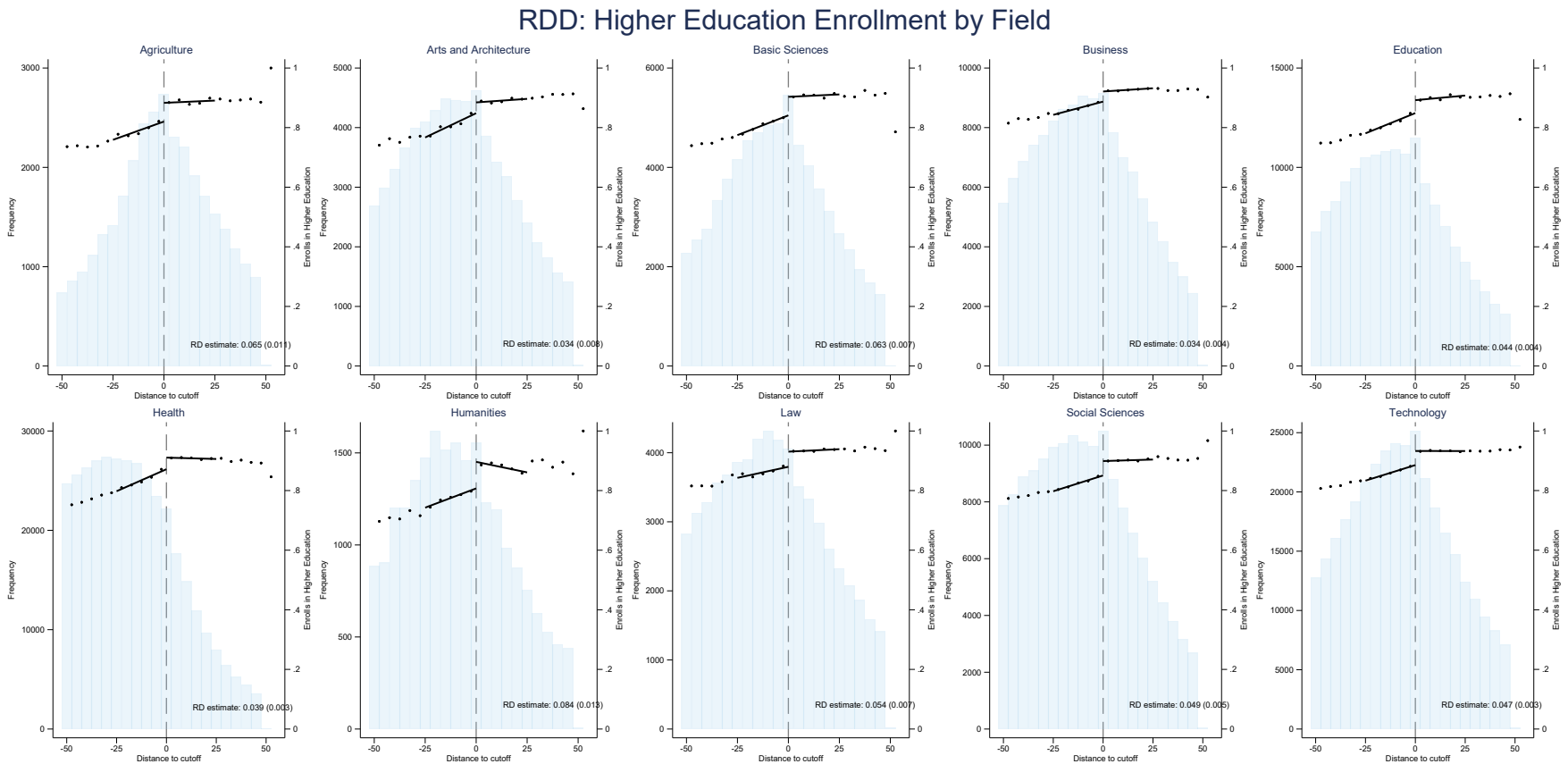
Target Enrollment Effects by Field

Field	Target	University	HE	N
Agriculture	0.432	0.083	0.065	21,102
Arts & Architecture	0.425	0.041	0.034	39,715
Basic Sciences	0.539	0.072	0.063	43,849
Business	0.513	0.051	0.034	79,964
Education	0.418	0.057	0.044	95,508
Health	0.473	0.043	0.039	206,050
Humanities	0.462	0.092	0.084	13,470
Law	0.577	0.054	0.054	37,000
Social Sciences	0.474	0.057	0.049	90,504
Technology	0.602	0.063	0.047	211,145

University enrollment effects by field.



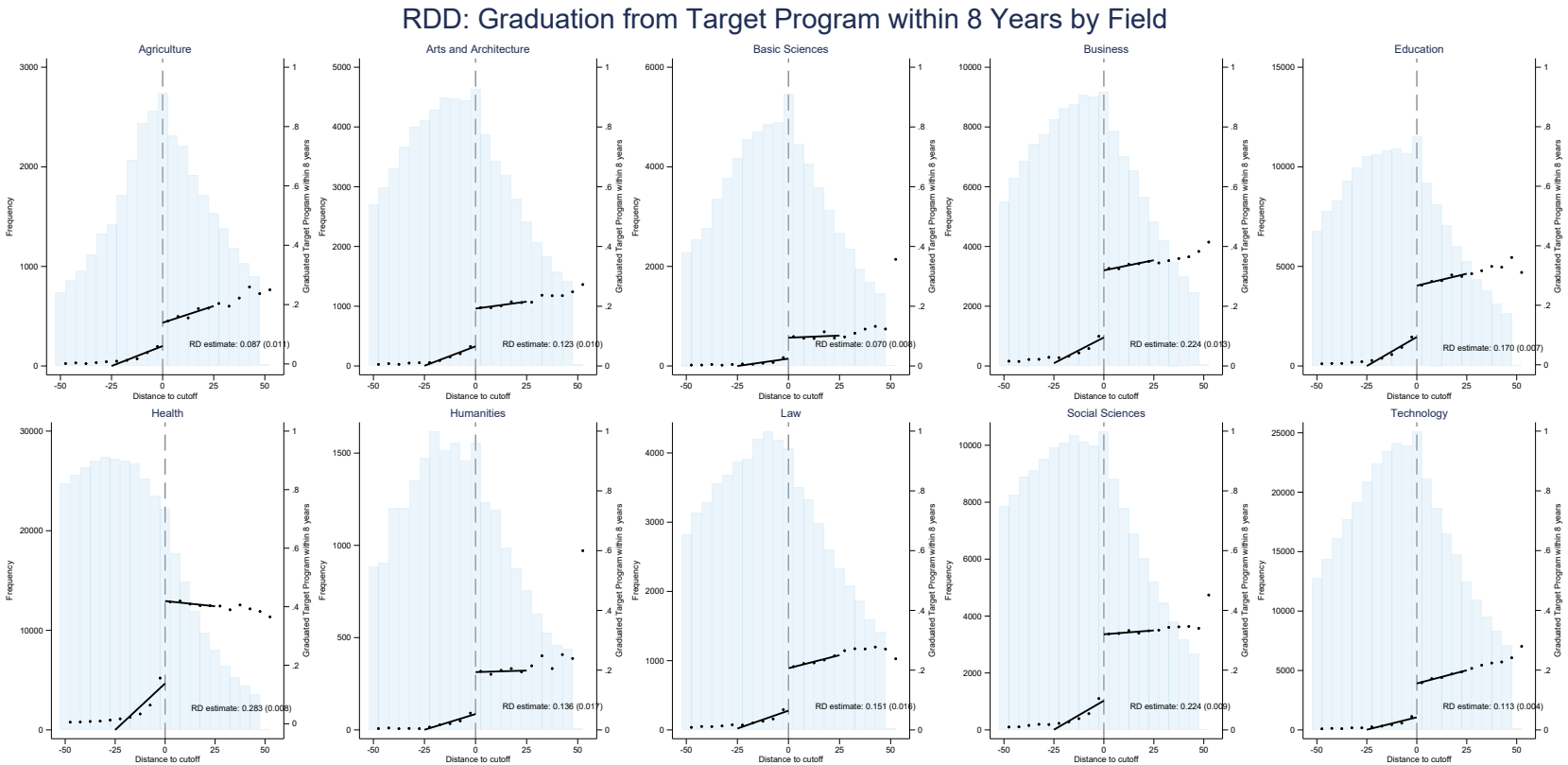
Higher-education enrollment effects by field



RDD estimates by field: university and HE enrollment

Field	Outcome	Estimate	SE	N	Baseline
Agriculture	University	0.083	0.012	21,102	0.729
	HE	0.065	0.011	21,102	0.793
Arts & Architecture	University	0.041	0.009	39,715	0.723
	HE	0.034	0.008	39,715	0.808
Basic Sciences	University	0.072	0.007	43,849	0.771
	HE	0.063	0.007	43,849	0.809
Business	University	0.051	0.005	79,964	0.802
	HE	0.034	0.004	79,964	0.865
Education	University	0.057	0.005	95,508	0.757
	HE	0.044	0.004	95,508	0.815
Health	University	0.043	0.003	206,050	0.815
	HE	0.039	0.003	206,050	0.834
Humanities	University	0.092	0.014	13,470	0.713
	HE	0.084	0.013	13,470	0.775
Law	University	0.054	0.008	37,000	0.849
	HE	0.054	0.007	37,000	0.862
Social Sciences	University	0.057	0.005	90,504	0.782
	HE	0.049	0.005	90,504	0.824
Technology	University	0.063	0.003	211,145	0.790
	HE	0.047	0.003	211,145	0.860

Target Graduation Effects by Field

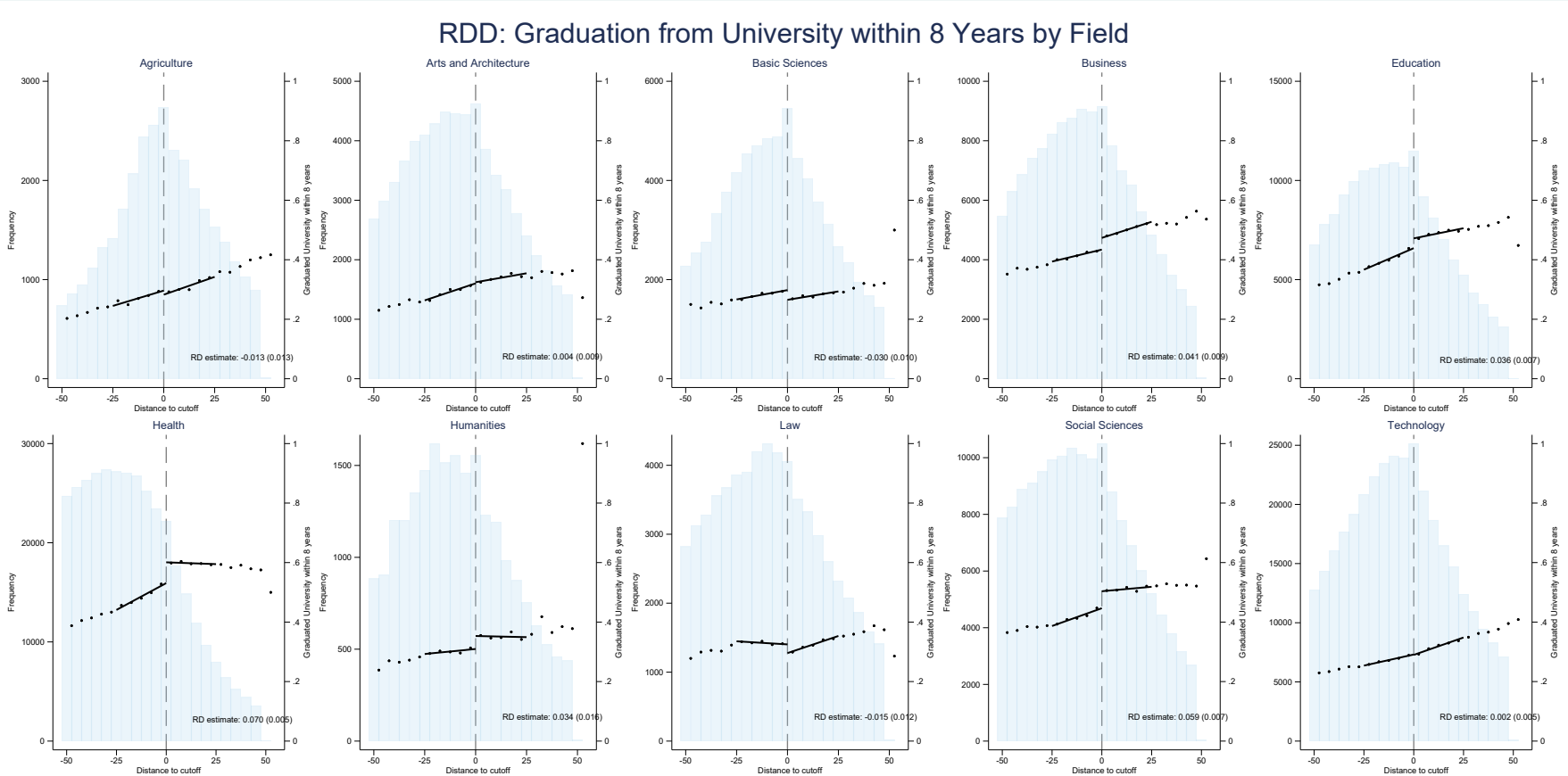


- Crossing the cutoff increases target-program graduation across all fields.
- The largest effects appear in Health, Business, Social Sciences, Education, and Law.

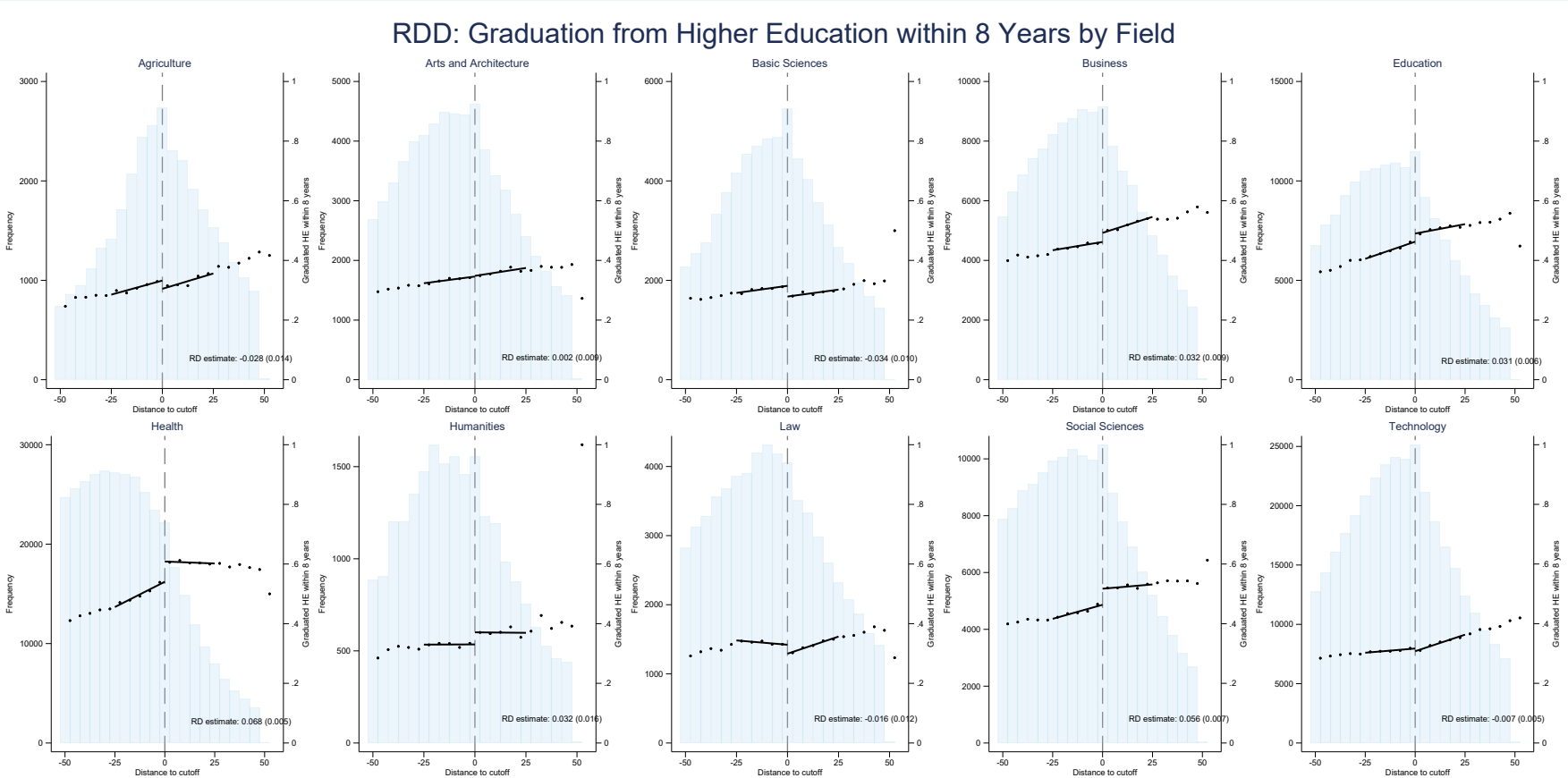
RDD estimates by field: target graduation

Field	Estimate	SE
Agriculture	0.087	0.011
Arts & Architecture	0.123	0.010
Basic Sciences	0.070	0.008
Business	0.224	0.013
Education	0.170	0.007
Health	0.283	0.008
Humanities	0.136	0.017
Law	0.151	0.016
Social Sciences	0.224	0.009
Technology	0.113	0.004

Graduation effects by field: university



Graduation effects by field: any higher education



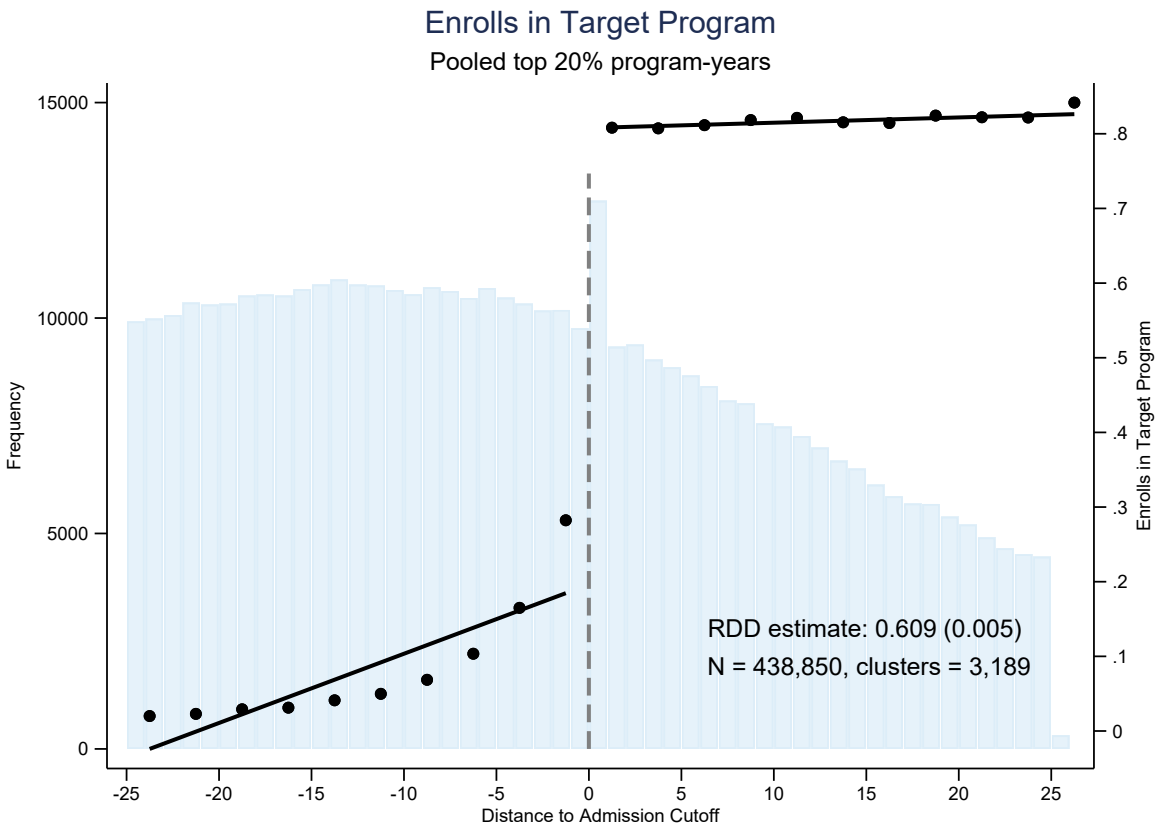
RDD estimates by field: university and HE graduation

Field	Outcome	Estimate	SE	N	Baseline
Agriculture	University	-0.013	0.013	21,102	0.274
	HE	-0.028	0.014	21,102	0.312
Arts & Architecture	University	0.004	0.009	39,715	0.292
	HE	0.002	0.009	39,715	0.335
Basic Sciences	University	-0.030	0.010	43,849	0.283
	HE	-0.034	0.010	43,849	0.303
Business	University	0.041	0.009	79,964	0.414
	HE	0.032	0.009	79,964	0.448
Education	University	0.036	0.007	95,508	0.403
	HE	0.031	0.006	95,508	0.434
Health	University	0.070	0.005	206,050	0.484
	HE	0.068	0.005	206,050	0.497
Humanities	University	0.034	0.016	13,470	0.301
	HE	0.032	0.016	13,470	0.330
Law	University	-0.015	0.012	37,000	0.330
	HE	-0.016	0.012	37,000	0.337
Social Sciences	University	0.059	0.007	90,504	0.417
	HE	0.056	0.007	90,504	0.440
Technology	University	0.002	0.005	211,145	0.273
	HE	-0.007	0.005	211,145	0.310

RDD: Threshold crossing effects

Selective Program-Years

Selective Programs: Target Enrollment

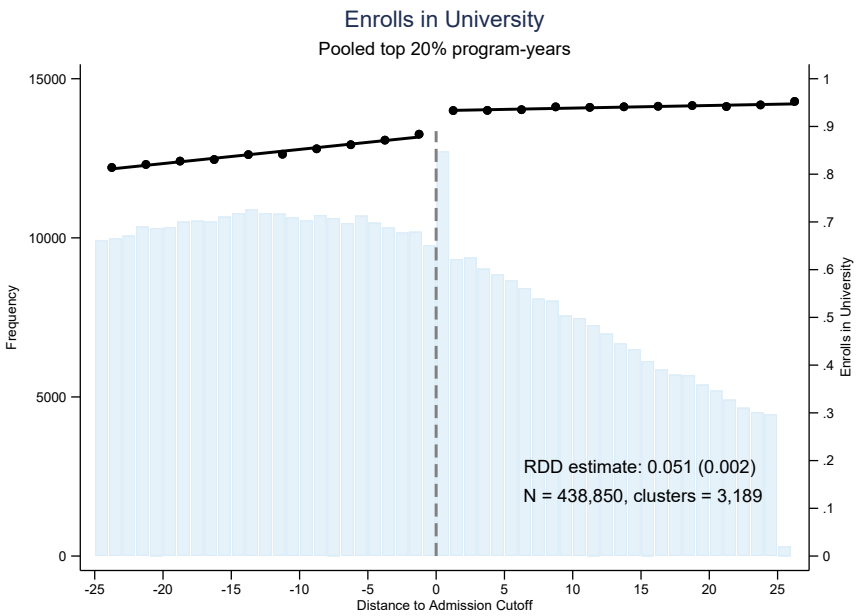


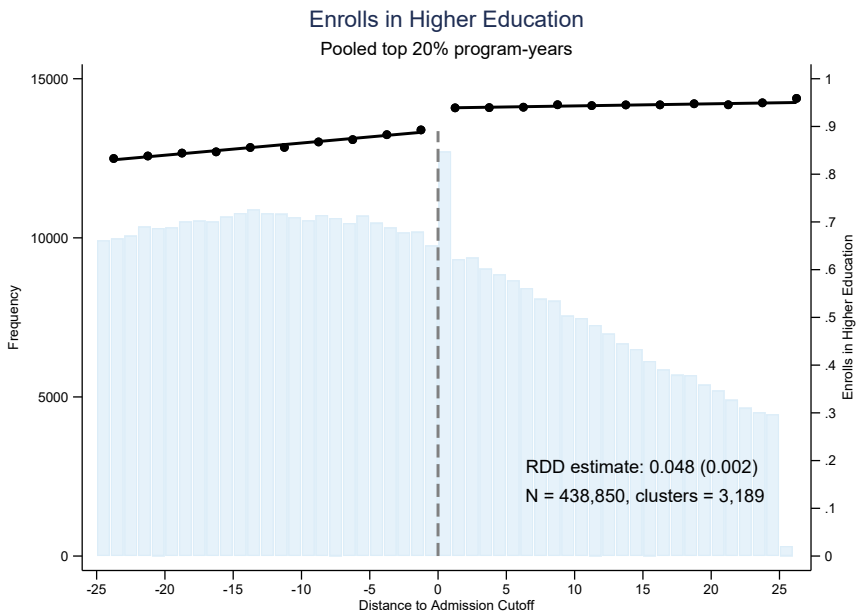
Selective programs

- Selectivity is based on the Language–Math score distribution of admitted students.
- Programs are classified as selective if they appear at least once in the top 20 % of the pooled post-2012 selectivity ranking.

Crossing the cutoff in selective programs increases target-program enrollment by approximately 61 percentage points.

Selective Programs: Broader Enrollment Outcomes





- Effects on university and higher-education enrollment remain much smaller than target-program enrollment.
- Even among selective programs, cutoff crossing mainly changes **where students enroll**, not overall higher-education participation.

RDD: Threshold crossing effects

Selectivity Gaps and Next-Best Alternatives

Construction of Delta Selectivity

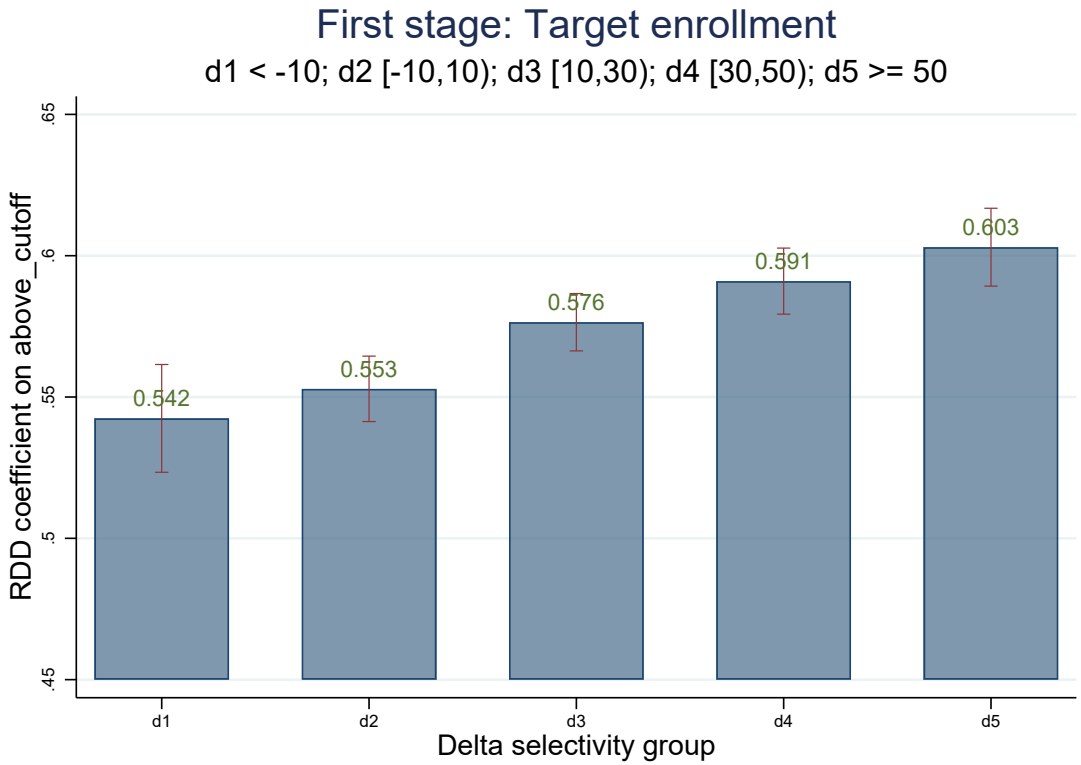
- For each target application, we identify the student's next-best feasible alternative.
- Program selectivity is measured by the average Language–Math score among regularly admitted students.
- We define:

$$\Delta Selectivity = Selectivity_{target} - Selectivity_{nextbest}.$$

- Positive values indicate that the target program is more selective than the next-best alternative.

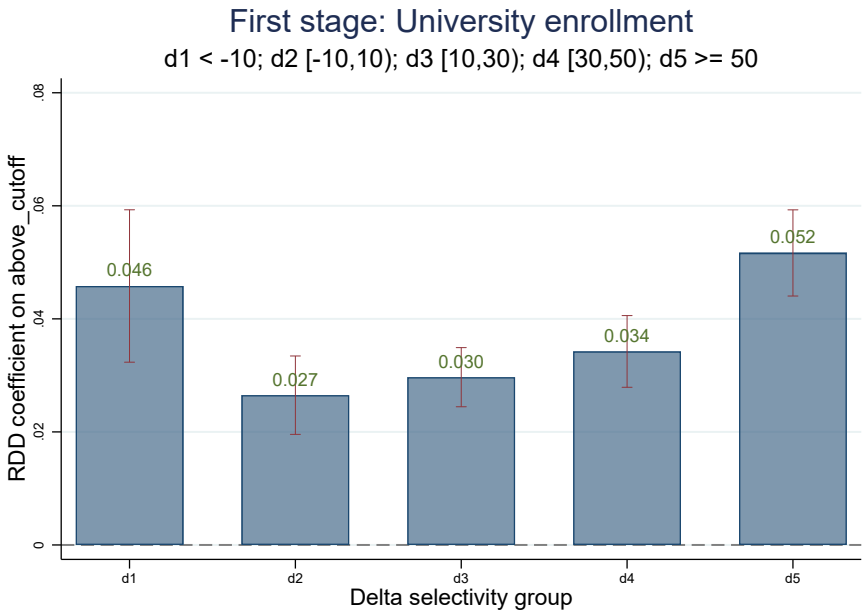
Group	Definition
d1	$\Delta < -10$
d2	$[-10, 10)$
d3	$[10, 30)$
d4	$[30, 50)$
d5	≥ 50

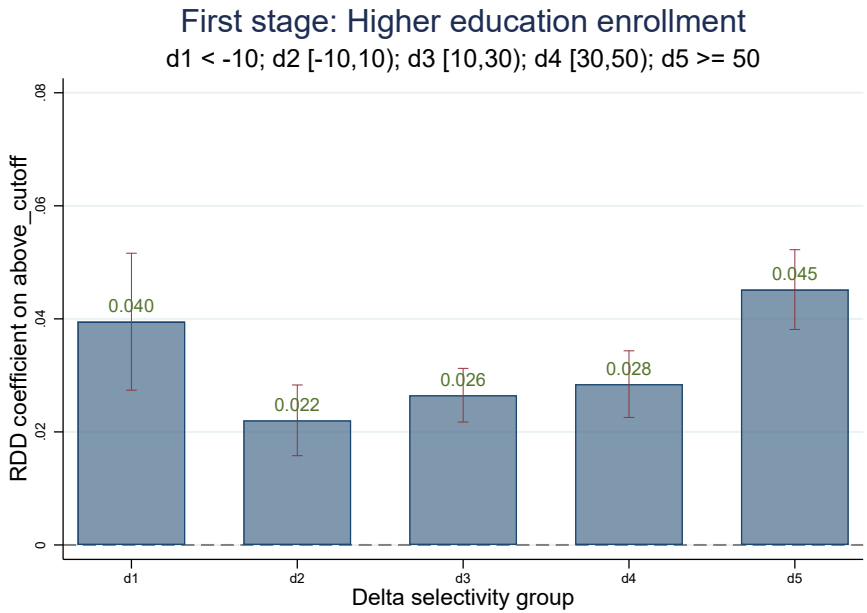
Target Enrollment by Selectivity Gap



- Crossing the cutoff increases target-program enrollment across all groups.
- Effects are strongest when the target program is substantially more selective than the next-best alternative.
- Admission matters most when students face large differences in program selectivity.

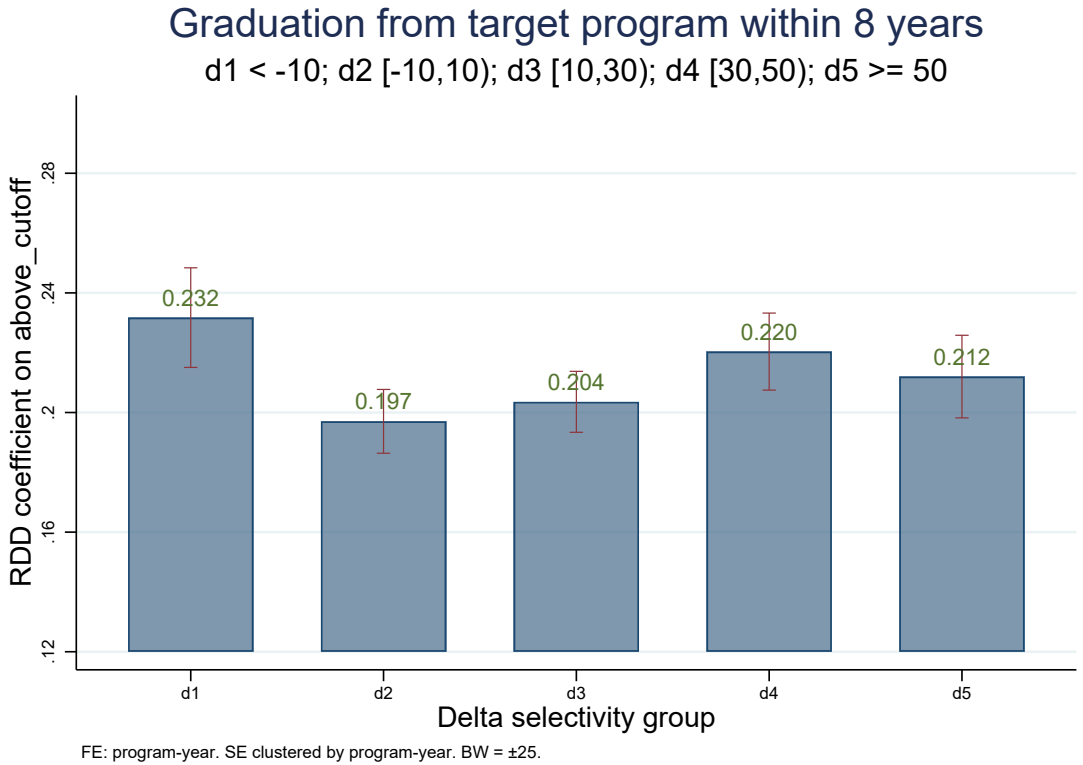
Broader Enrollment Outcomes by Selectivity Gap





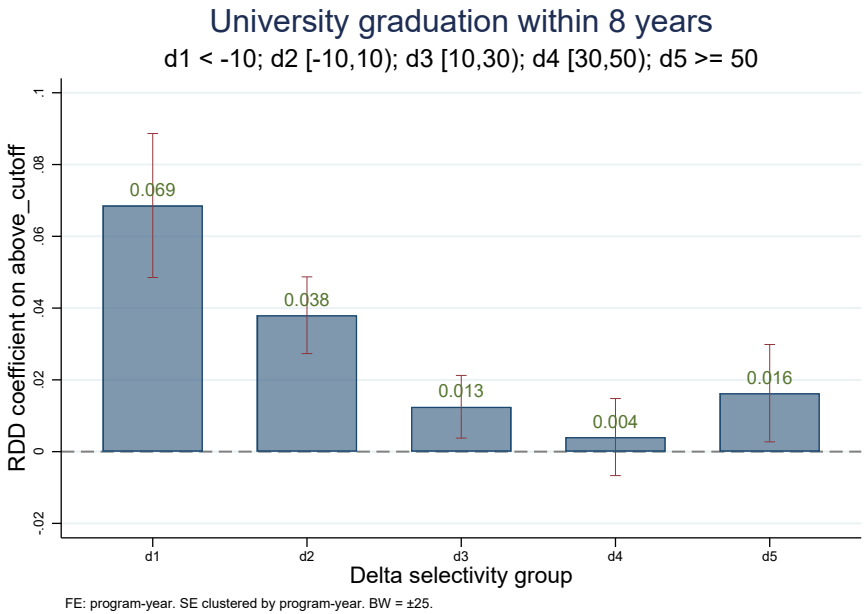
- Effects on university and higher-education enrollment remain relatively small across groups.
- Differences across selectivity-gap groups are much less pronounced than for target-program enrollment.
- The main response margin remains the specific program attended rather than overall higher-education participation.

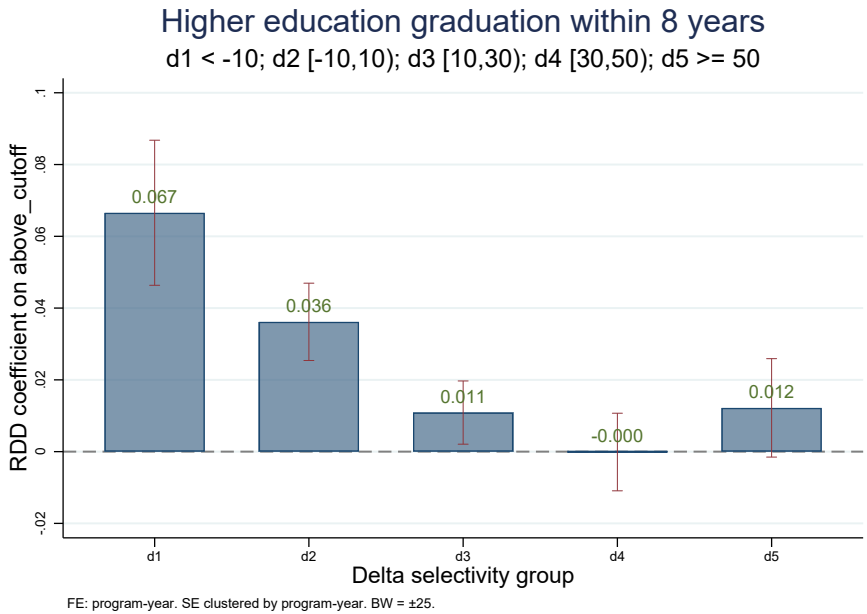
Target Graduation by Selectivity Gap



- Crossing the cutoff increases target-program graduation across all selectivity-gap groups.
- Target-graduation effects remain large across the distribution of target versus next-best selectivity.

Broader Graduation Outcomes by Selectivity Gap

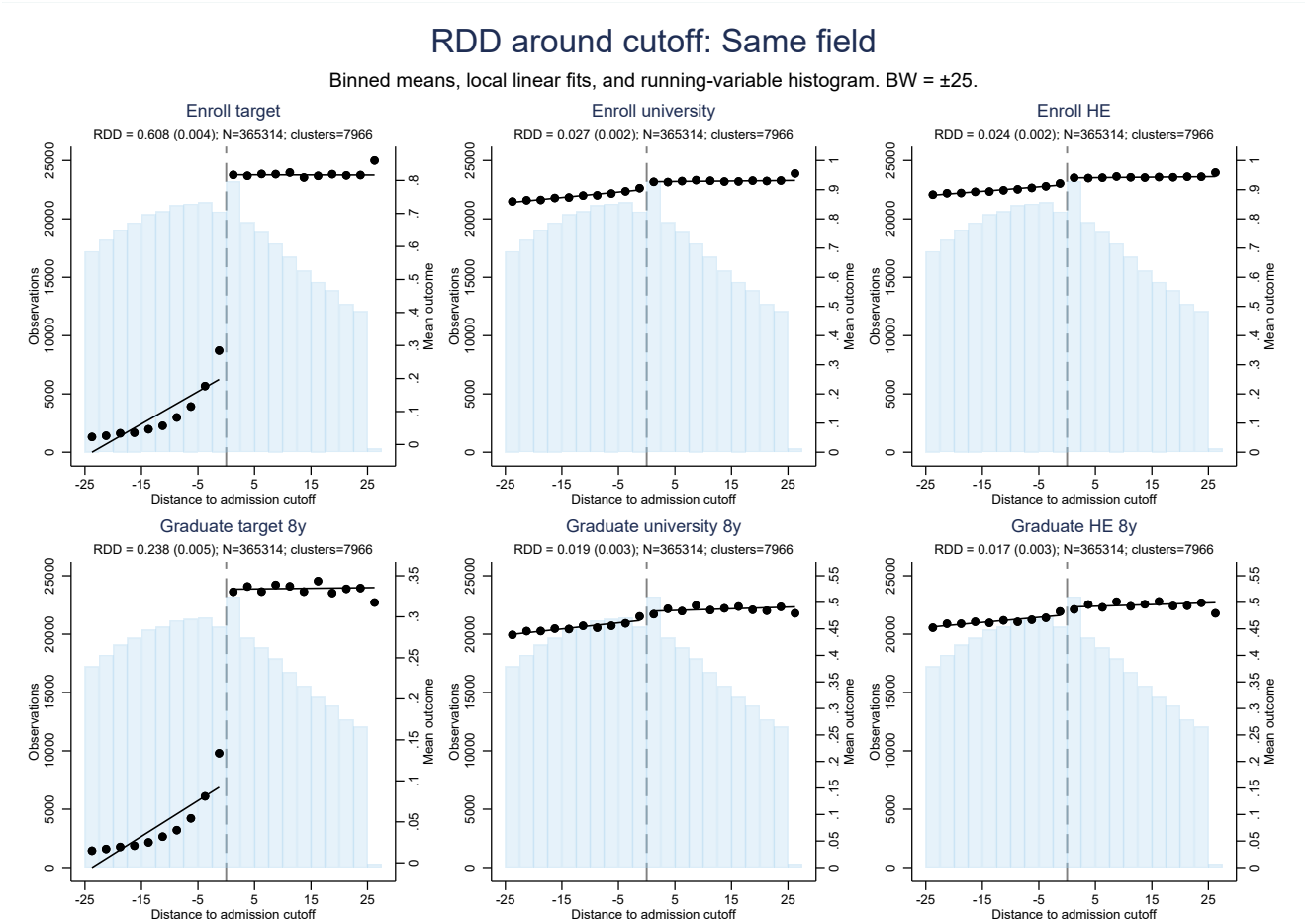




- Effects on university and higher-education graduation are smaller than target-program graduation effects.
- Broader graduation effects are more heterogeneous across selectivity-gap groups.
- Crossing the cutoff mainly changes where students graduate, rather than whether they complete higher education.

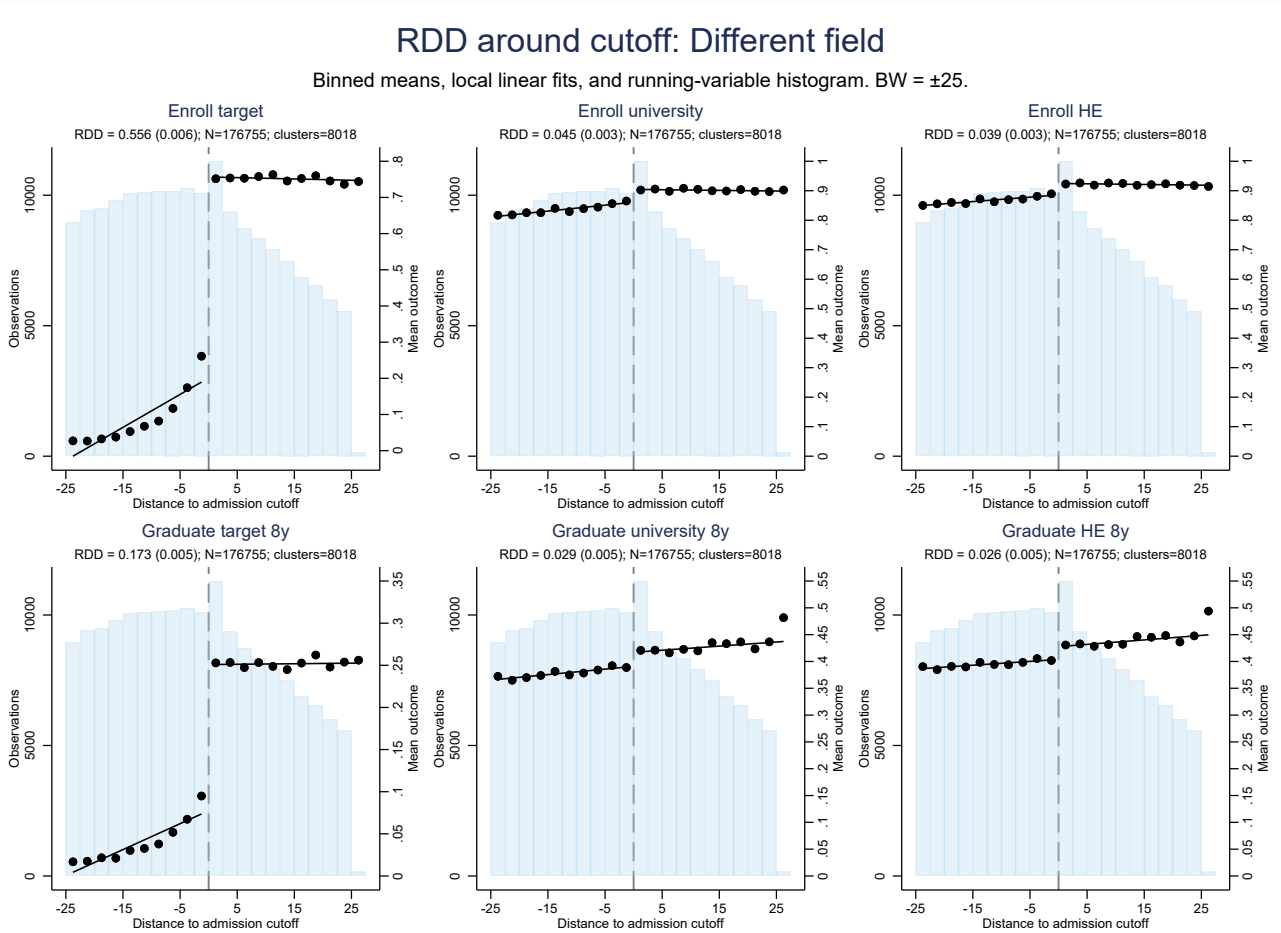
Same-field vs Different-field Alternatives

RDD around cutoff: same field



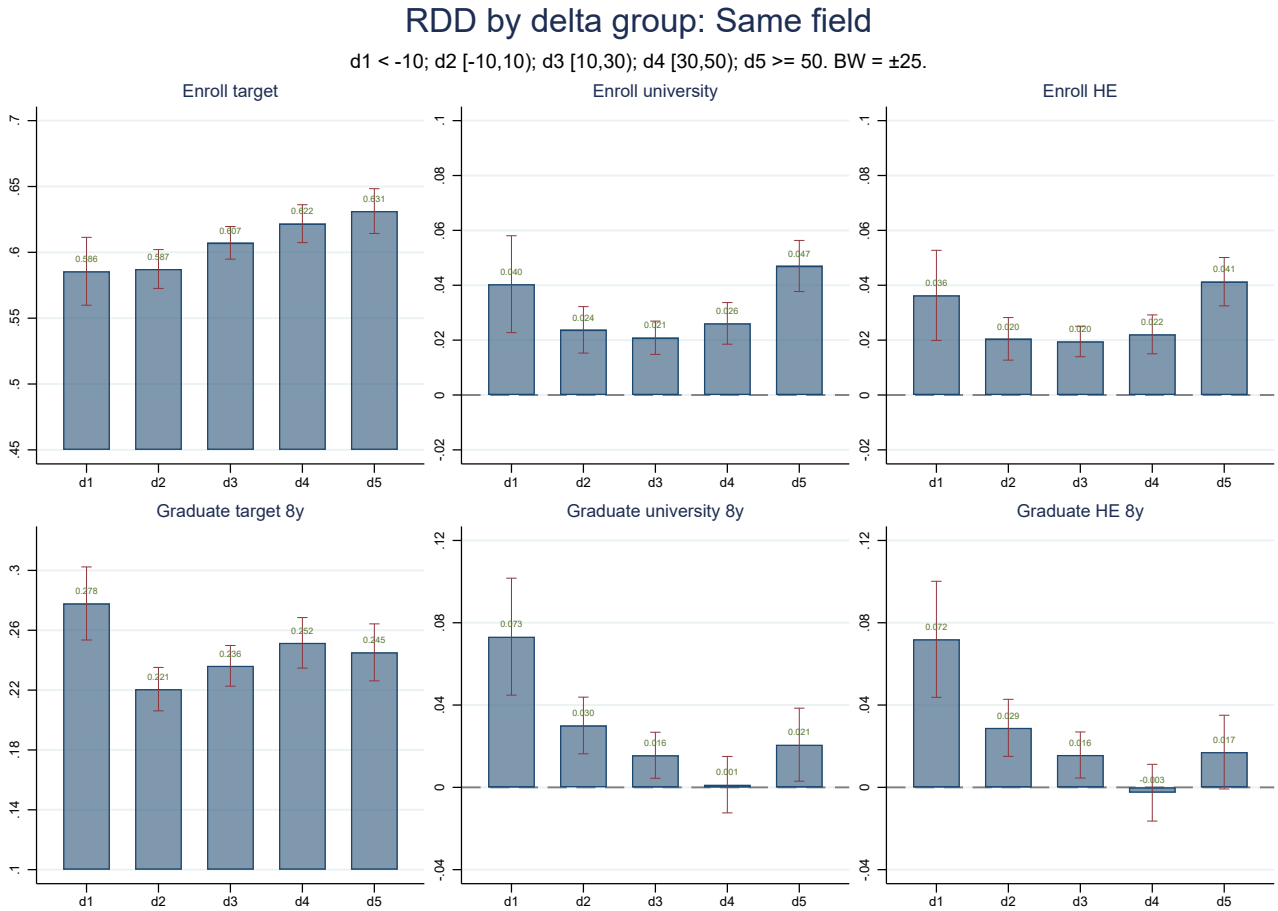
- Crossing the cutoff generates large increases in target-program enrollment and graduation when the next-best alternative belongs to the same field.
- Effects on broader university and higher-education outcomes remain small, suggesting that admission mainly affects program choice within a field.

RDD around cutoff: different field



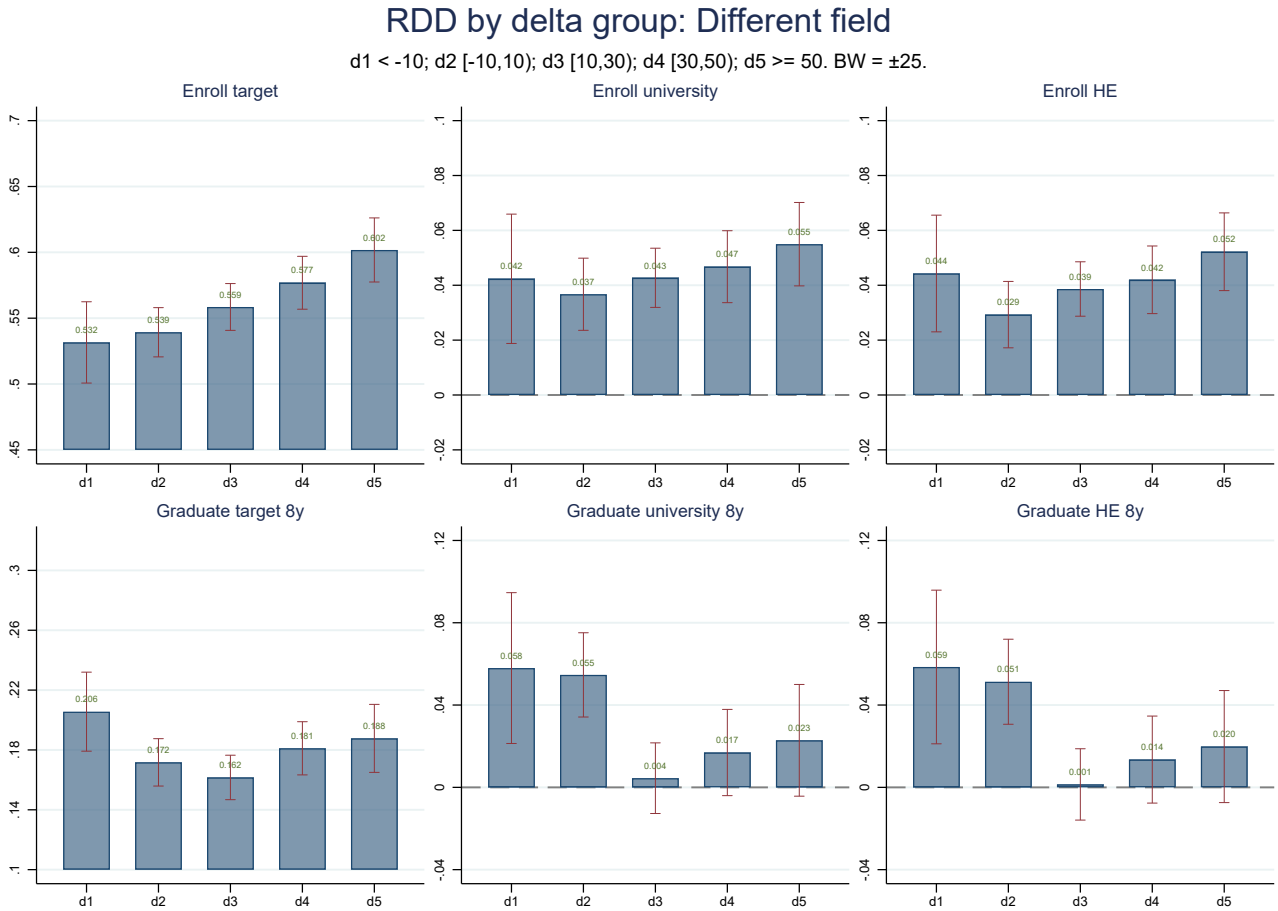
- Target-program enrollment and graduation effects remain large when the next-best alternative belongs to a different field.
- Effects on broader university and higher-education outcomes are somewhat larger than in the same-field sample, consistent with admission affecting both program and field choice.

RDD by selectivity gap: same field



- Within same-field alternatives, target-enrollment effects increase with the selectivity gap between the target and next-best program.
- Broader university and higher-education graduation effects remain comparatively small across groups.

RDD by selectivity gap: different field



- Within different-field alternatives, target-enrollment effects also increase with the selectivity gap.
- Broader graduation effects are less stable across groups, suggesting more heterogeneous substitution patterns when the fallback option is in another field.

Inframarginal Program-Year Analysis

Inframarginal Design No. 1

Design I: the question and the three ingredients

Question. When a program admits a larger entering cohort, do the students who would have been admitted anyway (the *inframarginal* students) become less likely to graduate?

Why inframarginals. The RDD above identifies effects for applicants *at* the admission cutoff, a very particular group. Inframarginal students are ranked well inside the admitted group. A small change in capacity does not decide whether they get in; it only changes *who their classmates are and how crowded the program is*. Two mechanisms:

- **Congestion**: larger classes, fewer resources per student.
- **Peer composition**: expansions admit weaker entrants (see the composition slide).

The exercise has three ingredients, defined one by one on the next slides:

1. **Who** is inframarginal in program p: applicants whose admission rank is at most a fixed threshold T_p.
2. **Treatment and instrument**: the size of the entering cohort N_{pt}, instrumented by the vacancies offered Z_{pt}.
3. **Outcome**: the share of those inframarginal applicants who graduate within 8 years.

Step 1. Who is inframarginal?

Admission rank. $Rank_{ipt}$ is the position of applicant i in the selection list of program p in year t (DEMRE `LUGAR`). It exists only for applicants who were admitted or wait-listed, so only they can be inframarginal.

Threshold. Each program gets *one* number T_p, the same for all years 2007–2016, built from its *enrolled* cohort sizes N_{pt} (students who actually enrolled in p):

- **First Cohort** (main): T_p = size of the first enrolled cohort observed for p.
- **Minimum Cohort** (robustness): T_p = size of the *smallest* enrolled cohort of p over 2007–2016.

Always $T_p^{\text{Min}} \leq T_p^{\text{First}}$: the Minimum Cohort definition is stricter and keeps only applicants who rank inside the entering class of *every* observed year, even the smallest.

Definition. $\text{Inframarginal}_{ipt} = \mathbf{1}\{Rank_{ipt} \leq T_p\}$, applied to every ranked applicant, whether or not i ends up enrolling in p.

Example. Program p enrolled 60 students in its first observed year, then 80, 55, 70 (2007–2010).

	T_p	Rank 50	Rank 58	Rank 65
First Cohort	60	inframarginal	inframarginal	not
Minimum Cohort	55	inframarginal	not	not

T_p does not move with a given year's vacancies, so a capacity change in t cannot change who counts as inframarginal. **Caveat**: ranks count admitted plus wait-listed applicants and T_p is based on enrollment, so "inframarginal" means ranked inside a typical entering class, not a guarantee of admission.

Step 2. The dependent variable depends on the definition

For program p and year t, let $\mathcal{I}_{pt}(T_p) = \{i : \ i \text{ applied to } p \text{ in } t, \ Rank_{ipt} \leq T_p\}$ be its inframarginal applicants and $|\mathcal{I}_{pt}|$ their number. The outcome is

$$y_{pt}(T_p) = \frac{1}{|\mathcal{I}_{pt}(T_p)|} \sum_{i \in \mathcal{I}_{pt}(T_p)} \mathbf{1}\{i \text{ graduates within 8 years of } t \text{ from the program where } i \text{ first enrolled}\}.$$

- **The sample behind each y_{pt} is set by T_p.** First Cohort and Minimum Cohort contain different applicants, so they give different outcome values, different weights and different estimates. Each panel of the results table is a separate run with its own y.
- **Non-enrollees count as zeros.** An inframarginal applicant who never enrolls (or leaves higher education) counts as not graduating. y therefore mixes take-up and completion. The population slide below restricts to enrollees as a check.
- **Which program?** Graduation is measured in the program where the student *actually enrolled* first (MINEDUC, then DEMRE records). For most students this is p; an applicant ranked in p who enrolls elsewhere counts as a graduate if they finish *that* program.
- **Secondary outcomes** (same applicants): graduation from *any university*, and from *any higher-education institution*, within 8 years. These separate dropping out of the system from switching programs.
- **8 years** after the admission year, using titulados through 2024, so all cohorts 2007–2016 are fully observed.

Step 3. Treatment, instrument and model

Treatment. N_{pt}: total entering cohort of p in year t (all students enrolled in p, not only the inframarginal ones). **Instrument.** Z_{pt}: total vacancies offered by p in t. Neither depends on the inframarginal definition.

First differences within program, 2008–2016 (this removes any permanent program characteristic):

$$\text{First stage:} \quad \Delta N_{pt} = \pi \, \Delta Z_{pt} + \alpha_t \; [+ \, \gamma_{u(p)t}] + \nu_{pt}$$

$$\text{Second stage:} \quad \Delta y_{pt} = \beta \, \widehat{\Delta N}_{pt} + \alpha_t \; [+ \, \gamma_{u(p)t}] + \varepsilon_{pt}$$

- β: change in the graduation rate of inframarginal applicants (pp, results $\times 100$) per additional entering student that is induced by a change in vacancies.
- Two specifications: (1) year FE α_t; (2) adds university $\times$ year FE $\gamma_{u(p)t}$, comparing programs of the same university in the same year. This absorbs university-wide shocks (budget, demand, policy) that move both vacancies and graduation.
- **Identifying assumption**: conditional on the FE, changes in vacancies affect inframarginal graduation only through the size of the cohort. The placebo on leads of ΔZ checks this.
- Weights: average number of inframarginal applicants of p in t and $t-1$ (same weights in first stage, reduced form and 2SLS). SE clustered by program.

Design I: data and sample

	Mean	SD
Entering cohort, N_{pt}	66.4	54.4
Vacancies, Z_{pt}	80.2	59.4
Inframarginal applicants per program-year	61.7	47.3
ΔZ_{pt}		12.8
ΔN_{pt}		11.5
Graduation from the program, 8y (%)	34.4	21.4
Graduation from any university, 8y (%)	46.3	18.4
Universities		33
Programs		1,307
Program-years, 2007–2016		8,218
Inframarginal applicants		507,169
First-difference observations		6,529

- **Unit**: program × admission year, 2007–2016.
- **Applications and rank**: DEMRE Formulario C and the admission rank (`lugar`).
- **Enrollment**: Formulario D and MINEDUC enrollment, N_{pt}.
- **Vacancies**: total vacancies offered by the program, Z_{pt}.
- **Graduation**: MINEDUC titulados, 2007–2024.
- Table shown for the First Cohort definition.

Notes: Means and SD over program-years. First-difference observations are program-years with the previous year observed (2008–2016).

Design I: results

	(1)	(2)
Panel A. First Cohort threshold		
First stage: ΔN on ΔZ	0.735***	0.764***
	(0.052)	(0.046)
Kleibergen–Paap F	197.1	273.4
Graduation from the program, 8y ($\times$100)		
Reduced form	0.004	-0.025**
	(0.017)	(0.012)
2SLS	0.005	-0.033**
	(0.023)	(0.015)
Secondary outcomes, 2SLS ($\times$100)		
Graduation from any university, 8y	0.022	0.025
	(0.016)	(0.016)
Graduation from any HE, 8y	0.018	0.024
	(0.016)	(0.016)
FD observations	6,529	6,528
Programs	1,258	1,258
Mean program graduation, 8y (%)	36.9	
Panel B. Minimum Cohort threshold		
First stage	0.698***	0.732***
	(0.050)	(0.048)
2SLS, graduation from the program ($\times$100)	0.025	-0.024
	(0.023)	(0.017)
Year FE	✓	✓
University $\times$ year FE		✓

- Ten more vacancies bring in about 7–8 more students ($F > 190$).
- With year FE only, the effect on program graduation is zero.
- Within university and year, ten more entrants lower inframarginal graduation from the program by **0.33 pp** (mean 36.9 %).
- Graduation from any university does not fall: the loss looks like switching programs, not dropping out of higher education.
- Panel B repeats the whole exercise with the Minimum Cohort definition (smaller group of applicants, new y and weights): same sign and size, not significant.

Notes: Panel A: First Cohort threshold. Panel B: Minimum Cohort threshold (Step 1). Outcome: 8-year graduation of inframarginal applicants (Step 2). RF and 2SLS $\times$100: pp of graduation per vacancy (RF) or per entering student (2SLS). SE clustered by program. *** $p < 0{,}01$, ** $p < 0{,}05$, * $p < 0{,}10$.

Design I: first differences of logs

	(1)	(2)
First stage: $\Delta \ln N$ on $\Delta \ln Z$	0.814***	0.832***
	(0.047)	(0.044)
Kleibergen–Paap F	297.4	350.5
$\Delta \ln$ graduation from the program, 8y		
Reduced form	-0.024	-0.194
	(0.159)	(0.150)
2SLS	-0.031	-0.242
	(0.200)	(0.180)
Secondary outcomes, 2SLS, $\Delta \ln y$		
Graduation from any university, 8y	0.027	0.026
	(0.041)	(0.044)
Graduation from any HE, 8y	0.013	0.022
	(0.038)	(0.042)
Observations, program outcome	5,895	5,895
Programs	1,258	1,258
Observations lost to $y = 0$, program outcome	634	
Year FE	X	X
University × year FE		X

Notes: First Cohort threshold. $\Delta \ln X_{pt} = \ln X_{pt} - \ln X_{p,t-1}$ for N (entering cohort), Z (vacancies) and y (graduation rate). Program-years with $y = 0$ in t or $t-1$ drop out of the $\Delta \ln y$ regressions. 2SLS: $\Delta \ln N$ instrumented by $\Delta \ln Z$. Weights and clustering as in the previous slide. Coefficients are elasticities: the 2SLS is the percent change in the graduation rate per 1 % change in the entering cohort. The first stage uses the full sample (6,529 / 6,528 obs.). *** $p < 0{,}01$, ** $p < 0{,}05$, * $p < 0{,}10$.

- A 1 % increase in vacancies raises the entering cohort by about 0.8 %.
- In logs, the effect on program graduation is imprecise: an elasticity of −0,24 (s.e. 0.18) with university × year FE, i.e. 10 % more entrants lower graduation by 2.4 % (about 0.9 pp), but zero is not rejected. About 10 % of program-years are lost because the graduation rate is zero.

Design I: robustness to the threshold and the horizon

Share of the threshold

	First Cohort		Minimum Cohort	
Share of threshold	(1)	(2)	(1)	(2)
100 %	0.005	-0.033**	0.025	-0.024
	(0.023)	(0.015)	(0.023)	(0.017)
90 %	0.008	-0.031**	0.029	-0.021
	(0.023)	(0.015)	(0.024)	(0.018)
80 %	0.011	-0.029*	0.030	-0.021
	(0.022)	(0.015)	(0.025)	(0.019)
70 %	0.008	-0.029*	0.022	-0.026
	(0.021)	(0.015)	(0.025)	(0.019)
60 %	0.006	-0.032**	0.022	-0.029
	(0.021)	(0.015)	(0.026)	(0.020)
University × year FE		✓		✓

8 vs. 10 years, cohorts 2007–2014

Threshold	Horizon	(1)	(2)
First Cohort	8 years	0.002	-0.056***
		(0.021)	(0.014)
	10 years	0.031	-0.045*
		(0.033)	(0.027)
Minimum Cohort	8 years	0.011	-0.062***
		(0.027)	(0.015)
	10 years	0.046	-0.048
		(0.043)	(0.033)
First stage		0.774***	0.803***
FD observations		4,886	4,885
University × year FE			✓

- 2SLS, program graduation, ×100. Left: inframarginal = $Rank \leq s \cdot T_p$ with s from 60 % to 100 % (100 % is the main definition; lower s keeps only applicants ranked further inside the class). Right: same students observed at both horizons.
- With university × year FE the effect is negative and of similar size under every threshold (significant for First Cohort). It persists at 10 years, so it is not only delayed graduation.

Design I: heterogeneity by field of study

			(1) Year FE		(2) + Univ. × year FE	
Field	Programs	FD obs.	First stage	2SLS	First stage	2SLS
Business and Law	130	651	0.625***	0.152**	0.643***	0.149**
			(0.065)	(0.062)	(0.089)	(0.068)
Health	221	1,290	0.408***	0.012	0.373***	0.071
			(0.033)	(0.058)	(0.034)	(0.067)
STEM	433	2,168	0.857***	-0.027**	0.872***	-0.051***
			(0.040)	(0.013)	(0.027)	(0.012)
Education and Humanities	372	1,913	0.698***	-0.007	0.723***	-0.056
			(0.097)	(0.058)	(0.066)	(0.045)
Arts, Agriculture and Other	102	507	0.634***	0.058	0.686***	0.098
			(0.059)	(0.111)	(0.064)	(0.079)

Notes: First Cohort. 2SLS $\times 100$, pp of program graduation per entering student. Field group fixed at the program level. SE clustered by program. *** $p < 0{,}01$, ** $p < 0{,}05$, * $p < 0{,}10$.

- The negative effect concentrates in STEM, where the first stage is strongest ($-0{,}05$ pp per student with university $\times$ year FE).
- Business and Law moves the other way.

Design I: selectivity heterogeneity

Panel A. Joint quartile 2SLS (×100) with university × year FE				
Outcome	Q1	Q2	Q3	Q4
Program	0.012	-0.086	0.032	-0.034**
	(0.075)	(0.055)	(0.058)	(0.016)
Any university	0.088	-0.036	0.075*	0.024
	(0.067)	(0.035)	(0.040)	(0.018)
Any HE	0.049	-0.034	0.080**	0.024
	(0.068)	(0.036)	(0.040)	(0.018)
Panel B. Continuous interaction with university × year FE, program graduation (×100)				
2SLS: ΔN	-0.032	(0.024)		
2SLS: $\Delta N \times \tilde{S}$	-0.001	(0.013)		
RF: $\Delta Z \times \tilde{S}$	-0.005	(0.008)		
Sanderson–Windmeijer F (ΔN; $\Delta N \times \tilde{S}$)	372.3	282.7		

Notes: First Cohort, first differences, SE clustered by program. Panel A: joint 2SLS, $\Delta N \times Q_q$ instrumented by $\Delta Z \times Q_q$, year × quartile and university × year FE. Panel B: both ΔN and $\Delta N \times \tilde{S}$ are instrumented, by ΔZ and $\Delta Z \times \tilde{S}$ ($\tilde{S}$ = standardized predetermined selectivity), year and university × year FE. Sanderson–Windmeijer F is the first-stage F of each endogenous regressor. Coefficients ×100 (pp per student). *** $p < 0{,}01$, ** $p < 0{,}05$, * $p < 0{,}10$.

- With university × year FE, only Q4 is significant for program graduation ($-0{,}034^{**}$, $-0{,}34$ pp per 10 students). Q1–Q3 are imprecise.
- The continuous interaction does not confirm a gradient: $\Delta N \times \tilde{S}$ is $-0{,}001$ (s.e. 0.013), against $-0{,}029^{*}$ with year FE only. The more negative effects in selective programs reflect differences across universities, not selectivity within a university.

Design I: selectivity heterogeneity by inframarginal population

Population	Mean	Pooled	Q1	Q2	Q3	Q4	Q4, univ.×year FE	$\Delta N \times \tilde{S}$
Graduation from the program, 8 years								
All ranked inframarginals	36.9	0.005	0.044	0.061	0.046	-0.022	-0.034**	-0.029*
		(0.023)	(0.071)	(0.083)	(0.054)	(0.021)	(0.016)	(0.015)
Enrolled in program	44.5	0.004	0.025	0.121	-0.025	-0.017	-0.033**	-0.026
		(0.021)	(0.105)	(0.116)	(0.070)	(0.016)	(0.016)	(0.020)
Enrolled, top half of ranks	47.7	-0.002	0.058	0.089	-0.038	-0.020	-0.040**	-0.020
		(0.023)	(0.127)	(0.127)	(0.081)	(0.018)	(0.018)	(0.023)
Enrolled, near threshold	41.1	0.023	-0.005	0.172	0.006	-0.005	-0.025	-0.037*
		(0.025)	(0.129)	(0.114)	(0.070)	(0.019)	(0.019)	(0.020)
Graduation from any university, 8 years								
All ranked inframarginals	48.4	0.022	0.070	-0.034	0.067*	0.024	0.024	0.009
		(0.016)	(0.063)	(0.030)	(0.035)	(0.020)	(0.018)	(0.013)
Enrolled in program	53.0	0.029*	0.068	-0.014	0.016	0.034*	0.035*	0.015
		(0.017)	(0.071)	(0.035)	(0.043)	(0.020)	(0.019)	(0.017)
Enrolled, top half of ranks	56.8	0.025	0.016	-0.044	0.009	0.036	0.032	0.023
		(0.021)	(0.121)	(0.063)	(0.064)	(0.025)	(0.024)	(0.023)
Enrolled, near threshold	49.0	0.038**	0.135	0.033	0.024	0.033	0.040*	0.003
		(0.019)	(0.085)	(0.045)	(0.048)	(0.022)	(0.021)	(0.014)

Notes: 2SLS, pp of graduation per additional student; First Cohort; year FE unless noted; SE clustered by program. "All ranked inframarginals": every applicant with $Rank_{ipt} \leq T_p$ (non-enrollees count as non-graduates). "Enrolled": inframarginals enrolled in the program (conditions on take-up). Top half and near threshold split enrolled inframarginals at half the threshold. *** $p < 0{,}01$, ** $p < 0{,}05$, * $p < 0{,}10$.

- Restricting to enrollees does not sharpen the effect. In selective programs (Q4, university × year FE), program graduation falls by about 0.03–0.04 pp per student in every population, and it is not concentrated near the threshold.
- Graduation from *any* university moves the other way (Q4: $+0{,}035^{*}$). Inframarginals in expanding selective programs seem to switch programs rather than drop out.

Composition effects: who enters when capacity changes?

- The designs ask whether capacity changes move *how many* students enter a program. Here we ask whether they change *who* enters.
- The dependent variable is the mean characteristic of the entering cohort (Formulario D entrants, program $\times$ year), not enrollment:
  - Mean and SD of LM PSU, mean NEM.
  - LM PSU of the **last admitted** (lowest application score among regular admits, status 24).
  - Share female; shares from private-paid and municipal schools (SES proxy).
- **Income is not available**: Formulario A has no income and the DEMRE socioeconomic questionnaire is not in the data. Income of the last admitted is replaced by the school type of the last admitted.
- Each design keeps its own specification, with only the outcome replaced. The same outcomes are used in Designs I–IV.

Design I: composition of the entering cohort

$$\Delta y_{pt}^{comp} = \pi \, \Delta Z_{pt} + \alpha_t \, [+ \, \gamma_{u(p)t}] + \varepsilon_{pt}$$

Outcome (Δ)	(1)	(2)
First-year enrollment	6.584***	6.802***
	(0.371)	(0.396)
Mean LM PSU, entrants	-1.268***	-1.307***
	(0.116)	(0.127)
LM PSU of the last admitted	-2.386***	-2.324***
	(0.357)	(0.381)
SD of LM PSU, entrants	0.017	0.096
	(0.055)	(0.059)
Mean NEM, entrants	-2.015***	-1.893***
	(0.204)	(0.198)
Female, entrants, pct.	-0.017	0.015
	(0.073)	(0.082)
Private-paid school, entrants, pct.	-0.180***	-0.118*
	(0.062)	(0.062)
Municipal school, entrants, pct.	-0.015	0.004
	(0.073)	(0.083)
Observations (FD)	6,524	6,523
Year FE	✓	✓
University × year FE		✓

Notes: Z: total vacancies, in tens, so each coefficient is the effect of 10 more seats. First differences, 2008–2016, SE clustered by program. The first row (first-year enrollment) is shown for reference. Ten more seats bring in about 7 more entrants and lower mean entrant PSU by 1.3 points and the last admitted's PSU by 2.3–2.4 points, with or without university × year FE: programs that expand reach further down the score distribution.

Design I with a continuous-treatment DiD

Why. Design I's first stage and reduced form are TWFE regressions on a *continuous* dose, ΔZ_{pt}. Callaway, Goodman-Bacon and Sant'Anna (2024) show that the TWFE slope:

- is a weighted average of causal responses, with weights that need not match the dose distribution (and can be driven by unusual doses);
- mixes the *level* effect of being treated with the *slope* of the dose-response;
- under plain parallel trends it also picks up selection bias: comparing doses requires the stronger assumption that every dose group has the same potential trends.

Parameters. With dose D and untreated group $D = 0$:

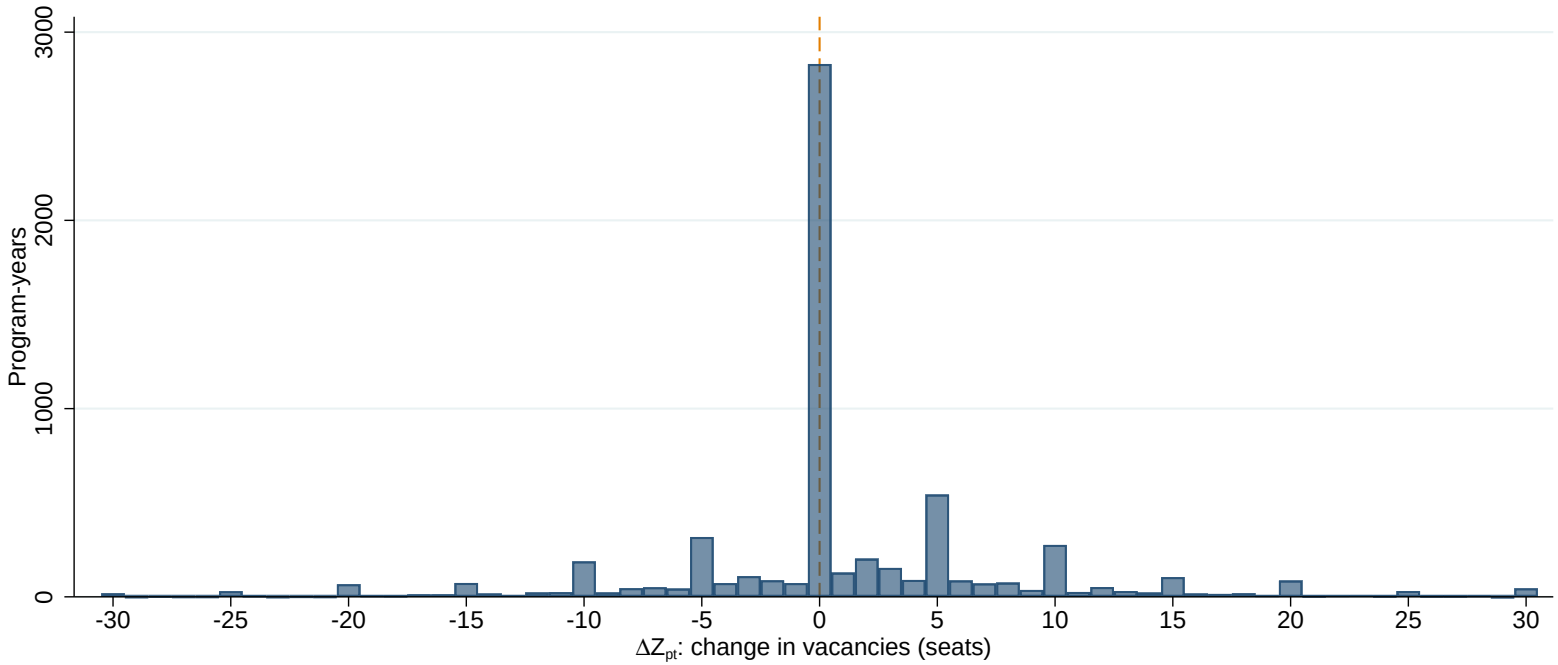
$$ATT(d) = E[\Delta Y \mid D = d] - E[\Delta Y \mid D = 0], \qquad ACRT(d) = \frac{\partial\, ATT(d)}{\partial d}$$

$$ATT^o = E[ATT(D) \mid D > 0], \qquad ACRT^o = E[ACRT(D) \mid D > 0]$$

- $ATT(d)$ needs only parallel trends between dose d and the untreated.
- $ACRT(d)$, the *average causal response* (comparing doses), needs **strong parallel trends**: no selection on the gains from a given dose.
- **Capacity effects should scale with the dose**: congestion and peer composition depend on how many students enter. A level effect that does not grow with d is hard to read as a capacity effect.

Continuous DiD: mapping to Design I

- Each program-year is a 2×2 comparison of $t-1$ and t. **Dose**: $D_{pt} = \Delta Z_{pt}$. **Untreated**: programs with $\Delta Z_{pt} = 0$ (43 % of program-years).
- **Expansions** ($\Delta Z > 0$) and **contractions** ($\Delta Z < 0$, dose $|\Delta Z|$) are estimated separately, each against the same untreated programs. TWFE forces them to be symmetric.
- **Comparison programs**: untreated programs in the *same year* (analog of FE (1)) or in the *same university and year* (analog of FE (2); 96 % of program-years have one).
- **Estimation**: subtract the untreated mean of the cell, then regress on a cubic in the dose (B-spline of degree 3, no interior knots, the `contdid` default). $ACRT(d)$ is its derivative. Robustness: one interior knot at the median dose. Doses ≤ 30 seats ($\approx$P95).
- Same outcomes, sample and weights as Design I (First Cohort, inframarginal graduation within 8 years, weights avg. inframarginal cohort).
- **Per entrant**: $ACRT^{o}_{RF}/ACRT^{o}_{FS}$, the continuous-DiD analog of the 2SLS; compared with the TWFE ratio on the same sample.
- **Pre-trend**: same estimator on Δy_{t-1}. **Inference**: bootstrap clustered by program (499 reps), pointwise and uniform (sup-t) 95 % bands.



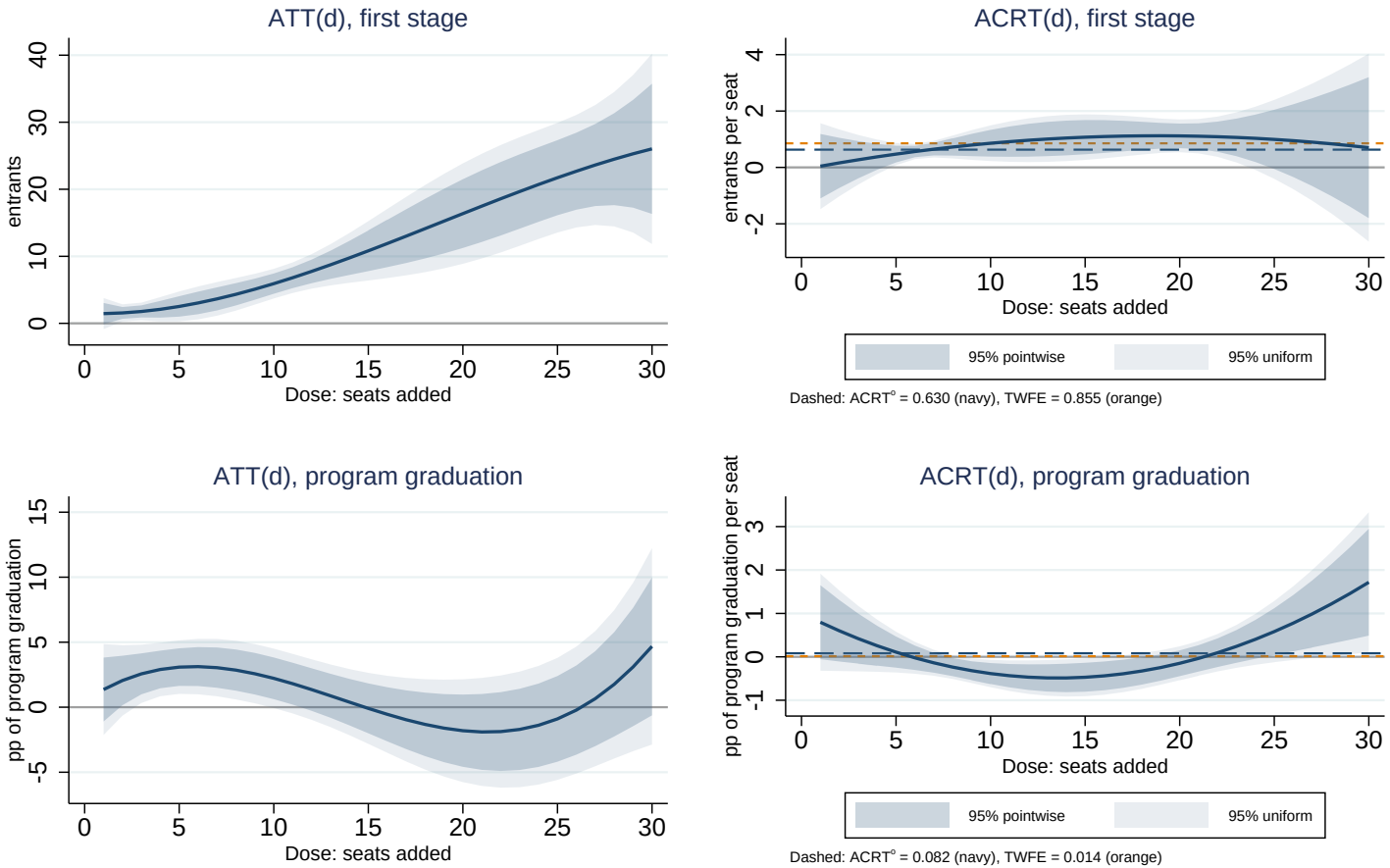
Continuous DiD: aggregate parameters

	Expansions ($\Delta Z > 0$)		Contractions ($\Delta Z < 0$)	
Comparison programs ($\Delta Z = 0$)	Same year	Same univ. × year	Same year	Same univ. × year
Panel A. First stage: ΔN, entrants				
ATT^o	6.388***	6.306***	-6.722***	-6.099***
	(0.568)	(0.602)	(0.476)	(0.516)
$ACRT^o$ (per seat)	0.663***	0.630***	-0.641***	-0.469***
	(0.106)	(0.118)	(0.076)	(0.107)
$ACRT^o$, spline with knot at median	0.728***	0.661***	-0.595***	-0.406***
	(0.069)	(0.082)	(0.066)	(0.097)
TWFE slope (per seat)	0.756***	0.855***	-0.689***	-0.659***
	(0.065)	(0.081)	(0.040)	(0.045)
Panel B. Reduced form: program graduation, 8y (pp)				
ATT^o	1.786***	1.901***	-0.637	0.265
	(0.540)	(0.638)	(0.598)	(0.694)
$ACRT^o$ (per seat)	-0.046	0.082	0.085	0.038
	(0.097)	(0.123)	(0.107)	(0.135)
$ACRT^o$, spline with knot at median	0.001	-0.004	0.131	0.123
	(0.109)	(0.130)	(0.102)	(0.132)
TWFE slope (per seat)	0.040	0.014	-0.000	0.083
	(0.037)	(0.050)	(0.048)	(0.055)
Panel C. Per entrant: RF / FS (pp of program graduation)				
$\mathbf{ACRT^o_{RF}/ACRT^o_{FS}}$	-0.069	0.130	-0.133	-0.080
	(0.152)	(0.216)	(0.174)	(0.310)
$ACRT^o$ ratio, knot at median	0.002	-0.007	-0.220	-0.303
	(0.149)	(0.198)	(0.176)	(0.358)
ATT^o_{RF}/ATT^o_{FS}	0.280***	0.301***	0.095	-0.043
	(0.087)	(0.098)	(0.091)	(0.114)
TWFE ratio (= 2SLS on this sample)	0.052	0.016	0.000	-0.125
	(0.050)	(0.060)	(0.070)	(0.084)
Panel D. Per entrant: graduation from any university (pp)				
$\mathbf{ACRT^o_{RF}/ACRT^o_{FS}}$	0.094	0.056	0.038	0.218
	(0.107)	(0.159)	(0.133)	(0.225)
TWFE ratio (= 2SLS on this sample)	0.051*	0.038	-0.019	-0.002
	(0.029)	(0.032)	(0.040)	(0.053)
Panel E. Pre-trend: same estimator on Δy_{t-1} (pp)				
ATT^o	-0.693	-0.816	-1.275*	-0.441
	(0.565)	(0.931)	(0.761)	(1.031)
$ACRT^o$ (per seat)	-0.049	0.028	-0.230*	-0.301
	(0.122)	(0.210)	(0.131)	(0.221)
TWFE slope (per seat)	-0.008	-0.054	-0.144**	-0.116*
	(0.046)	(0.060)	(0.061)	(0.067)
Treated program-years	2,174	1,979	1,346	1,289
Comparison program-years	2,831	2,089	2,831	1,977
Mean dose (seats)	9.2	9.1	9.5	9.5

- **First stage**: 0.63–0.66 entrants per added seat (0.47–0.64 per seat cut), somewhat below TWFE.
- **No dose-response in graduation**: $ACRT^o$ is between −0,05 and +0,08 pp per seat, never significant. Per entrant, −0,13 to +0,13 pp, with SE of 0.15–0.31, 10–20 times the Design I 2SLS SE.
- **Level effect of expanding**: $ATT^o = +1{,}8$ to $+1{,}9$ pp***, but it does not grow with the dose (next slide).
- Expansions only, TWFE 2SLS is ≈ 0. The negative Design I (2) estimate leans on **contractions** (−0,125), where the pre-trend check fails (Panel E).

Continuous DiD: dose-response, expansions

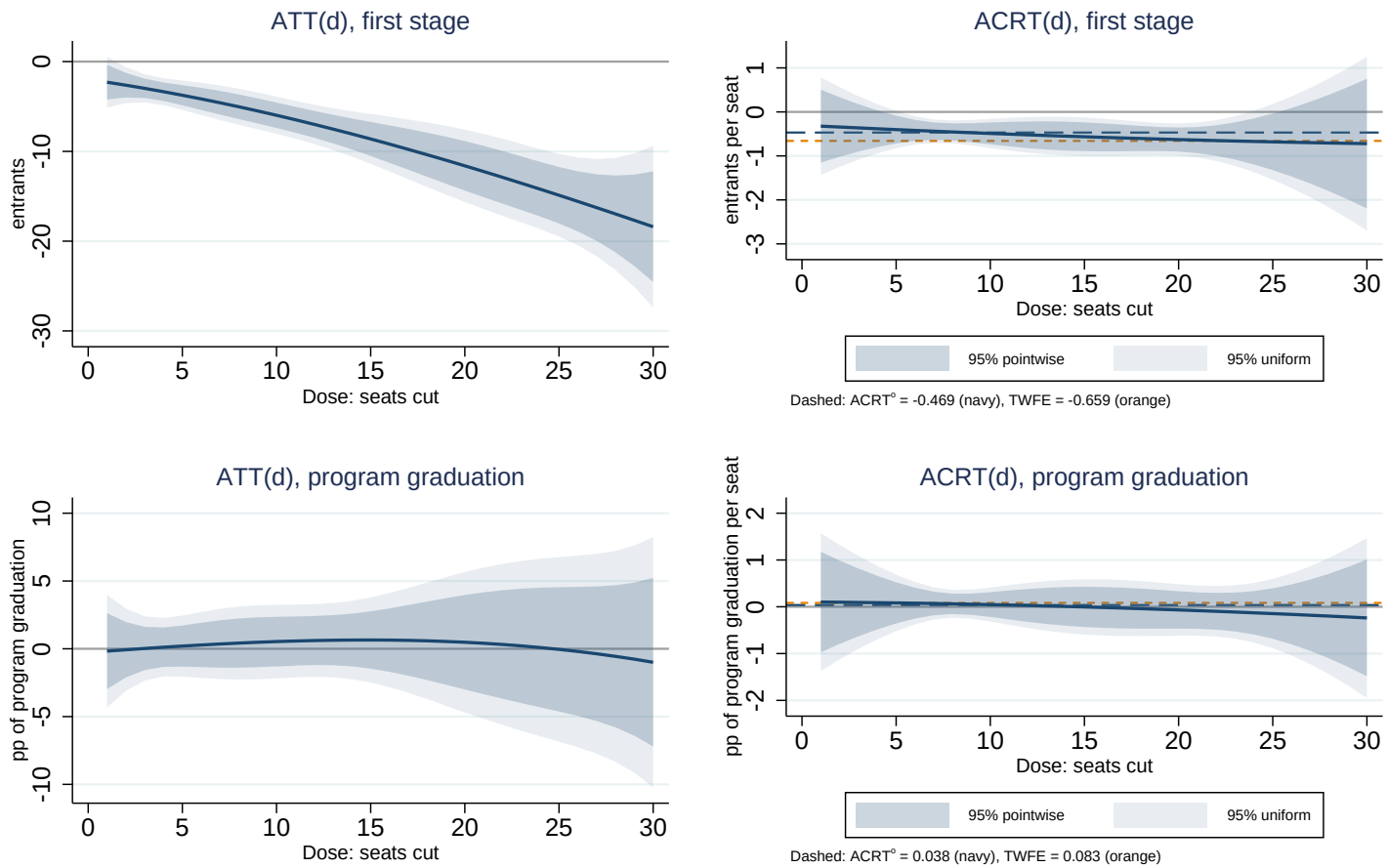
Comparison: untreated programs in the same university and year



Notes: Cubic in the dose; bands: 95 % pointwise (dark) and uniform (light), bootstrap clustered by program. The same-year comparison (FE (1) analog) gives the same picture. **Reading**: entrants rise almost linearly with the seats added, but program graduation of inframarginals does not: $ATT(d)$ is positive at small doses and indistinguishable from zero at large ones, and $ACRT(d)$ crosses zero. A capacity effect would make $ATT(d)$ fall steadily with d.

Continuous DiD: dose-response, contractions

Comparison: untreated programs in the same university and year



Notes: Dose = seats cut. Cuts lower entry roughly in proportion to the dose (about 0.5 entrants per seat). Inframarginal graduation is flat in the dose: $ATT(d) \approx 0$ and $ACRT(d) \approx 0$ over the whole range.

Design I: what the continuous DiD adds

1. The **first stage survives**: each seat added or cut moves entry by about two-thirds of an entrant, all along the dose range.
2. **No dose-response in inframarginal graduation.** $ACRT^o$ is close to zero per seat and per entrant, but imprecise: we cannot reject zero or the Design I estimate ($-0{,}033$ pp per entrant).
3. Programs that **expand at all** see inframarginal graduation rise by about 1.8 pp, whatever the size of the expansion. An effect that does not scale with the number of entrants does not look like congestion or peer effects. More likely, programs that decide to expand differ from those that do not (a violation of parallel trends on the extensive margin). The pre-period ATT^o is negative ($-0{,}7$ to $-0{,}8$, n.s.), consistent with mean reversion.
4. The negative Design I estimate with university $\times$ year FE **comes from contractions**, which follow a decline in graduation (pre-period TWFE slope significant at 5–10 %). Together with the lead placebo, this weakens a causal reading of the $-0{,}033$.

Inframarginal Design No. 2

Design II: what happens when a selective program expands its seats?

Research question

When a selective program suddenly increases its regular-admission vacancies, what happens to (a) its own first-year enrollment, (b) the composition of its entering cohort, and (c) the outcomes of its inframarginal students?

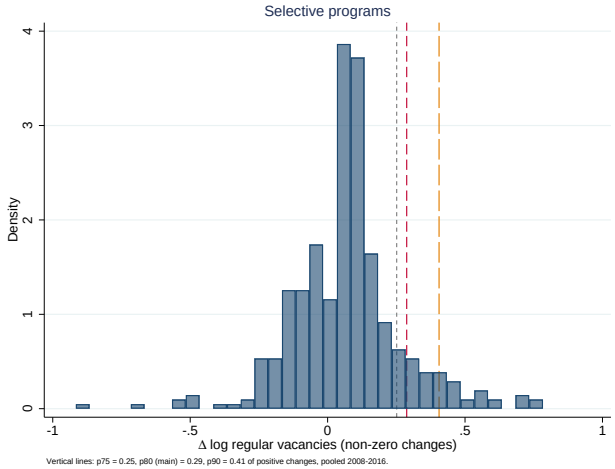
Approach

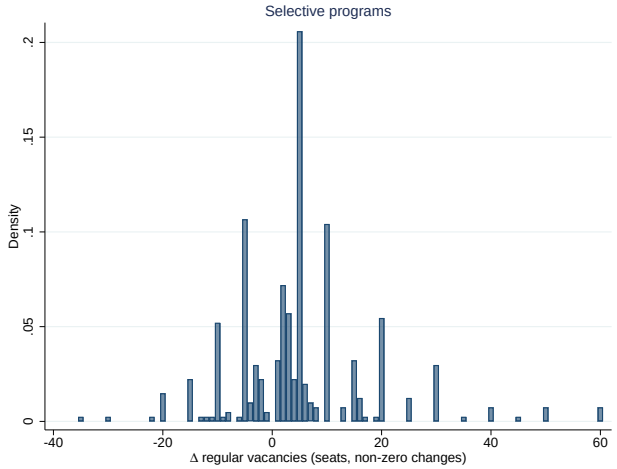
- A simple DiD built on Design I, but with a *binary* treatment: a large, sudden own expansion.
- Design I uses every year-to-year change in vacancies, most of which are small and routine. Here, identification comes from discrete capacity decisions, and the event study makes pre-trends testable.
- The shock is the program's own decision, so it is not exogenous to its demand. Leads in the event study are the main diagnostic.
- Questions (a) and (b) are DiD effects of the shock. Question (c) uses the shock as an *instrument* for the size of the entering cohort, with the inframarginal graduation outcome of Design I in the second stage (2SLS).

Sample

- Incumbent program $\times$ admission-year panel, 2007–2016 (programs with 2007–2009 entrants; SUA entrants excluded): 796 programs.
- **Selective programs**: top 20 % of 2007–2009 mean entrant PSU (predetermined; cutoff 633.6 points). 169 programs.

Design II: distribution of vacancy changes, selective programs





			Shocks, 2010–2016		Programs ever treated	
Definition	$\Delta \log V$ threshold	Increase	All	Selective	All	Selective
Top 25 %	0.251	+ 29 %	183	41	141	27
Top 20 % (main)	0.288	+ 33 %	159	33	119	20
Top 10 %	0.405	+ 50 %	73	16	58	10
Programs in sample					796	169

- **Large change** = Design IV shock: top 20 % of positive $\Delta \log V$ ($\geq +33$ %; many increases sit exactly there) and sudden.
- Only 20 selective programs are ever treated. The top 25 % and top 10 % thresholds serve as robustness checks.

Design II: construction of variables and models

Treatment

- $\text{shock}_{pt} = 1$ if $\Delta \log V_{pt}$ is in the top 20 % of positive changes, $V_{p,t-2}$ is observed, p was not shocked in $t-1$, and the increase is not reversed in $t+1$. Shocks are dated 2010–2016 (identical to Design IV's source shocks).
- g_p: first shock year (staggered). $\text{Treated}_{pt} = \mathbf{1}\{t \geq g_p\}$, absorbing.

Outcomes: regular vacancies V_{pt} (mechanical check), first-year enrollment N_{pt} (Formulario D), and composition of entrants.

(A) TWFE, two sets of fixed effects, reported as columns (1) and (2) of the tables:

(1) $y_{pt} = \beta\, \text{Treated}_{pt} + \mu_p + \lambda_t + \varepsilon_{pt}$ (2) $y_{pt} = \beta\, \text{Treated}_{pt} + \mu_p + \lambda_{u(p)t} + \varepsilon_{pt}$

μ_p: program FE. λ_t: year FE. $\lambda_{u(p)t}$: university × year FE (as in Design I, column (2)).

(B) Event study, FE of (2)

$$y_{pt} = \sum_{\ell \neq -1} b_\ell\, \mathbf{1}\{t - g_p = \ell\} + \mu_p + \lambda_{u(p)t} + e_{pt}, \qquad \ell \in \{\leq -5, \ldots, \geq 4\}$$

(C) Callaway & Sant'Anna: never treated only, and not yet treated plus never treated. Balanced panel; program and year only.

- SE clustered by program.
- (D) Inframarginal outcomes, question (c): 2SLS with Treated_{pt} as the instrument for N_{pt} and the event study (B) of its reduced form. The 2SLS results close this design.

Design II (D): 2SLS for inframarginal graduation

Idea. The sudden expansion raises the size of the entering cohort (first stage = result (a)). We ask how much the graduation of the students who would have been admitted anyway changes per additional entrant.

Variables

- Endogenous regressor: N_{pt}, first-year enrollment of p in t (all entrants, Formulario D).
- Instrument: $\text{Treated}_{pt} = \mathbf{1}\{t \geq g_p\}$, after the first own sudden expansion (same treatment as (A)).
- Outcome: y_{pt}, the graduation rate (%) of the *inframarginal applicants* of p in t, **exactly as in Design I**: applicants with $Rank_{ipt} \leq T_p$ (Step 1, First Cohort main, Minimum Cohort robustness), graduation from the program of first enrollment within 8 years, non-enrollees count as zeros (Step 2). Secondary: any university, any HE (8y), and program graduation at 10 years (cohorts ≤ 2014). T_p is fixed per program, so the shock does not mechanically change who is inframarginal.

Model, same FE sets (1) and (2) as the TWFE:

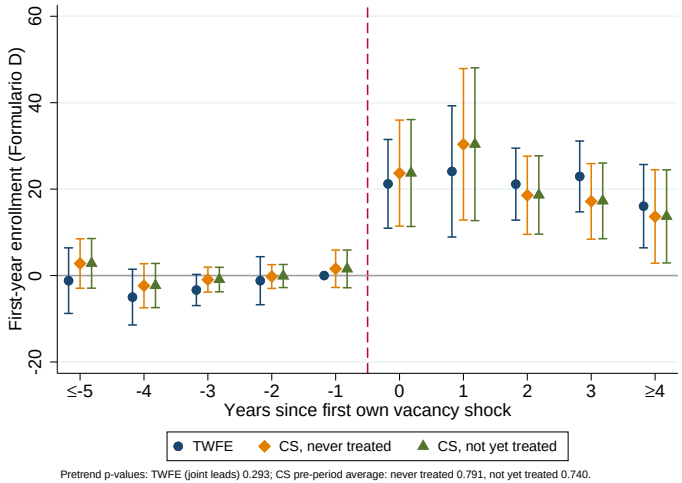
$$N_{pt} = \pi\, \text{Treated}_{pt} + \mu_p + \lambda + \nu_{pt}, \qquad y_{pt} = \beta\, \widehat{N}_{pt} + \mu_p + \lambda + \varepsilon_{pt}, \qquad \beta = \frac{\rho}{\pi}$$

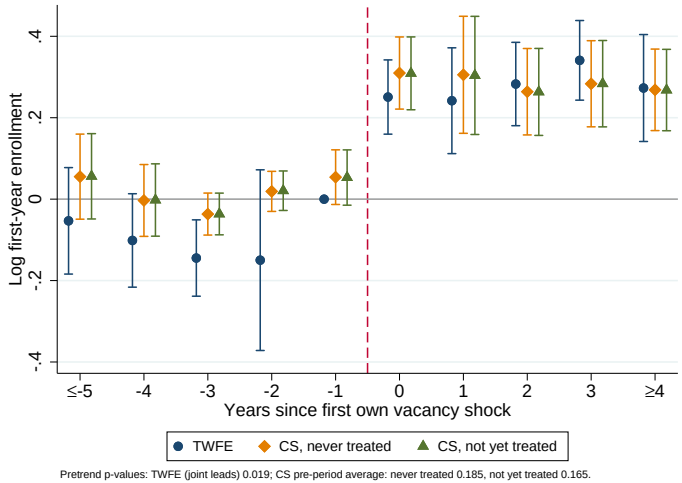
λ: year FE in (1); university $\times$ year FE in (2). ρ: reduced form (effect of Treated on y).

- β: pp of inframarginal graduation per additional entrant. Same units as Design I's 2SLS.
- Differences with Design I: binary, large and discrete instrument instead of every yearly ΔZ; levels with program FE instead of first differences; unweighted. FS, RF and 2SLS use the same program-years (non-missing y). SE clustered by program.
- **Exclusion**: the shock also changes *who* enters (entrant PSU falls, see composition). β is the effect of an expansion of one entrant, size and composition together, not of class size alone.

Design II: results, own enrollment

	Selective programs				All programs	
	TWFE (1)	TWFE (2)	CS never	CS not yet	TWFE (1)	TWFE (2)
Regular vacancies	24.237***	24.481***	22.784***	22.818***	18.196***	18.162***
	(4.337)	(4.448)	(4.219)	(4.217)	(1.573)	(1.640)
Pre-mean treated; N		49.87; 1,883			40.37; 8,915	
First-year enrollment (Formulario D)	22.215***	23.381***	19.541***	19.592***	14.565***	15.703***
	(4.422)	(4.521)	(4.792)	(4.829)	(1.767)	(1.774)
Pre-mean treated; N		54.54; 1,883			42.55; 8,915	
Log first-year enrollment	0.370***	0.364***	0.276***	0.275***	0.275***	0.299***
	(0.049)	(0.044)	(0.039)	(0.039)	(0.040)	(0.037)
Pre-mean treated; N		3.91; 1,882			3.54; 8,865	





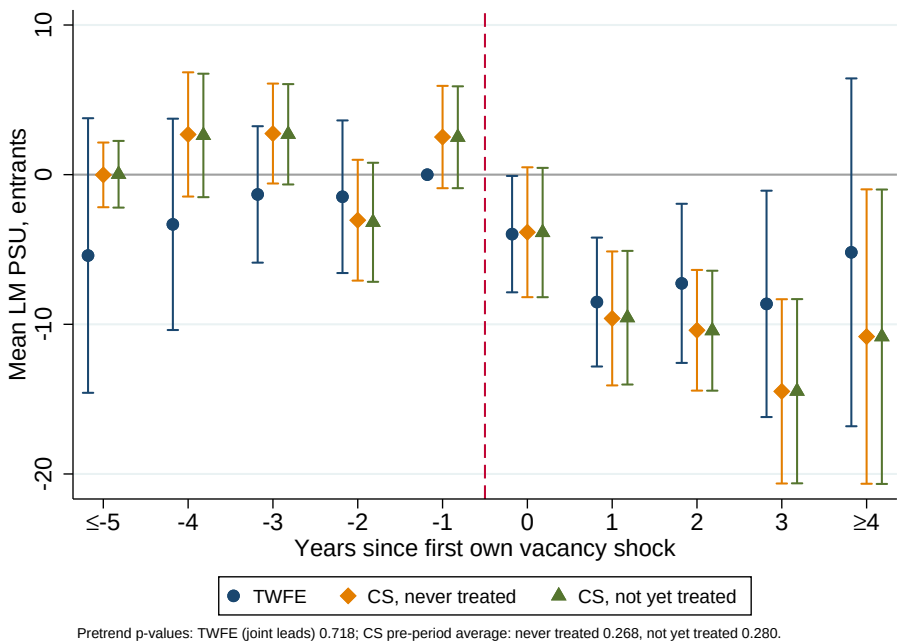
Notes: TWFE (1): program and year FE. TWFE (2): program and university × year FE. CS: Callaway–Sant'Anna simple ATT. SE clustered by program. Event studies: selective programs. *** $p < 0{,}01$, ** $p < 0{,}05$, * $p < 0{,}10$. After a sudden expansion, selective programs add about 24 seats and 23 entrants (pass-through $\approx 0{,}96$, +36 log points), with flat leads in levels (TWFE (2) joint-leads $p = 0{,}29$); the leads of log enrollment reject ($p = 0{,}02$).

Design II: composition of the entering cohort

	TWFE (1)	TWFE (2)
Mean LM PSU, entrants	-5.071***	-5.128***
	(1.424)	(1.421)
Pre-mean treated; N	584.11; 8,865	
SD of LM PSU, entrants	-0.103	-0.391
	(0.621)	(0.595)
Pre-mean treated; N	34.03; 8,726	
Mean NEM, entrants	-7.695***	-6.589***
	(2.175)	(2.184)
Pre-mean treated; N	598.05; 8,865	
Female, entrants, pct.	-0.510	-0.345
	(0.752)	(0.807)
Pre-mean treated; N	50.29; 8,865	
Private-paid school, entrants, pct.	-0.383	-0.181
	(0.385)	(0.429)
Pre-mean treated; N	10.97; 8,865	
Municipal school, entrants, pct.	1.584**	1.532**
	(0.731)	(0.726)
Pre-mean treated; N	31.12; 8,865	
Private-paid school, last admitted, pct.	3.453**	4.433**
	(1.628)	(1.730)
Pre-mean treated; N	9.07; 8,847	

Notes: All incumbent programs. TWFE (1): program and year FE. TWFE (2): program and university $\times$ year FE. SE clustered by program. Shares are in percentage points. Last admitted = lowest application score among regular admits. Income is not available in the data (Formulario A has no income; the socioeconomic questionnaire is not in `data/`), so school dependency is the SES proxy.

Design II: composition event studies and threshold robustness



Pretrend p-values: TWFE (joint leads) 0.718; CS pre-period average: never treated 0.268, not yet treated 0.280.

Outcome	Top 25 %	Top 20 % (main)	Top 10 %
Regular vacancies	31.032***	24.481***	30.683***
	(7.391)	(4.448)	(7.128)
First-year enrollment (Formulario D)	29.600***	23.381***	28.790***
	(7.484)	(4.521)	(6.981)
Log first-year enrollment	0.333***	0.364***	0.434***
	(0.037)	(0.044)	(0.070)
Mean LM PSU, entrants	-7.353***	-4.642*	-11.847***
	(2.286)	(2.557)	(3.297)
Private-paid school, entrants, pct.	0.312	0.880	-1.580
	(1.147)	(1.424)	(1.976)

- Event study and table: selective programs. Mean entrant PSU falls by 5 points (TWFE (2)) to 9 points (CS), about 0.1–0.15 SD of program-year means, with flat leads ($p = 0{,}72$).
- Results are robust to the top 25 % and top 10 % shock definitions.
- The table uses program and university $\times$ year FE.

Design II (D): reduced form, inframarginal graduation

	Selective programs			All programs	
Outcome	TWFE (1)	TWFE (2)	CS not yet	TWFE (1)	TWFE (2)
Grad. program 8y, First Cohort (pct.)	-6.48***	-1.46	-1.30	0.50	0.71
	(1.94)	(2.45)	(1.86)	(1.14)	(1.19)
Grad. university 8y, First Cohort (pct.)	-2.65	-1.52		0.21	-0.04
	(1.65)	(1.73)		(0.93)	(0.93)
Grad. HE 8y, First Cohort (pct.)	-2.74*	-1.60		-0.00	-0.28
	(1.64)	(1.72)		(0.93)	(0.93)
Grad. program 10y, First Cohort (pct.)	-0.64	3.78		-0.68	-0.22
	(4.59)	(4.34)		(2.09)	(2.14)
Grad. program 8y, Minimum Cohort (pct.)	-6.64***	-1.22	-2.63	0.57	0.69
	(2.00)	(2.50)	(2.03)	(1.16)	(1.20)
Grad. university 8y, Minimum Cohort (pct.)	-2.80	-1.49		0.27	-0.06
	(1.71)	(1.82)		(0.98)	(0.97)
Grad. HE 8y, Minimum Cohort (pct.)	-2.90*	-1.58		0.17	-0.19
	(1.70)	(1.81)		(0.98)	(0.96)
Grad. program 10y, Minimum Cohort (pct.)	-0.33	4.24		-0.22	0.34
	(4.51)	(4.23)		(2.14)	(2.17)
Observations (8y, First Cohort)	1,842	1,825		6,628	6,628

Notes: Reduced form of the 2SLS below: effect of Treated_{pt} on the inframarginal graduation outcomes of Design I (Steps 1–2), for both thresholds. Graduation uses titulados through 2024; 10-year outcomes cover cohorts 2007–2014. TWFE (1): program and year FE. TWFE (2): adds university $\times$ year FE. CS: Callaway–Sant'Anna, not yet treated. SE clustered by program. Caveat: the shock can change *who* applies, hence who is inframarginal, even though T_p is fixed; these are effects on that group's graduation rate. *** $p < 0{,}01$, ** $p < 0{,}05$, * $p < 0{,}10$.

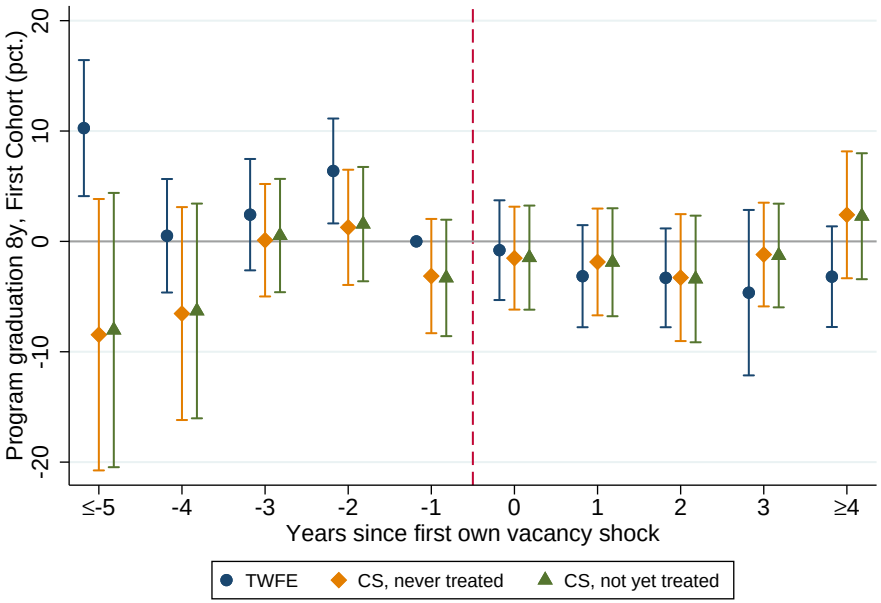
- Among selective programs, program graduation falls by 6.5 pp with program and year FE, but the estimate shrinks and loses significance with university $\times$ year FE ($-1{,}5$) and with Callaway–Sant'Anna ($-1{,}3$). No effect across all programs. The event studies follow.

Design II (D): reduced-form event study, inframarginal graduation

Outcome = second-stage y_{pt} (program graduation 8y of inframarginal applicants, First Cohort); event time around the first own sudden expansion g_p; selective programs:

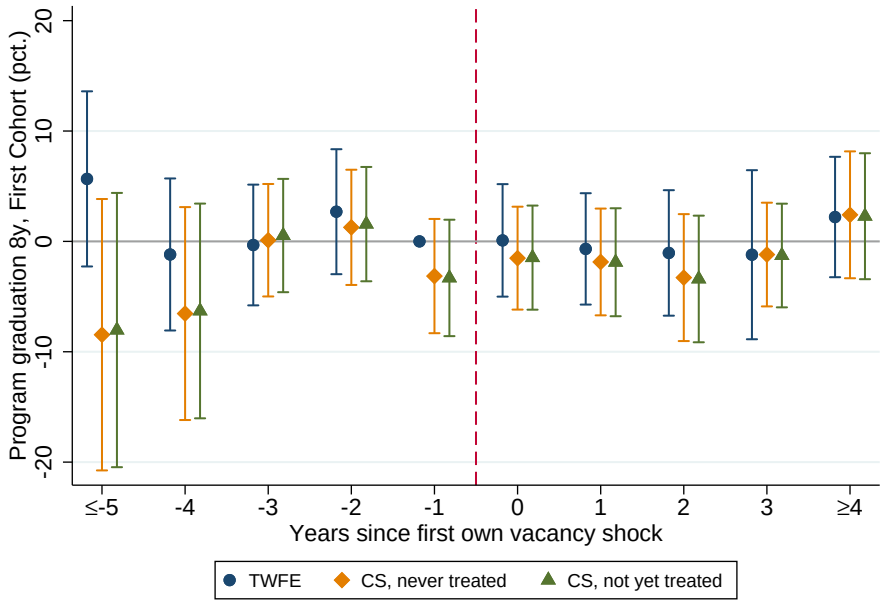
$$y_{pt} = \sum_{\ell \neq -1} b_\ell \, \mathbf{1}\{t - g_p = \ell\} + \mu_p + \lambda + e_{pt}, \qquad \ell \in \{\leq -5, \ldots, \geq 4\}$$

(1) Program and year FE



Pretrend p-values: TWFE (joint leads) 0.001; CS pre-period average: never treated 0.008, not yet treated 0.012.

(2) Program and university × year FE

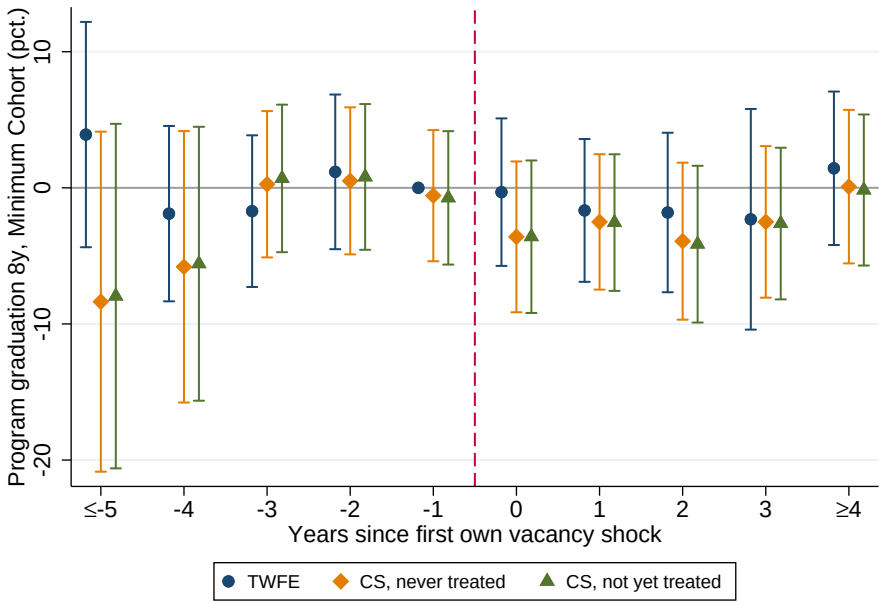


Pretrend p-values: TWFE (joint leads) 0.364; CS pre-period average: never treated 0.008, not yet treated 0.012.

- With FE (1) the leads reject (joint $p = 0{,}001$): graduation is already falling before the shock ($+10$ pp at ≤ -5, $+6$ pp at -2). The $-6{,}5^{***}$ reduced form, and the $-0{,}28^{***}$ 2SLS, compare post years with a high pre-period.
- With FE (2) the leads pass ($p = 0{,}36$) and the post coefficients are about -1 to $+2$ pp, none significant. CS (program and year only) also fails its pre-period test ($p = 0{,}008$ never treated, $0{,}012$ not yet) and gives post effects near -2 pp.

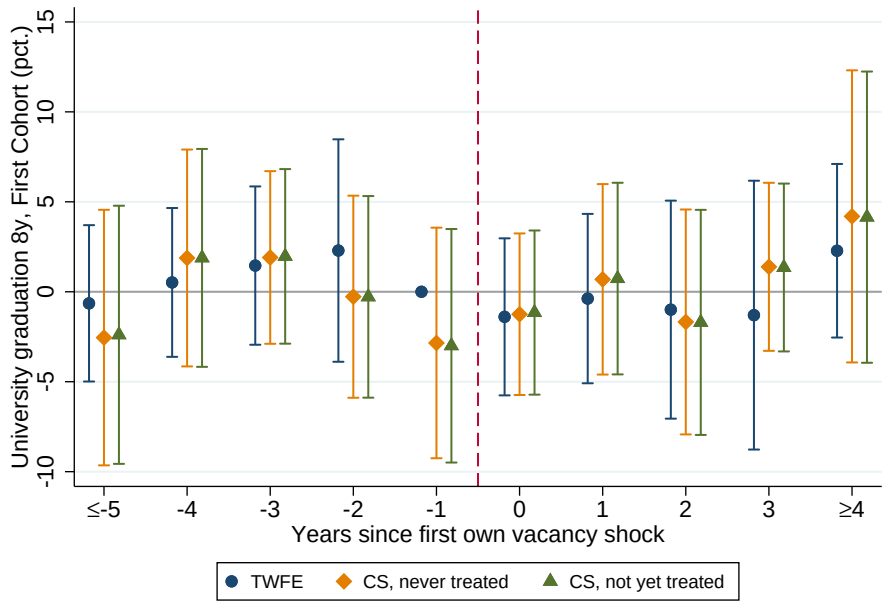
Design II (D): reduced-form event study, other outcomes

Program graduation 8y, Minimum Cohort



Pretrend p-values: TWFE (joint leads) 0.591; CS pre-period average: never treated 0.038, not yet treated 0.054.

University graduation 8y, First Cohort



Pretrend p-values: TWFE (joint leads) 0.794; CS pre-period average: never treated 0.584, not yet treated 0.591.

- Selective programs, TWFE with FE (2); CS with program and year. Minimum Cohort: same pattern as First Cohort (TWFE leads $p = 0{,}59$; CS pre-period $p = 0{,}04$ / $0{,}05$).
- Graduation from any university: flat leads (TWFE $p = 0{,}79$; CS $p = 0{,}58$) and no visible post effect.
- Bottom line: the reduced form behind the Design II 2SLS does not survive the event study; the negative estimate under FE (1) reflects a pre-existing decline, and with university $\times$ year FE it is small and not significant.

Design II (D): results, inframarginal graduation

	Selective programs		All programs	
	(1)	(2)	(1)	(2)
Panel A. First Cohort threshold				
First stage: N_{pt} on Treated$_{pt}$	23.45***	25.04***	17.90***	18.94***
	(4.47)	(4.50)	(1.93)	(1.96)
Kleibergen–Paap F	27.5	30.9	85.7	93.6
Reduced form: program graduation, 8y	-6.48***	-1.46	0.50	0.71
	(1.94)	(2.45)	(1.14)	(1.19)
2SLS: pp of graduation per additional entrant				
Graduation from the program, 8y	-0.276***	-0.058	0.028	0.037
	(0.100)	(0.098)	(0.064)	(0.062)
Graduation from any university, 8y	-0.113	-0.061	0.012	-0.002
	(0.073)	(0.070)	(0.052)	(0.049)
Graduation from any HE, 8y	-0.117	-0.064	-0.000	-0.015
	(0.073)	(0.070)	(0.052)	(0.049)
Graduation from the program, 10y	-0.029	0.171	-0.040	-0.013
	(0.212)	(0.197)	(0.122)	(0.125)
Pre-shock mean, program graduation 8y, treated (%)	42.8	42.8	32.0	32.0
Program-years (8y)	1,842	1,825	6,628	6,628
Panel B. Minimum Cohort threshold				
First stage: N_{pt} on Treated$_{pt}$	23.45***	25.04***	17.90***	18.94***
	(4.47)	(4.50)	(1.93)	(1.96)
Kleibergen–Paap F	27.5	30.9	85.7	93.6
Reduced form: program graduation, 8y	-6.64***	-1.22	0.57	0.69
	(2.00)	(2.50)	(1.16)	(1.20)
2SLS: pp of graduation per additional entrant				
Graduation from the program, 8y	-0.283***	-0.049	0.032	0.036
	(0.103)	(0.100)	(0.065)	(0.063)
Graduation from any university, 8y	-0.120	-0.059	0.015	-0.003
	(0.076)	(0.073)	(0.055)	(0.051)
Graduation from any HE, 8y	-0.124	-0.063	0.009	-0.010
	(0.076)	(0.073)	(0.055)	(0.051)
Graduation from the program, 10y	-0.015	0.192	-0.013	0.020
	(0.208)	(0.192)	(0.124)	(0.126)
Pre-shock mean, program graduation 8y, treated (%)	43.2	43.2	33.3	33.3
Program-years (8y)	1,842	1,825	6,628	6,628
Program and year FE	✓		✓	
Program, university × year FE		✓		✓

- First stage: a sudden expansion adds 23–25 entrants in selective programs (F 28–31), 18–19 in all programs.
- Selective, program + year FE: $-0{,}28^{***}$ pp of program graduation per entrant, about -6 pp for a typical shock (mean 42.8 %).
- With university × year FE: $-0{,}06$, not significant.
- Any university / any HE: about $-0{,}11$, not significant. Part of the loss is switching or delay.
- At 10 years the effect disappears ($-0{,}03$ / $+0{,}17$).
- All programs: zero. Both inframarginal definitions give the same estimates.

Notes: (1) program and year FE; (2) program and university × year FE. Unweighted, SE clustered by program. 2SLS: pp of graduation per additional entrant (Design I, with university × year FE: $-0{,}033$). **Caveat**: the reduced form fails the CS pre-period test ($p = 0{,}008$) and CS not-yet-treated gives $-1{,}3$ (1.9), so the 2SLS inherits a fragile reduced form. *** $p < 0{,}01$, ** $p < 0{,}05$, * $p < 0{,}10$.

Inframarginal Design No. 3

Entry of private universities into the SUA

Research question

Did first-year enrollment in incumbent CRUCH programs change after eight private universities entered the SUA in 2012, particularly in programs with greater pre-existing exposure to these entrants?

Sample

- Incumbent CRUCH program $\times$ admission-year panel, 2007–2016.
- Outcome: number of students entering the first year of program p in year t.

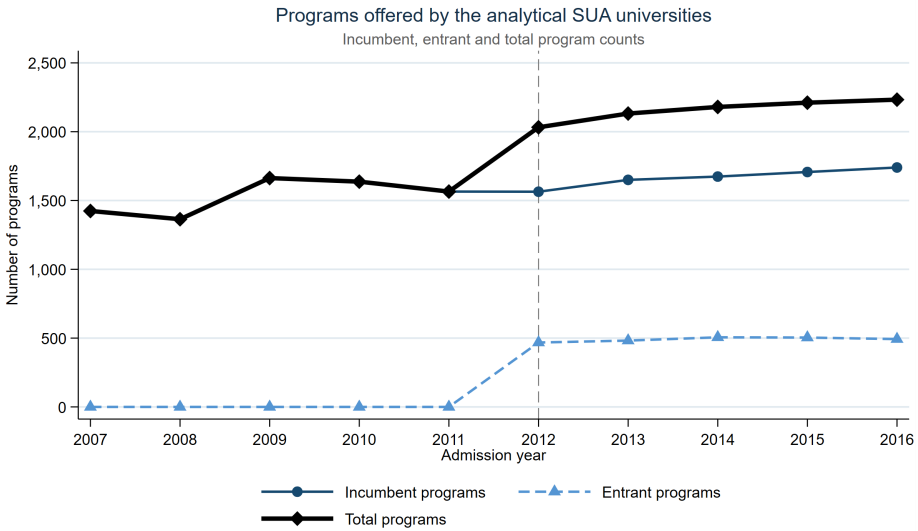
Exposure design

- Markets are defined by pre-treatment region $\times$ academic field.
- Exposure measures the 2009–2011 enrollment presence of the future entrant universities in the incumbent program's market.
- Total exposure uses the entrant enrollment share; Triangular and Gaussian measures additionally weight entrants by PSU-score proximity.
- Exposure is fixed over time and interacted with the post-entry period, 2012–2016.

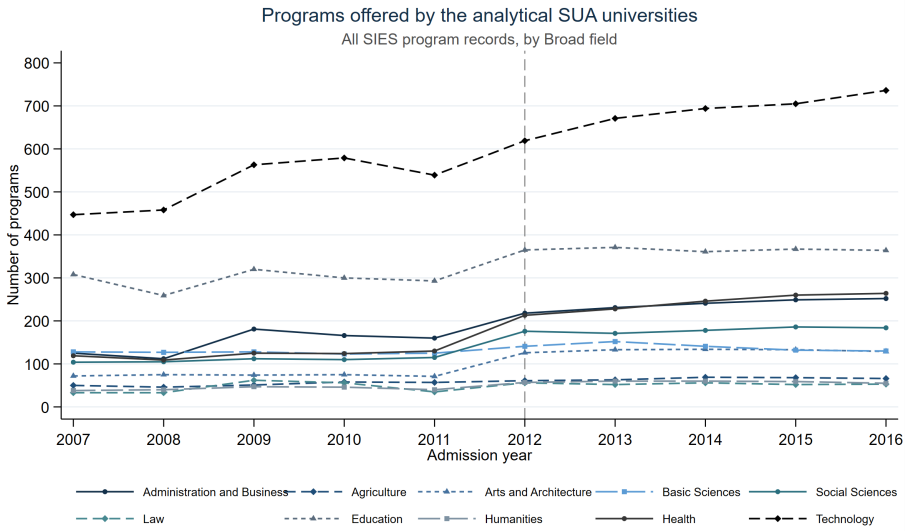
The design compares pre/post-2012 enrollment changes across incumbent programs with different levels of exposure to the future entrants.

Evolution of the number of SUA programs

Programs by university status



Programs by Broad field



- The incorporation of the eight entrant universities adds roughly 470 programs in 2012. Their program supply remains relatively stable afterward, while incumbents account for most subsequent growth.
- Technology represents the largest field and an important share of program expansion.

Construction of the exposure measures

1. Pre-reform enrollment in market m

$$A_m = \sum_{\substack{q \in \mathcal{E} \\ m(q)=m}} \overline{N}^{\text{first}}_{q,2009\text{–}2011}, \qquad T_m = \sum_{\substack{q \\ m(q)=m}} \overline{N}^{\text{first}}_{q,2009\text{–}2011}$$

- $\mathcal{E}$ is the set of entrant programs and $m(q)$ is program q's pre-treatment region $\times$ field market.
- $\overline{N}^{\text{first}}_{q,2009\text{–}2011}$ is average first-year enrollment: A_m is entrant and T_m total enrollment in m.

2. Total exposure

$$E_p = \frac{A_{m(p)}}{T_{m(p)}}$$

- Entrant share of pre-reform first-year enrollment in the market of incumbent program p.
- Incumbents in the same region $\times$ field market share the same E_p.

3. Selectivity-weighted exposure

$$E_p^k = \frac{\sum_{\substack{q \in \mathcal{E} \\ m(q)=m(p)}} k(|S_p - S_q|)\, \overline{N}^{\text{first}}_{q,2009\text{–}2011}}{T_{m(p)}}$$

- S_p, S_q: pre-reform mean PSU scores of incumbent p and entrant q; $|S_p - S_q|$ is their selectivity distance.
- $k \in \{\text{Triangular}, \text{Gaussian}\}$ gives greater weight to closer entrants, so E_p^k varies within a market.

4. Kernel-weighted denominator

$$E_p^{k,\text{KD}} = \frac{\sum_{\substack{q \in \mathcal{E} \\ m(q)=m(p)}} k(|S_p - S_q|)\, \overline{N}^{\text{first}}_{q,2009\text{–}2011}}{\sum_{\substack{\ell \\ m(\ell)=m(p)}} k(|S_p - S_\ell|)\, \overline{N}^{\text{first}}_{\ell,2009\text{–}2011}}$$

- Same numerator as in 3; the denominator weights every program ℓ in the market (incumbents, including p, and entrants) by its selectivity distance to p.
- $E_p^{k,\text{KD}} \in [0,1]$ is the entrant share of the enrollment that is close to p in selectivity.

SUA exposure: samples and descriptive statistics

Panel A: Analytical samples

Variable	Broad	ISCED-97	Generic
Programs	949	949	945
Program-year observations	8,041	8,030	8,002
Incumbent universities	25	25	25
Pre-treatment markets	130	215	638
Mean annual enrollment, 2009–2011	53.2	53.2	53.2
Mean PSU score, 2009–2011	588.8	588.8	589.0

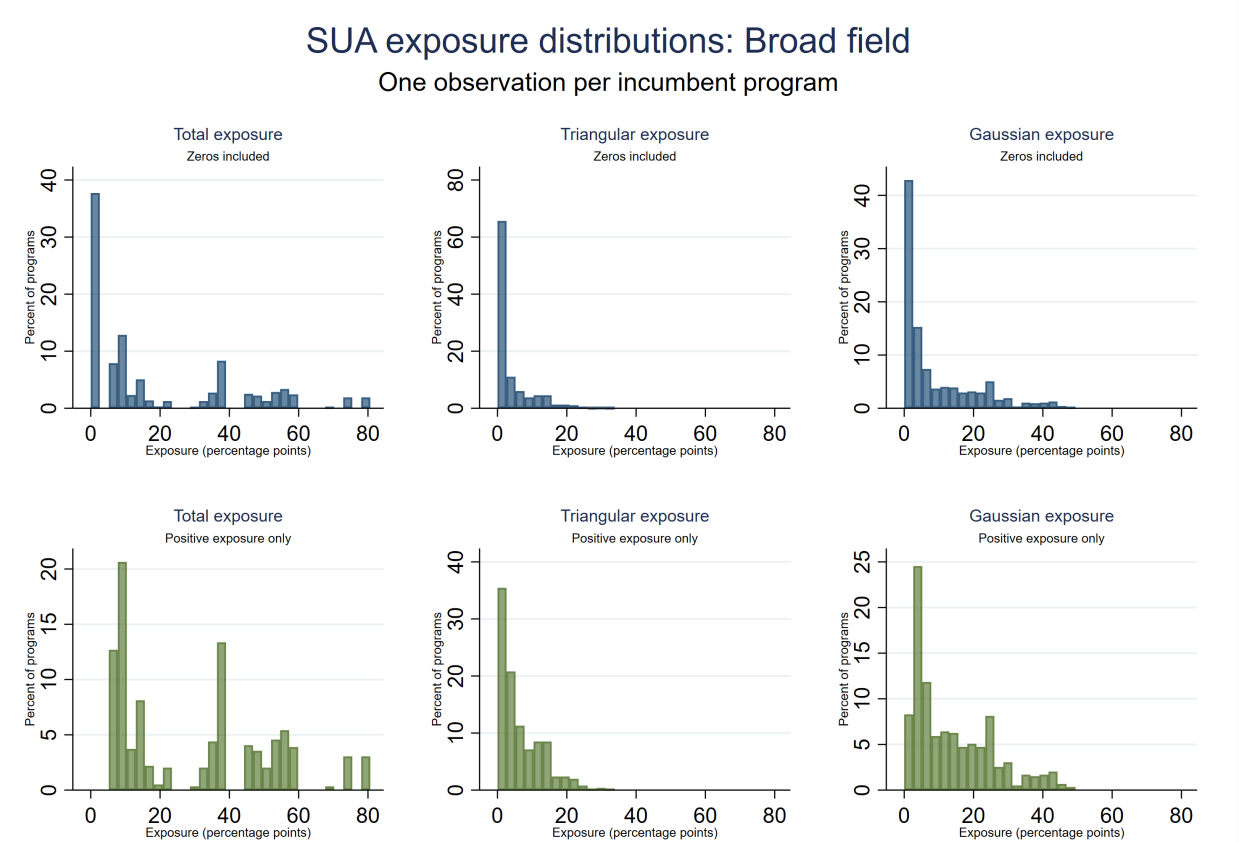
Panel B. Exposure by academic-field definition

Field definition	Exposure measure	Mean	Std. dev.	Share zero
Broad	Total	18.53	22.21	37.7 %
	Triangular	3.48	5.57	46.8 %
	Gaussian	8.58	11.26	37.7 %
ISCED-97	Total	17.88	22.96	43.0 %
	Triangular	3.16	5.79	56.9 %
	Gaussian	8.20	11.70	43.0 %
Generic	Total	13.27	23.40	66.2 %
	Triangular	1.83	6.13	83.4 %
	Gaussian	5.83	11.90	67.4 %

Notes: Panel A reports the analytical sample under each field definition. Enrollment is the unweighted mean across programs of average annual first-year enrollment in 2009–2011; PSU is the program's mean score over the same period. Panel B reports unweighted exposure moments in percentage points, including zero-exposure programs. Total is the entrant share of market enrollment; Triangular and Gaussian additionally weight entrant enrollment by PSU proximity using a 50-point bandwidth. Markets combine pre-treatment region with the corresponding field definition.

Distribution of SUA exposure measures

Alternative fields →



- Each incumbent program appears once. The horizontal axis reports exposure in percentage points and the vertical axis the share of programs.
- The upper row includes all programs; the lower row excludes zeros and shows the distribution conditional on positive exposure.
- Total exposure has the largest values. Triangular weighting is the most restrictive, while Gaussian weighting produces an intermediate distribution.

First-stage specification

$$N_{pt}^{\text{first}} = \pi_k \left(E_p^k \times Post_t \right) + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \nu_{pt}.$$

- p indexes incumbent programs and t years. N_{pt}^{first} is first-year enrollment of program p in year t, while E_p^k is its pre-reform exposure and $Post_t = 1$ in 2012–2016.
- $f(p)$ denotes the pre-treatment field of program p and $r(p)$ its pre-treatment region. Accordingly, $\alpha_{f(p),t}$ are field $\times$ year fixed effects and $\gamma_{r(p),t}$ are region $\times$ year fixed effects.
- μ_p denotes program fixed effects, and π_k is the post-2012 enrollment difference associated with a 10 percentage-point increase in exposure measure k.

Exposure is positive in only four regions

- In the 2009–2011 analytical sample, entrant universities offered regular undergraduate programs only in Metropolitana, Valparaíso, Biobío and La Araucanía.
- Thus, these are the only regions containing entrant programs and positive entrant enrollment before integration into the SUA.
- In every other region, the entrant-enrollment numerator is zero: $A_m = 0$, and therefore $E_p = A_{m(p)} / T_{m(p)} = 0$.

First stage: exposure to entrant universities

Event studies →

$$N_{pt}^{\text{first}} = \pi \left(E_p^{k,m} \times Post_t \right) + \mu_p + \alpha_{f_m(p),t} + \gamma_{r(p),t} + \nu_{pt}.$$

	Field × year FE			+ Region × year FE		
Exposure measure	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
Total	1.967***	1.893***	1.872***	0.798	0.904*	0.324
	(0.336)	(0.335)	(0.289)	(0.662)	(0.510)	(0.338)
Wald *F*	34.36	31.92	41.90	1.45	3.14	0.92
Triangular	0.935	1.392	2.185**	−5.252**	−3.415***	−0.975
	(0.972)	(0.934)	(0.927)	(2.115)	(1.310)	(0.829)
Wald *F*	0.93	2.22	5.56	6.17	6.80	1.39
Gaussian	1.775***	1.939***	2.342***	−2.767*	−1.447*	−0.558
	(0.439)	(0.450)	(0.533)	(1.401)	(0.832)	(0.571)
Wald *F*	16.36	18.60	19.33	3.90	3.03	0.96
Observations	8,041	8,030	7,756	8,038	8,027	7,753
Programs	949	949	922	948	948	921
Markets	130	215	615	129	214	614
Program FE	Yes	Yes	Yes	Yes	Yes	Yes
Field × year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region × year FE	No	No	No	Yes	Yes	Yes

Notes: The dependent variable is first-year enrollment in incumbent CRUCH programs. Total exposure is entrants' pre-treatment enrollment share; Triangular and Gaussian additionally weight entrant enrollment by PSU-score proximity using a 50-point bandwidth. Markets combine pre-treatment region with the indicated field definition. Exposure is interacted with $Post_t$, equal to one in 2012–2016, and measured in 10 percentage-point units. All regressions absorb program and corresponding field × year fixed effects; the last three columns additionally absorb region × year fixed effects. Standard errors are clustered by the corresponding pre-treatment market and reported in parentheses. Wald *F* statistics test the exposure coefficient. Sample differences reflect singleton removal by `reghdfe`. $^{***}p < 0{,}01$, $^{**}p < 0{,}05$, $^{*}p < 0{,}10$. Robustness: regions with entrant universities only → positive exposure only →

SUA exposure: log-log first stages

Event studies →

$$\log N_{pt}^{\text{first}} = \beta_k \left[\widetilde{\log E_p^k} \times Post_t\right] + \theta_k \left[D_p^k \times Post_t\right] + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \varepsilon_{pt},$$

$$\widetilde{\log E_p^k} = \begin{cases} \log(E_p^k), & E_p^k > 0, \\ 0, & E_p^k = 0, \end{cases} \qquad D_p^k = \mathbf{1}\{E_p^k > 0\}.$$

	Field × year FE			+ Region × year FE		
Exposure measure	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
Total	0.100*** (0.034)	0.072** (0.035)	0.021 (0.030)	−0.005 (0.039)	−0.024 (0.032)	−0.022 (0.027)
F	8.84	4.24	0.48	0.01	0.55	0.69
Triangular	−0.013 (0.015)	0.004 (0.016)	0.016 (0.021)	−0.050*** (0.017)	−0.024 (0.016)	−0.005 (0.017)
F	0.81	0.07	0.59	8.95	2.37	0.07
Gaussian	−0.026 (0.032)	−0.024 (0.030)	0.001 (0.018)	−0.107*** (0.030)	−0.079*** (0.026)	−0.025 (0.017)
F	0.68	0.64	0.01	12.72	9.35	2.19
Observations	8,041	8,030	7,756	8,038	8,027	7,753
Programs	949	949	922	948	948	921

Notes: Each coefficient reports β_k, the post-2012 elasticity with respect to exposure intensity. Zero-exposure programs are retained by setting $\widetilde{\log E_p^k} = 0$ when $E_p^k = 0$ and including $D_p^k \times Post_t$ separately; the coefficient on this indicator is not reported. Thus, β_k captures variation in exposure intensity among positively exposed programs while allowing zero-exposure programs to remain in the estimation sample. All specifications absorb program and corresponding field × year fixed effects; the last three columns additionally absorb pre-treatment region × year fixed effects. Standard errors are clustered at the pre-treatment market level. Observations with zero first-year enrollment are excluded because the dependent variable is in logs. $^{*}p < 0{,}10$, $^{**}p < 0{,}05$, $^{***}p < 0{,}01$. Robustness: regions with entrant universities only → positive exposure only →

First stage: kernel-weighted denominator

Event studies →

$$N_{pt}^{\text{first}} = \pi \left(E_p^{k,\text{KD}} \times Post_t \right) + \mu_p + \alpha_{f_m(p),t} + \gamma_{r(p),t} + \nu_{pt}.$$

	Field × year FE			+ Region × year FE		
Exposure measure	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
Triangular KD	0.659**	0.717**	1.371***	-1.726**	-1.286***	-0.208
	(0.299)	(0.292)	(0.481)	(0.737)	(0.427)	(0.470)
Wald *F*	4.86	6.03	8.13	5.48	9.08	0.20
Programs with zero exposure	444	540	767	443	539	766
Gaussian KD	1.326***	1.420***	1.913***	-1.919*	-1.049*	-0.218
	(0.346)	(0.350)	(0.394)	(1.051)	(0.600)	(0.472)
Wald *F*	14.70	16.42	23.56	3.33	3.06	0.21
Programs with zero exposure	358	408	616	357	407	615
Observations	8,041	8,030	7,756	8,038	8,027	7,753
Programs	949	949	922	948	948	921
Markets	130	215	615	129	214	614
Program FE	Yes	Yes	Yes	Yes	Yes	Yes
Field × year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region × year FE	No	No	No	Yes	Yes	Yes

Notes: Exposure uses the kernel-weighted denominator (point 4 of the exposure construction): the denominator weights every program in the market by its selectivity distance to p, with the same 50-point Triangular and Gaussian kernels as the numerator. Sample and fixed effects as in the main first stage; SE clustered by pre-treatment market. Exposure in 10 percentage-point units. As in the main first stage, coefficients are positive with field × year FE and negative with region × year FE. Magnitudes are smaller because KD exposure takes larger values (Broad, 2011 mean: 0.11 for Triangular KD vs. 0.03 for Triangular). $^{***}p < 0{,}01$, $^{**}p < 0{,}05$, $^{*}p < 0{,}10$.

Log first stage: kernel-weighted denominator

Event studies →

$$\log N_{pt}^{\text{first}} = \beta_k \left[\widetilde{\log E_p^{k,\text{KD}}} \times Post_t \right] + \theta_k \left[D_p^k \times Post_t \right] + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \varepsilon_{pt}.$$

	Field × year FE			+ Region × year FE		
Exposure measure	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
Triangular KD	-0.001	0.014	0.022	-0.052***	-0.032*	-0.014
	(0.014)	(0.018)	(0.023)	(0.018)	(0.017)	(0.018)
Wald *F*	0.00	0.60	0.96	8.86	3.83	0.60
Programs with zero exposure	444	540	767	443	539	766
Gaussian KD	-0.012	-0.018	0.009	-0.138***	-0.095***	-0.030*
	(0.036)	(0.032)	(0.019)	(0.037)	(0.028)	(0.018)
Wald *F*	0.12	0.32	0.24	13.59	11.31	3.01
Programs with zero exposure	358	408	616	357	407	615
Observations	8,041	8,030	7,756	8,038	8,027	7,753
Programs	949	949	922	948	948	921
Markets	130	215	615	129	214	614
Program FE	Yes	Yes	Yes	Yes	Yes	Yes
Field × year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region × year FE	No	No	No	Yes	Yes	Yes

Notes: Exposure uses the kernel-weighted denominator (point 4 of the exposure construction): the denominator weights every program in the market by its selectivity distance to p, with the same 50-point Triangular and Gaussian kernels as the numerator. Sample and fixed effects as in the main first stage; SE clustered by pre-treatment market. $\widetilde{\log E} = \log E$ if $E > 0$ and 0 otherwise; $D_p^k = \mathbf{1}\{E_p^{k,\text{KD}} > 0\}$. Observations with zero first-year enrollment are excluded. Compare with the main log first stage. $^{***}p < 0{,}01$, $^{**}p < 0{,}05$, $^{*}p < 0{,}10$.

First-stage heterogeneity by incumbent selectivity

	PSU $<$ 550		PSU 550–649		PSU $\geq$ 650	
	Tri.	Gau.	Tri.	Gau.	Tri.	Gau.
SUA exposure						
Coefficient	−1,61	−0,54	−3,17**	−0,98	−15,75*	−5,01
	(2,78)	(2,10)	(1,54)	(0,83)	(8,25)	(4,38)
F-statistic	0.34	0.07	4.26	1.41	3.65	1.31
Observations	2,215		4,609		1,188	
Programs	272		531		143	
Clusters	89		100		25	

Notes: Standard errors clustered by pre-treatment market in parentheses. All regressions include program, Broad-field × year, and region × year FE. Coefficients correspond to a 10 percentage-point increase in exposure. The same split for conditional similarity is in the appendix. $^{*}p < 0{,}10$, $^{**}p < 0{,}05$, $^{***}p < 0{,}01$.

Cosine exposure

Each program is represented by a vector describing the distribution of its 2011 first-year students. Each coordinate counts students with a particular PSU profile and, depending on the definition, campus region and Broad field. Programs are more similar when their students are distributed across the same vector cells.

1. Program vectors

$$\mathbf{v}_p^{PSU} = (v_{p,b})_{b\in\mathcal{B}}, \qquad \mathbf{v}_p^{PSU\times R} = (v_{p,rb})_{\substack{r\in\mathcal{R}\\ b\in\mathcal{B}}}, \qquad \mathbf{v}_p^{PSU\times R\times F} = (v_{p,rfb})_{\substack{r\in\mathcal{R},\ f\in\mathcal{F}\\ b\in\mathcal{B}}}.$$

Here, b denotes a 25-point PSU bin, r campus region, and f Broad field. Each vector cell contains the number of first-year students in that combination. Entrant-program vectors are constructed analogously.

2. Similarity between incumbent p and entrant j

$$\cos\theta_{j,p}^{d} = \frac{\mathbf{v}_j^d \cdot \mathbf{v}_p^d}{\|\mathbf{v}_j^d\|\,\|\mathbf{v}_p^d\|}, \qquad d \in \{PSU,\ PSU\times R,\ PSU\times R\times F\}.$$

The cosine compares the **shape of the student distributions**, rather than program size. Under interaction-cell definitions, similarity is zero when programs do not share the indicated region or field.

3. Additive similarity measures

$$\cos\theta_{j,p}^{PSU+R} = \frac{\cos\theta_{j,p}^{PSU} + \mathbf{1}\{r_j = r_p\}}{2}, \qquad \cos\theta_{j,p}^{PSU+R+F} = \frac{\cos\theta_{j,p}^{PSU} + \mathbf{1}\{r_j = r_p\} + \mathbf{1}\{f_j = f_p\}}{3}.$$

These measures give equal weight to PSU-profile similarity and coincidence in campus region and, where applicable, Broad field.

4. Exposure of incumbent program p

$$E_p^d = \sum_{j\in\mathcal{J}} w_j \cos\theta_{j,p}^{d}, \qquad w_j = \frac{N_{j,2011}^{\text{first}}}{\sum_{\ell\in\mathcal{J}} N_{\ell,2011}^{\text{first}}}.$$

Exposure is higher when incumbent p resembles entrant programs that account for a larger share of entrant enrollment.

Notes: $\mathcal{J}$ contains programs from the eight entrant universities. Campus region is the university program's location in 2011. Student school region and SES are not used.

Cosine exposure: samples and descriptive statistics

Panel A: Analytical samples

Variable	Full sample	$N_{p,2011}^{\text{first}} \geq 10$
Programs	996	939
Program-year observations	9,105	8,618
Incumbent universities	25	25
Total first-year enrollment, 2011	54,659	54,303
Mean enrollment per program, 2011	54.9	57.8
Mean annual enrollment, 2007–2011	54.8	57.3
Mean PSU score, 2011	578.0	580.9

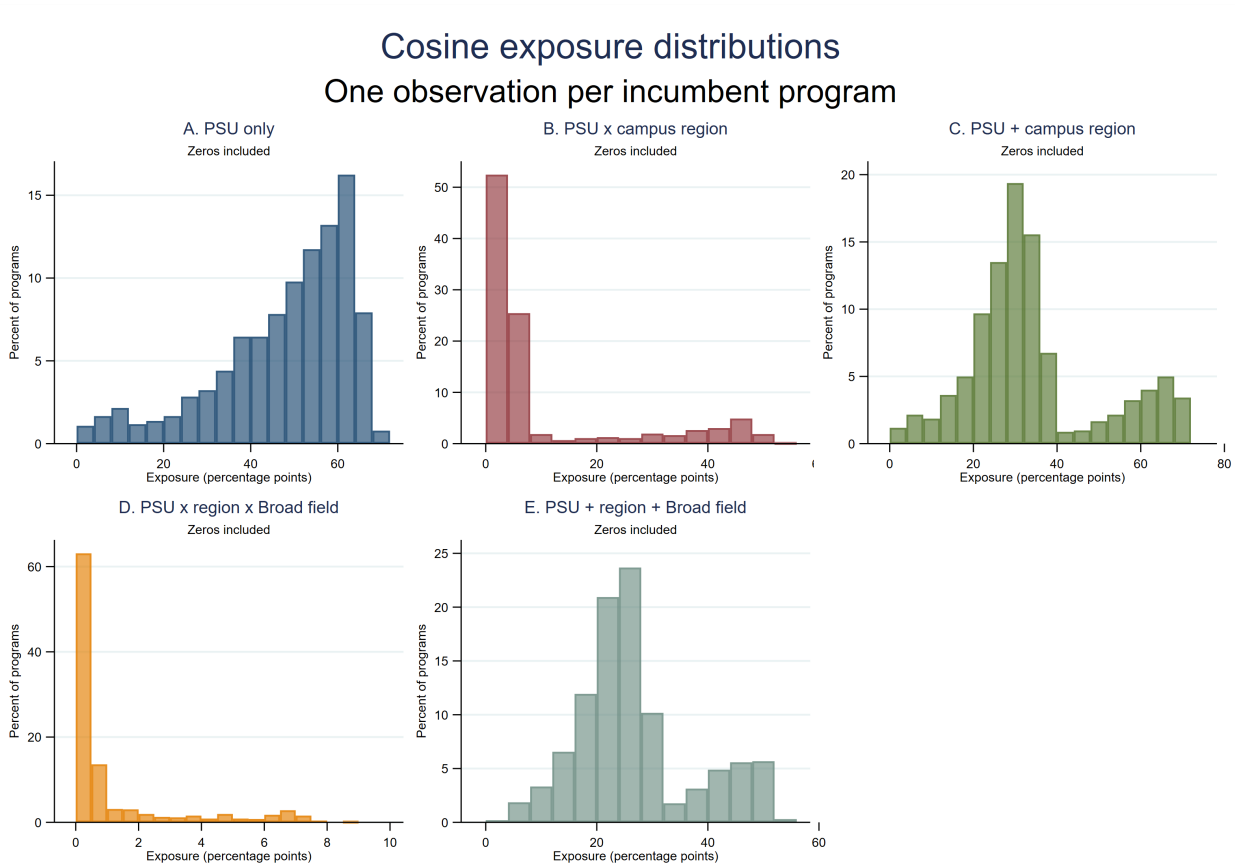
Panel B: Distribution of raw cosine exposure

	Full sample			$N_{p,2011}^{\text{first}} \geq 10$		
Exposure measure	Mean	Std. dev.	Share zero	Mean	Std. dev.	Share zero
PSU bins	0.4762	0.1559	0.0 %	0.4842	0.1540	0.0 %
PSU bins × campus region	0.0977	0.1495	37.0 %	0.1024	0.1521	34.7 %
PSU bins + campus region	0.3405	0.1614	0.0 %	0.3490	0.1604	0.0 %
PSU bins × campus region × Broad field	0.0113	0.0196	38.3 %	0.0119	0.0200	35.8 %
PSU bins + campus region + Broad field	0.2681	0.1078	0.0 %	0.2745	0.1064	0.0 %

Notes: The unit is an incumbent SUA program and exposure moments are unweighted across programs. The restricted sample excludes 57 programs with fewer than 10 first-year students in 2011. Multiplicative measures require coincidence in the indicated dimensions; additive measures assign equal weight to PSU similarity, campus-region coincidence and, where applicable, Broad-field coincidence. The restriction is applied after exposure construction, so entrant profiles, weights, and exposure scaling remain unchanged. Campus region is the university program's location; student school region and SES are not used.

Distribution of cosine exposure measures

Positive exposure only →



Zeros included. Exposure measured in percentage points; each panel uses its own x-axis scale.

- Each incumbent program appears once. The horizontal axis reports cosine exposure in percentage points and the vertical axis the share of programs.
- Multiplicative definitions are more restrictive. In particular, PSU × campus region × Broad field places substantial mass at zero and has a much smaller support than the PSU-only and additive measures.

Cosine exposure: first-stage results

Event studies →

$$N_{pt}^{\text{first}} = \beta_k(E_p^k \times Post_t) + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \varepsilon_{pt}.$$

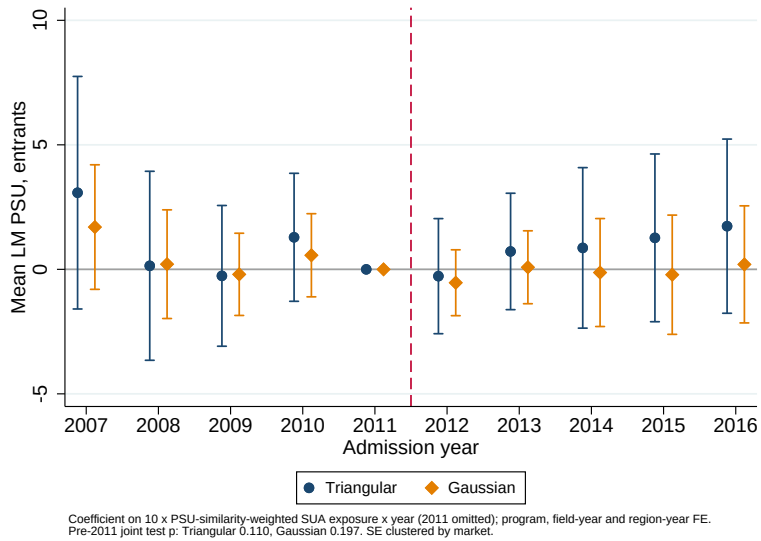
	Full sample			$N_{p,2011}^{\text{first}} \geq 10$		
	(1)	(2)	(3)	(4)	(5)	(6)
Exposure measure	+ year	+ field×year	+ region×year	+ year	+ field×year	+ region×year
PSU bins	−1,811*	−1,403	−0,716	−2,116**	−1,597	−0,630
	(0,948)	(1,018)	(1,002)	(1,017)	(1,098)	(1,075)
Wald F	3,65	1,90	0,51	4,33	2,11	0,34
PSU bins × region	3,976***	4,126***	−5,174	3,903***	4,119***	−5,267
	(0,601)	(0,598)	(3,736)	(0,611)	(0,611)	(3,797)
Wald F	43,82	47,55	1,92	40,82	45,41	1,92
PSU bins × region × field	3,680***	3,624***	−3,922*	3,590***	3,612***	−4,023*
	(0,658)	(0,647)	(2,075)	(0,666)	(0,655)	(2,140)
Wald F	31,33	31,36	3,57	29,09	30,44	3,53
PSU bins + region	4,299***	4,416***	−1,476	4,344***	4,577***	−1,298
	(0,643)	(0,641)	(2,066)	(0,667)	(0,668)	(2,217)
Wald F	44,76	47,54	0,51	42,36	47,02	0,34
PSU bins + region + field	4,639***	4,429***	−1,480	4,728***	4,590***	−1,302
	(0,656)	(0,642)	(2,071)	(0,690)	(0,669)	(2,224)
Wald F	50,07	47,54	0,51	46,97	47,02	0,34
Observations	9,105	9,105	9,103	8,618	8,618	8,610
Programs	996	996	996	939	939	938

Notes: The dependent variable is incumbent-program first-year enrollment. Coefficients are reported first, program-clustered standard errors are in parentheses, and Wald F statistics appear in the corresponding labeled row. Exposure is measured in SD units and $Post_t = 1$ in 2012–2016. Columns (1) and (4) include program and year FE; Columns (2) and (5) use Broad-field × year FE; Columns (3) and (6) add campus-region × year FE. Multiplicative measures require coincidence in the indicated dimensions; additive measures combine them with equal weights. Columns (4)–(6) exclude programs with fewer than 10 first-year students in 2011. $^{***}p < 0{,}01$, $^{**}p < 0{,}05$, $^{*}p < 0{,}10$.

Design III: composition of the entering cohort

	Field × year FE			+ Region × year FE		
Outcome	Total	Triangular	Gaussian	Total	Triangular	Gaussian
First-year enrollment	1.967***	0.935	1.775***	0.798	-5.252**	-2.767*
	(0.336)	(0.972)	(0.439)	(0.662)	(2.115)	(1.401)
Mean LM PSU, entrants	-2.160***	-5.454***	-3.794***	0.590	0.162	-0.475
	(0.390)	(1.435)	(0.573)	(0.662)	(1.496)	(0.769)
LM PSU of the last admitted	-3.063***	-10.221***	-5.990***	0.991	-2.931	-1.749
	(0.513)	(2.356)	(0.918)	(0.964)	(2.627)	(1.281)
SD of LM PSU, entrants	0.670***	2.180***	1.343***	-0.579**	0.362	0.267
	(0.152)	(0.402)	(0.236)	(0.245)	(0.483)	(0.307)
Mean NEM, entrants	0.376	-5.254*	-2.090	1.925	-8.043**	-5.279**
	(0.659)	(2.868)	(1.396)	(1.324)	(3.815)	(2.142)
Female, entrants, pct.	0.025	-0.498	-0.166	0.131	-0.682	-0.369
	(0.107)	(0.773)	(0.336)	(0.257)	(0.930)	(0.517)
Private-paid school, entrants, pct.	-0.203**	-0.065	-0.151	-0.120	0.657	0.401*
	(0.094)	(0.281)	(0.136)	(0.163)	(0.414)	(0.205)
Municipal school, entrants, pct.	0.500***	0.763	0.636**	0.323	-0.767	-0.345
	(0.163)	(0.594)	(0.322)	(0.317)	(0.470)	(0.345)
Observations	8,041	8,041	8,041	8,038	8,038	8,038

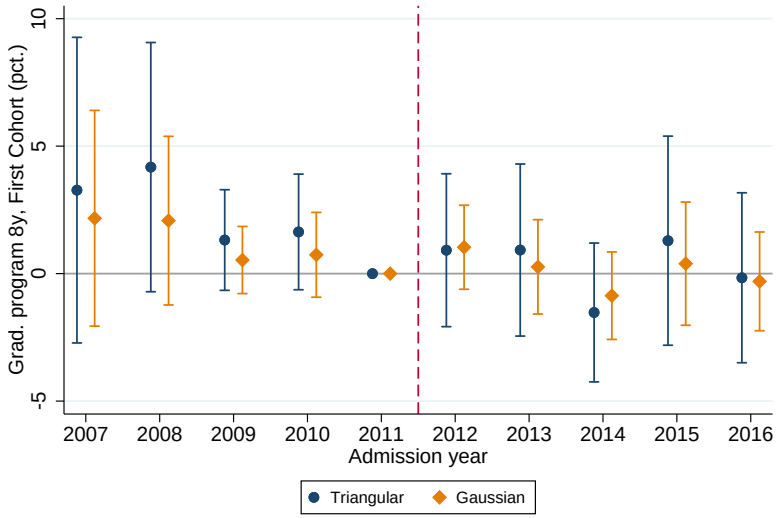
Notes: Levels first stage of Design III ($10 \times E_p \times \text{Post}_t$), broad area × region markets, SE clustered by market. The enrollment row reproduces the Design III estimate ($\pi = 1{,}97$).



- Event study: PSU-similarity-weighted exposure (Triangular, Gaussian); program, field- and region-year FE.
- Mean entrant PSU is flat before and after 2012 (pre-tests $p = 0{,}11$, $0{,}20$).
- The field-year-FE declines in the table vanish with region-year FE.

Design III: reduced form, inframarginal graduation

	Field × year FE			+ Region × year FE		
Outcome	Total	Triangular	Gaussian	Total	Triangular	Gaussian
Grad. program 8y, First Cohort (pct.)	-0.27	-1.61*	-0.72	-0.53	-1.58*	-0.88
	(0.28)	(0.94)	(0.51)	(0.54)	(0.83)	(0.60)
Grad. university 8y, First Cohort (pct.)	-0.64***	-2.14***	-1.13***	-0.15	-0.87	-0.40
	(0.15)	(0.76)	(0.33)	(0.34)	(0.77)	(0.41)
Grad. HE 8y, First Cohort (pct.)	-0.65***	-2.15***	-1.13***	-0.17	-0.84	-0.36
	(0.15)	(0.76)	(0.33)	(0.34)	(0.79)	(0.41)
Grad. program 10y, First Cohort (pct.)	-0.77*	-4.01***	-1.95***	-0.65	-3.11***	-1.76**
	(0.42)	(1.35)	(0.74)	(0.74)	(1.04)	(0.78)
Grad. program 8y, Minimum Cohort (pct.)	-0.33	-1.50	-0.73	-0.58	-1.23	-0.72
	(0.29)	(0.96)	(0.53)	(0.58)	(0.89)	(0.64)
Grad. university 8y, Minimum Cohort (pct.)	-0.67***	-2.15**	-1.16***	-0.15	-0.81	-0.40
	(0.16)	(0.86)	(0.38)	(0.35)	(0.82)	(0.45)
Grad. HE 8y, Minimum Cohort (pct.)	-0.66***	-2.13**	-1.13***	-0.21	-0.81	-0.37
	(0.16)	(0.85)	(0.38)	(0.36)	(0.83)	(0.44)
Grad. program 10y, Minimum Cohort (pct.)	-0.78*	-4.10***	-1.96**	-0.55	-3.13**	-1.68*
	(0.42)	(1.42)	(0.76)	(0.77)	(1.21)	(0.87)
Observations (8y, First Cohort)	5,970	5,970	5,970	5,967	5,967	5,967



- Outcome: inframarginal graduation as defined in Design I (program of first enrollment, any university, any HE; 8 and 10 years). No 2SLS.
- Levels specification of this design, $10 \times E_p \times \text{Post}_t$, broad area × region markets, SE clustered by market.
- The PSU-weighted measures are negative, especially for 10-year graduation (Triangular $-4{,}0^{***}$, $-3{,}1^{***}$ with region × year FE).
- The event study shows no break at 2012: 2007–2008 sit above 2011 and the post period is flat. The static estimate reflects a pre-existing trend.

Inframarginal Design No. 4

Design IV: what happens when a close competitor expands its seats?

Research question

When a close competitor j of program k suddenly increases its regular-admission vacancies, what happens to (a) k's first-year enrollment, (b) the composition of k's entering cohort, and (c) the graduation of k's inframarginal students?

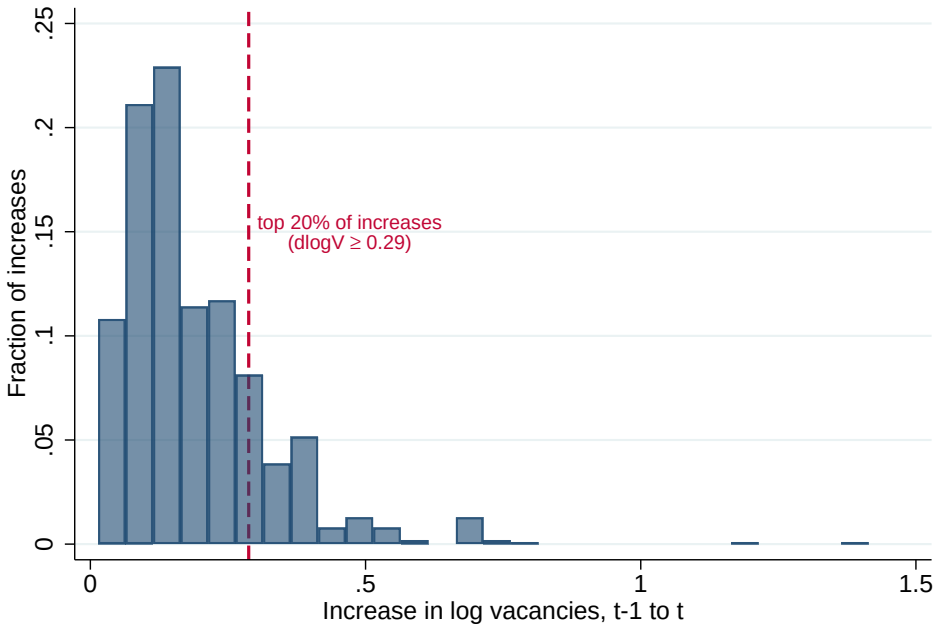
Approach

- A binary DiD, built like Design II: the treatment is an event at a *competitor*, not at k itself.
- Shock at j: the Design II shock ($\Delta \log V_{jt}$ in the top 20 % of positive changes, sudden; dated 2010–2016; 179 shocks).
- **Close competitor**: j is in k's market (campus region $\times$ broad field) and its 2007–2009 entrant PSU is within 50 points of k's ($|S_k - S_j| < 50$, the support of the triangular kernel).
- k is treated from the first year a close competitor has a shock, and stays treated (absorbing).

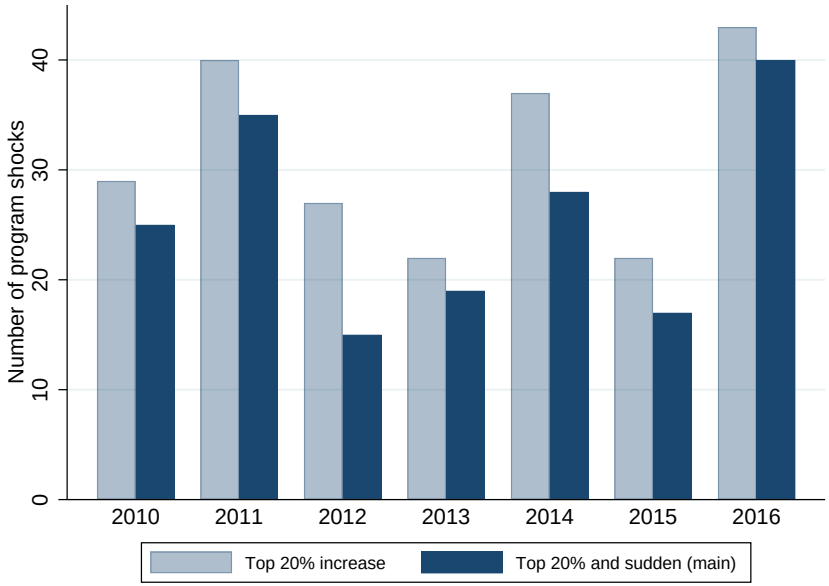
Sample

- Incumbent program $\times$ admission-year panel, 2007–2016 (programs with 2007–2009 entrants; SUA entrants excluded): 796 programs.
- Selective vs selective: the 169 selective incumbents (top 20 % of 2007–2009 entrant PSU), treated only by shocks at selective close competitors.

Design IV: distribution of vacancy changes



Program-years 2010-2016 with vacancies in t-1 and t: N = 6876. 71% have no change, 15% increase (1008, shown; trimmed at 1.5). Threshold: 80th percentile of positive changes, pooled 2008-2016.



- 71 % of program-years have no change in regular vacancies and 15 % increase. The top-20 % threshold is $\Delta \log V \geq 0{,}29 = \log(4/3)$: vacancies are round numbers, so many increases are exactly $+33$ %.
- The sudden conditions remove about a fifth of top-20 % increases. Shocks are spread over 2010–2016.

Design IV: where are the shocks?

	Program-years	Shocks	Shock rate (%)	Share of shocks (%)
Pre-period selectivity (entrant PSU quintile)				
No PSU data	809	20	2.5	11.2
Q1 (least selective)	976	31	3.2	17.3
Q2	1,150	26	2.3	14.5
Q3	1,288	29	2.3	16.2
Q4	1,331	40	3.0	22.3
Q5 (most selective)	1,322	33	2.5	18.4
Field of study				
Tecnología	2,015	55	2.7	30.7
Educación	1,441	36	2.5	20.1
Administración y Comercio	509	19	3.7	10.6
Salud	886	19	2.1	10.6
Ciencias Básicas	565	14	2.5	7.8
Arte y Arquitectura	347	13	3.7	7.3
Ciencias Sociales	625	12	1.9	6.7
Humanidades	157	10	6.4	5.6
Derecho	142	1	0.7	0.6
Agropecuaria	189	0	0.0	0.0
University				
Private (CRUCH)	2,563	59	2.3	33.0
State (CRUCH)	4,313	120	2.8	67.0
Year				
2010	929	25	2.7	14.0
2011	936	35	3.7	19.6
2012	946	15	1.6	8.4
2013	980	19	1.9	10.6
2014	1,026	28	2.7	15.6
2015	999	17	1.7	9.5
2016	1,060	40	3.8	22.3
Total	6,876	179	2.6	100.0

Shocks per program, 2010–2016	Programs
0	1,196
1	149
2	15
Total	1,360

- 88 % of programs are never shocked; shocks are almost always one-off.
- Shocks are spread across the selectivity distribution.
- More frequent at state universities and in Humanities; almost absent in Law and Agriculture.

Design IV: treatment, cohorts and models

Treatment

- $\text{close}_{kt} = 1$ if some $j \neq k$ in k's market with $|S_k - S_j| < 50$ has a shock in t.
- g_k: first year with $\text{close}_{kt} = 1$. $\text{Treated}_{kt} = \mathbf{1}\{t \geq g_k\}$, absorbing.

Models, as in Design II

$$(1) \quad y_{kt} = \beta\, \text{Treated}_{kt} + \mu_k + \lambda_t + \varepsilon_{kt}$$

$$(2) \quad y_{kt} = \beta\, \text{Treated}_{kt} + \mu_k + \lambda_{f(k)t} + \varepsilon_{kt}$$

μ_k: program FE. λ_t: year FE. $\lambda_{f(k)t}$: broad field $\times$ year FE.

- Event study with the FE of (1) and (2), event time binned at ≤ -5 and ≥ 4, $t = g_k - 1$ omitted.
- Callaway & Sant'Anna: never treated, and not yet treated plus never treated. Balanced panel; program and year only.
- SE clustered by market (region $\times$ broad field).

Programs by first treatment year

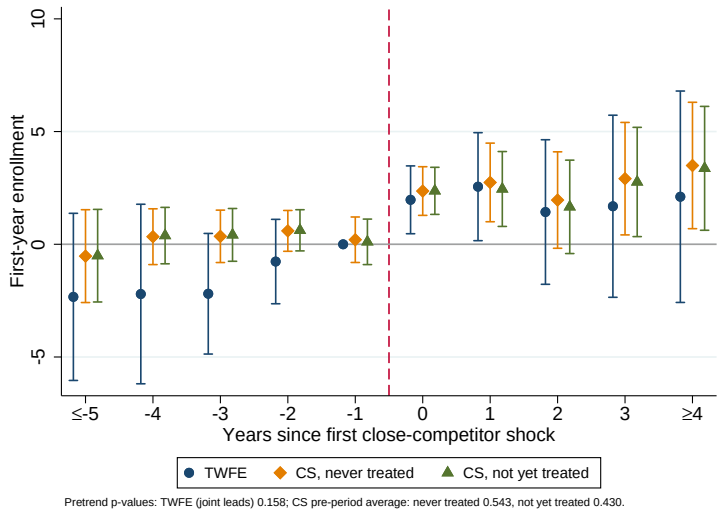
First treatment year	All incumbents	Selective vs selective
2010	153	22
2011	136	19
2012	42	9
2013	48	28
2014	23	1
2015	13	0
2016	34	4
Never treated	347	86
Programs	796	169

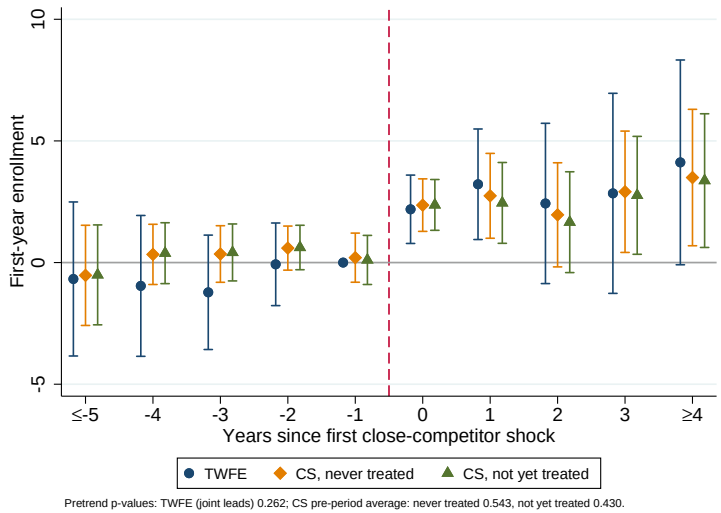
Most treatment comes early (2010–2011), and 44 % of incumbents are never treated, so both CS control groups are feasible.

Design IV: results, own enrollment

	TWFE (1) program, year	TWFE (2) program, field × year	CS never treated	CS not yet treated
First-year enrollment	2.964*	3.158**	2.673***	2.540***
	(1.602)	(1.332)	(0.988)	(0.955)
Pre-trend p	*0.158*	*0.262*	*0.543*	*0.430*
Pre-mean treated; N		52.02; 8,915		
Log first-year enrollment	0.066	0.071*	0.053**	0.046*
	(0.041)	(0.039)	(0.025)	(0.024)
Pre-trend p	*0.127*	*0.462*	*0.485*	*0.316*
Pre-mean treated; N		3.75; 8,865		

- After a close competitor expands, k's enrollment *rises* by about 3 students (5–7 %, pre-mean 52), in every specification.
- Pre-trends pass everywhere (joint $p \geq 0{,}13$).
- No sign of business stealing: more consistent with demand shocks shared by close programs in the same market.

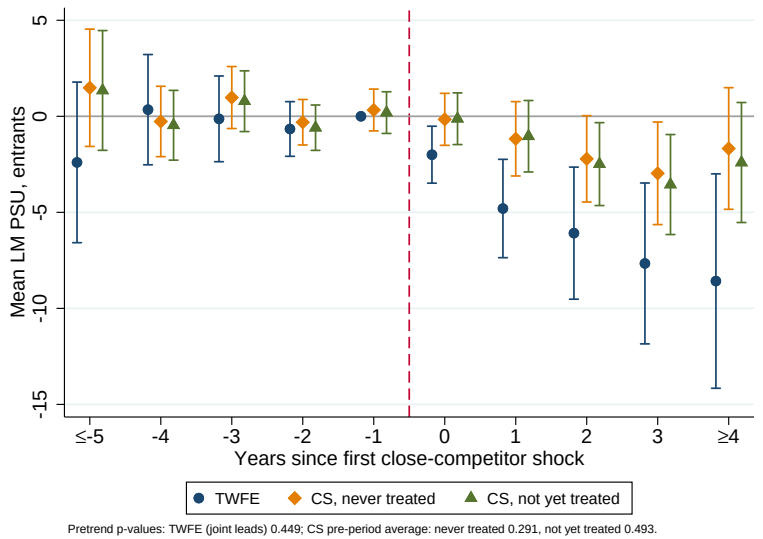




Notes: All incumbents. Event studies: FE (1) top, FE (2) bottom. Pre-trend p: TWFE joint leads; CS pre-period average. SE clustered by market. *** $p < 0{,}01$, ** $p < 0{,}05$, * $p < 0{,}10$.

Design IV: composition of the entering cohort

	TWFE (1) program, year	TWFE (2) program, field × year	CS never treated	CS not yet treated
Mean LM PSU, entrants	-1.927	-4.581***	-1.551	-1.813*
	(1.983)	(1.268)	(1.079)	(1.044)
Pre-trend p	*0.158*	*0.449*	*0.291*	*0.493*
Pre-mean treated; N		589.98; 8,865		
LM PSU of the last admitted	-3.964	-7.020***	-1.750	-2.049
	(2.501)	(1.954)	(1.915)	(1.929)
Pre-trend p	*0.241*	*0.126*	*0.761*	*0.776*
Pre-mean treated; N		567.61; 8,915		
SD of LM PSU, entrants	0.859*	1.058***	0.744*	0.788*
	(0.466)	(0.399)	(0.417)	(0.406)
Pre-trend p	*0.003*	*0.010*	*0.121*	*0.156*
Pre-mean treated; N		34.45; 8,726		
Mean NEM, entrants	-3.529**	-4.282***	-3.825**	-3.863**
	(1.629)	(1.627)	(1.579)	(1.536)
Pre-trend p	*0.728*	*0.802*	*0.278*	*0.298*
Pre-mean treated; N		606.37; 8,865		
Female, entrants, pct.	-0.724	-0.502	-1.449***	-1.350**
	(0.508)	(0.478)	(0.538)	(0.531)
Pre-trend p	*0.052*	*0.131*	*0.219*	*0.300*
Pre-mean treated; N		48.97; 8,865		
Private-paid school, entrants, pct.	-0.486	-0.620**	-0.098	-0.186
	(0.351)	(0.287)	(0.338)	(0.335)
Pre-trend p	*0.073*	*0.294*	*0.289*	*0.262*
Pre-mean treated; N		12.56; 8,865		
Municipal school, entrants, pct.	0.934	1.514***	0.099	0.200
	(0.607)	(0.558)	(0.574)	(0.567)
Pre-trend p	*0.198*	*0.147*	*0.031*	*0.040*
Pre-mean treated; N		33.01; 8,865		

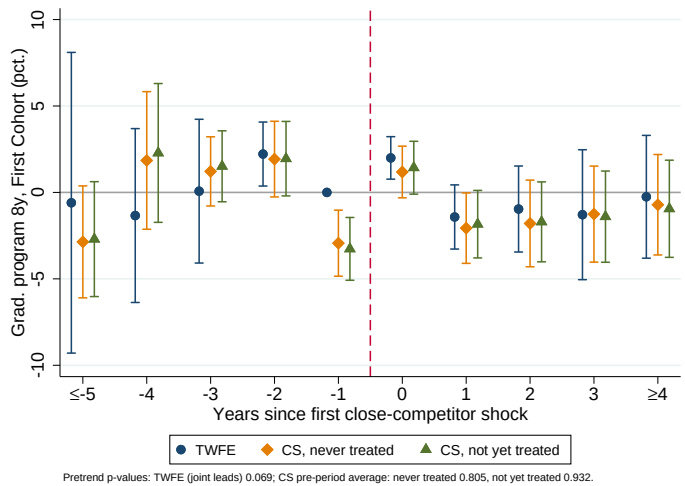


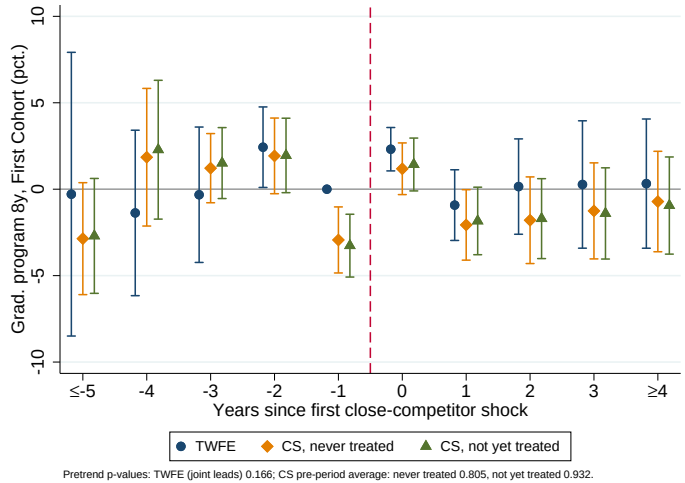
- Entrants are weaker after a close competitor expands: mean NEM falls by about 4 points in every specification, mean PSU by 2–5 and the PSU of the last admitted by 2–7 (significant with FE (2)).
- The PSU decline grows over time in the TWFE (2) event study (top), with flat leads ($p = 0{,}45$); CS is smaller (-2 to -3).
- The rise in the SD of PSU fails the TWFE pre-trend test.

Notes: All incumbents. Same specifications as the enrollment results. *** $p < 0{,}01$, ** $p < 0{,}05$, * $p < 0{,}10$.

Design IV: reduced form, inframarginal graduation

	TWFE (1) program, year	TWFE (2) program, field × year	CS never treated	CS not yet treated
Grad. program 8y, First Cohort (pct.)	-0.424	0.374	-0.698	-0.677
	(1.394)	(1.295)	(0.992)	(0.943)
Pre-trend p	*0.069*	*0.166*	*0.805*	*0.932*
Pre-mean treated; N		29.98; 6,628		
Grad. university 8y, First Cohort (pct.)	-0.051	-0.533	-0.170	-0.192
	(0.900)	(0.745)	(0.712)	(0.693)
Pre-trend p	*0.791*	*0.534*	*0.832*	*0.546*
Pre-mean treated; N		45.03; 6,628		
Grad. HE 8y, First Cohort (pct.)	-0.203	-0.629	-0.238	-0.255
	(0.861)	(0.722)	(0.707)	(0.688)
Pre-trend p	*0.893*	*0.763*	*0.930*	*0.850*
Pre-mean treated; N		46.59; 6,628		
Grad. program 10y, First Cohort (pct.)	-1.826	-0.595	-4.377***	-3.980***
	(1.738)	(2.023)	(1.481)	(1.371)
Pre-trend p	*0.943*	*0.937*	*0.472*	*0.537*
Pre-mean treated; N		38.99; 5,451		
Grad. program 8y, Minimum Cohort (pct.)	-0.491	0.377	-0.899	-0.895
	(1.449)	(1.318)	(1.033)	(0.984)
Pre-trend p	*0.080*	*0.213*	*0.839*	*0.987*
Pre-mean treated; N		30.76; 6,628		
Grad. university 8y, Minimum Cohort (pct.)	-0.026	-0.458	-0.170	-0.180
	(0.964)	(0.789)	(0.753)	(0.733)
Pre-trend p	*0.639*	*0.533*	*0.640*	*0.434*
Pre-mean treated; N		45.84; 6,628		
Grad. HE 8y, Minimum Cohort (pct.)	-0.089	-0.492	-0.199	-0.220
	(0.905)	(0.762)	(0.741)	(0.719)
Pre-trend p	*0.791*	*0.755*	*0.886*	*0.713*
Pre-mean treated; N		47.30; 6,628		
Grad. program 10y, Minimum Cohort (pct.)	-1.905	-0.718	-4.508***	-4.053***
	(1.744)	(2.026)	(1.510)	(1.399)
Pre-trend p	*0.912*	*0.932*	*0.409*	*0.480*
Pre-mean treated; N		39.65; 5,451		



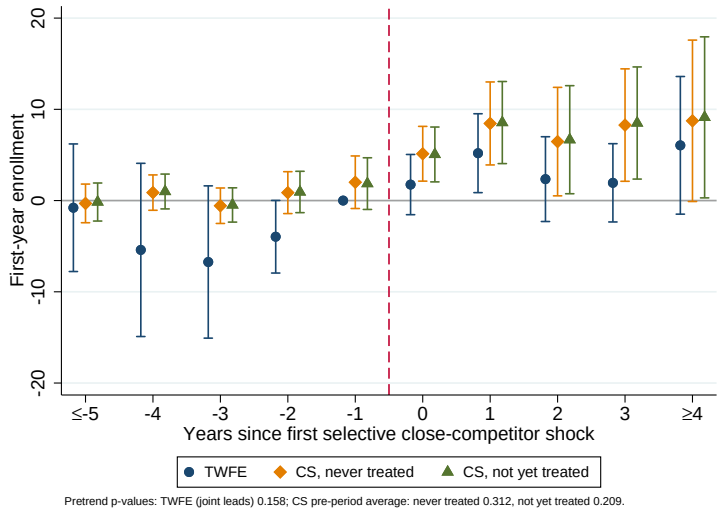


- Outcome: graduation (pct.) of k's inframarginal applicants, as in Design I (Steps 1–2); 10-year outcomes cover cohorts 2007–2014. Event studies: program graduation 8y, First Cohort; FE (1) top, FE (2) bottom.
- No effect on 8-year graduation from the program, any university or any HE. Program graduation at 10 years falls by about 4 pp in CS ($p < 0{,}01$), but not in TWFE.

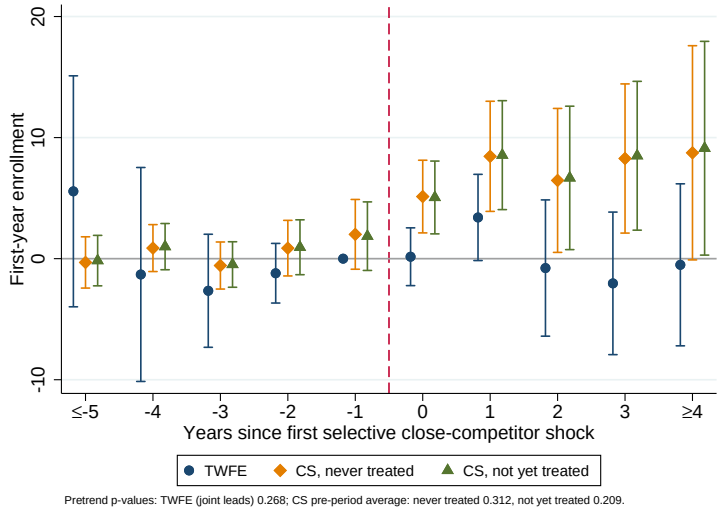
Design IV, selective vs selective: enrollment

	TWFE (1) program, year	TWFE (2) program, field × year	CS never treated	CS not yet treated
First-year enrollment	6.412**	1.505	7.932***	8.084***
	(2.962)	(2.138)	(2.744)	(2.738)
Pre-trend p	*0.158*	*0.268*	*0.312*	*0.209*
Pre-mean treated; N		76.65; 1,883		
Log first-year enrollment	0.069*	-0.004	0.069***	0.069***
	(0.036)	(0.035)	(0.022)	(0.022)
Pre-trend p	*0.060*	*0.017*	*0.987*	*0.639*
Pre-mean treated; N		4.11; 1,882		

- 169 selective incumbents; treated when a *selective* close competitor has a shock (35 shocks at selective programs). 83 are treated, mostly in 2010–2013.
- Enrollment rises by 6–8 students (about 9 %, pre-mean 77) with FE (1) and CS; with field × year FE the estimate is +1,5, not significant.
- The log-enrollment leads reject with FE (2) ($p = 0{,}02$).



Pretrend p-values: TWFE (joint leads) 0.158; CS pre-period average: never treated 0.312, not yet treated 0.209.



Pretrend p-values: TWFE (joint leads) 0.268; CS pre-period average: never treated 0.312, not yet treated 0.209.

Notes: Selective incumbents. Event studies: FE (1) top, FE (2) bottom. SE clustered by market. *** $p < 0{,}01$, ** $p < 0{,}05$, * $p < 0{,}10$.

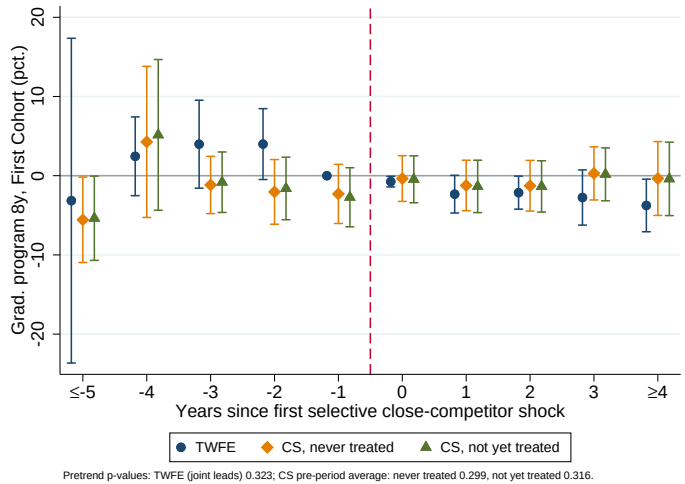
Design IV, selective vs selective: composition

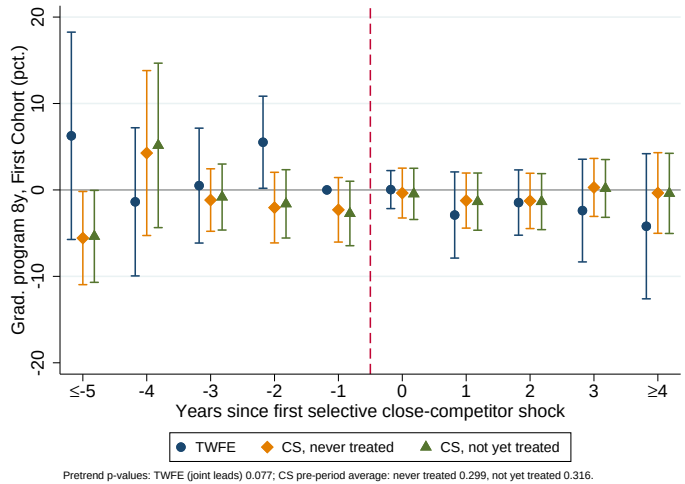
	TWFE (1) program, year	TWFE (2) program, field × year	CS never treated	CS not yet treated
Mean LM PSU, entrants	-1.553	-2.416	-2.741	-2.637
	(1.572)	(2.229)	(1.883)	(1.855)
Pre-trend p	*0.802*	*0.000*	*0.714*	*0.641*
Pre-mean treated; N		671.71; 1,882		
LM PSU of the last admitted	-4.256	-5.596*	-4.883	-3.606
	(2.512)	(3.260)	(3.976)	(4.007)
Pre-trend p	*0.369*	*0.040*	*0.167*	*0.082*
Pre-mean treated; N		651.90; 1,883		
SD of LM PSU, entrants	0.835	0.489	1.113	1.144
	(0.775)	(0.799)	(0.725)	(0.730)
Pre-trend p	*0.046*	*0.006*	*0.951*	*0.882*
Pre-mean treated; N		32.21; 1,881		
Mean NEM, entrants	-0.748	-4.879	-2.320	-2.018
	(2.745)	(3.129)	(2.741)	(2.705)
Pre-trend p	*0.237*	*0.744*	*0.579*	*0.451*
Pre-mean treated; N		678.55; 1,882		
Female, entrants, pct.	0.245	-0.097	0.328	0.213
	(0.837)	(1.128)	(1.016)	(1.020)
Pre-trend p	*0.049*	*0.001*	*0.173*	*0.088*
Pre-mean treated; N		45.56; 1,882		
Private-paid school, entrants, pct.	-1.132**	0.455	0.010	-0.046
	(0.543)	(0.851)	(0.947)	(0.947)
Pre-trend p	*0.075*	*0.150*	*0.163*	*0.140*
Pre-mean treated; N		33.90; 1,882		
Municipal school, entrants, pct.	1.158	0.836	2.131**	2.202**
	(0.858)	(0.579)	(1.069)	(1.075)
Pre-trend p	*0.021*	*0.591*	*0.427*	*0.434*
Pre-mean treated; N		21.23; 1,882		

Notes: Selective incumbents, same specifications as the enrollment results. Entrant PSU, NEM and the PSU of the last admitted point down (e.g. PSU of the last admitted −4 to −6), but only one estimate is significant at 10 %, and several TWFE pre-trend tests reject. CS shows a rise in the municipal-school share of about 2 pp. *** $p < 0{,}01$, ** $p < 0{,}05$, * $p < 0{,}10$.

Design IV, selective vs selective: inframarginal graduation

	TWFE (1) program, year	TWFE (2) program, field × year	CS never treated	CS not yet treated
Grad. program 8y, First Cohort (pct.)	-4.168*	-3.211	-0.544	-0.621
	(2.290)	(3.164)	(1.341)	(1.356)
Pre-trend p	*0.323*	*0.077*	*0.299*	*0.316*
Pre-mean treated; N		36.92; 1,842		
Grad. university 8y, First Cohort (pct.)	-0.501	0.257	-0.776	-0.662
	(1.010)	(1.215)	(1.228)	(1.238)
Pre-trend p	*0.545*	*0.051*	*0.760*	*0.935*
Pre-mean treated; N		52.09; 1,842		
Grad. HE 8y, First Cohort (pct.)	-0.503	0.240	-0.714	-0.595
	(1.020)	(1.216)	(1.228)	(1.240)
Pre-trend p	*0.597*	*0.050*	*0.745*	*0.920*
Pre-mean treated; N		52.24; 1,842		
Grad. program 10y, First Cohort (pct.)	-5.155	-5.386	-3.915**	-5.093***
	(3.198)	(5.516)	(1.860)	(1.934)
Pre-trend p	*0.302*	*0.669*	*0.880*	*0.976*
Pre-mean treated; N		41.90; 1,509		
Grad. program 8y, Minimum Cohort (pct.)	-4.322*	-3.166	-0.689	-0.833
	(2.289)	(3.242)	(1.376)	(1.396)
Pre-trend p	*0.438*	*0.178*	*0.316*	*0.329*
Pre-mean treated; N		37.33; 1,842		
Grad. university 8y, Minimum Cohort (pct.)	-0.513	0.305	-1.068	-1.077
	(1.010)	(1.203)	(1.269)	(1.281)
Pre-trend p	*0.360*	*0.004*	*0.536*	*0.642*
Pre-mean treated; N		52.46; 1,842		
Grad. HE 8y, Minimum Cohort (pct.)	-0.486	0.311	-0.992	-0.999
	(1.018)	(1.207)	(1.264)	(1.279)
Pre-trend p	*0.406*	*0.005*	*0.530*	*0.634*
Pre-mean treated; N		52.59; 1,842		
Grad. program 10y, Minimum Cohort (pct.)	-4.875	-4.859	-3.678**	-4.928**
	(3.160)	(5.482)	(1.871)	(1.946)
Pre-trend p	*0.237*	*0.698*	*0.881*	*0.989*
Pre-mean treated; N		42.04; 1,509		





- Program graduation at 8 years falls by about 4 pp with FE (1) ($-4{,}2^*$, flat leads $p = 0{,}32$; the event study declines gradually to about -4), $-3{,}2$ with FE (2), and less than 1 pp in CS.
- At 10 years CS gives -4 to -5 pp ($p < 0{,}05$); TWFE is similar in size but imprecise. Graduation from any university or any HE does not move.

Inframarginal Design No. 5

Design V: Design I at the individual level

Question. Same as Design I: among students who would have been admitted anyway, do those who enter when the cohort is larger graduate at lower rates?

What changes relative to Design I.

- **Same variation**: changes over time in a program's vacancies Z_{pt} around its historic mean, which move the size of the entering cohort n_{pt}.
- **Individual outcomes**: one row per inframarginal student, $Y_i = 1$ if i graduates within 8 years. We can control for PSU, NEM, sex and school type, which Design I cannot.
- **Application-portfolio fixed effects** $\phi_{ROL(i)}$: compare students who submitted the same list of programs (the ROL), which arguably holds fixed their preference for college and for this program.
- **Timing idea**: if students know the program's historic mean of vacancies but not the year-to-year deviation when they apply, the deviation is exogenous. **This does not hold as stated**: DEMRE publishes the offer of vacancies before applications, so we test it (last slide).

Design V: equations and sample

For inframarginal student i in program p, admission year t (the entering cohort of p in t is n_{pt}):

$$\text{First stage:} \quad n_{pt} = \alpha_1 Z_{pt} + x_i' \alpha_2 + \psi_p + \psi_t + \psi_{ROL(i)} + \nu_i$$

$$\text{Second stage:} \quad Y_i = \beta\, n_{pt} + x_i' \beta_2 + \phi_p + \phi_t + \phi_{ROL(i)} + \varepsilon_i$$

- **Sample**: inframarginal students as in Design I ($Rank_{ipt} \leq T_p$, T_p = first enrolled cohort). One row per student: the most preferred inframarginal application in the student's first inframarginal year (99 % of student-years have only one). **432,184 students, 1,307 programs, 8,218 program-years**, 2007–2016.
- **Outcome**: graduation within 8 years from the program where the student first enrolled (main); any university; any higher-education institution. Results $\times 100$.
- **Controls** x_i: PSU language+math, NEM, female, private-paid and municipal school (with missing indicators). *No* peer composition: it is a mechanism.
- **Fixed effects**: program and year always (no specification without year FE). ROL FE under several definitions (next slide).
- **Inference**: SE clustered by program-year, the level at which Z_{pt} and n_{pt} vary. β: pp change in graduation per additional entering student induced by vacancies.

Design V: how sparse is the ROL?

ROL definition	Groups	Singletons (%)	Spans $\geq$2 years (%)	Mean size
Exact ordered list	367,221	80.5	15.8	1.2
Unordered set of programs	330,329	70.1	21.8	1.3
Top 3 preferences	222,487	39.9	44.0	1.9
First preference + set of universities	122,781	21.0	62.3	3.5
Top 2 preferences	67,615	8.4	75.3	6.4
Set of universities	37,751	5.8	76.3	11.4
First preference	1,410	0.0	92.8	306.5

- The ROL is the ordered list of programs a student applied to. **The exact list is almost unique**: 80 % of students are alone in their group.
- With program and ROL FE, only students whose group spans $\geq$ 2 years of the same program contribute variation in Z ("Spans" column).
- So we report a ladder of coarser definitions. Main: **first preference + set of universities** (62 % contribute).

Notes: Main sample, 432,184 students. Singletons are dropped by the fixed effects. "Spans $\geq$2 years": share of students whose ROL $\times$ program group has the program in at least two years.

Design V: results

	(1)	(2)	(3)	(4)
ROL FE	–	–	Top 1 + univ.	Exact
Controls	No	Yes	Yes	Yes
First stage: n_{pt} on Z_{pt}	0.813***	0.812***	0.864***	0.927***
	(0.043)	(0.043)	(0.041)	(0.044)
Kleibergen–Paap F	361	362	435	444
2SLS, $\times 100$				
Graduation, enrolled program, 8y	0.075***	0.065**	0.064**	0.092**
	(0.026)	(0.026)	(0.030)	(0.041)
Graduation, any university, 8y	0.031***	0.019*	0.031**	0.034**
	(0.011)	(0.010)	(0.012)	(0.015)
Graduation, any higher education, 8y	0.031***	0.020**	0.029**	0.034**
	(0.010)	(0.010)	(0.012)	(0.015)
Students	432,184	432,184	341,430	84,087
Program-years (clusters)	8,218	8,218	8,218	7,735

- Strong first stage: ten more vacancies bring about 8–9 more entrants ($F > 350$).
- Graduation from the program **rises** by about 0.06–0.09 pp per additional entrant, not falls. This is the sign Design I gives *in levels*; column (1) reproduces it.
- Portfolio FE do not move the estimate. On a common sample (the 84,087 students of column 4) the 2SLS is 0.085–0.095 under every ROL definition; the drift across definitions in the full sample comes from the change of sample.
- Any university and any higher education: +0,02 to +0,03 pp.

Notes: 2SLS of graduation within 8 years ($\times 100$) on n_{pt} instrumented by Z_{pt}. All columns include program and year FE. SE clustered by program-year. *** $p < 0{,}01$, ** $p < 0{,}05$, * $p < 0{,}10$. Full ladder of seven ROL definitions and the common-sample panel in the issue-#8 tables.

Design V: is the variation in Z exogenous?

ROL FE	Mean	(1) –	(2) Top 1 + univ.
Panel A. Balance: covariate on Z_{pt} (per 10 vacancies)			
PSU language+math	612.19	0.218***	0.279***
		(0.077)	(0.067)
NEM	634.14	0.893***	0.966***
		(0.126)	(0.117)
Female	0.51	0.000	0.000
		(0.000)	(0.000)
Private-paid school	0.23	-0.000	-0.001
		(0.000)	(0.000)
Municipal school	0.26	0.001**	0.001***
		(0.000)	(0.000)
Applications in the ROL	5.04	0.002	-0.002
		(0.003)	(0.002)
Predicted graduation ($\times$100)	34.55	0.064***	0.076***
		(0.010)	(0.009)
Panel B. Placebo: next year's vacancies $Z_{p,t+1}$ (per 10)			
First stage, n_{pt}		0.239	0.207
		(0.355)	(0.331)
Graduation from the program ($\times$100)		0.377*	0.317
		(0.206)	(0.232)
Students (Panel B)		351,840	277,210

- **Balance (Panel A)**: with more vacancies the inframarginal students are slightly stronger (PSU, NEM). Their predicted graduation rises 0.06–0.08 pp per ten vacancies, about 13 % of the reduced form. The controls absorb it.
- **Placebo (Panel B)**: next year's vacancies do not predict this year's cohort size, but **predict this year's graduation as much as this year's vacancies do**.
- Reading: programs that expand are also programs whose graduation was improving. With program FE in levels, that trend loads on Z. It matches the sign gap between Design I in levels (+) and in first differences (−).
- Next: a specification that absorbs program trends.

Notes: Z in units of ten vacancies. Panel B uses cohorts with $Z_{p,t+1}$ observed. SE clustered by program-year.

Appendix

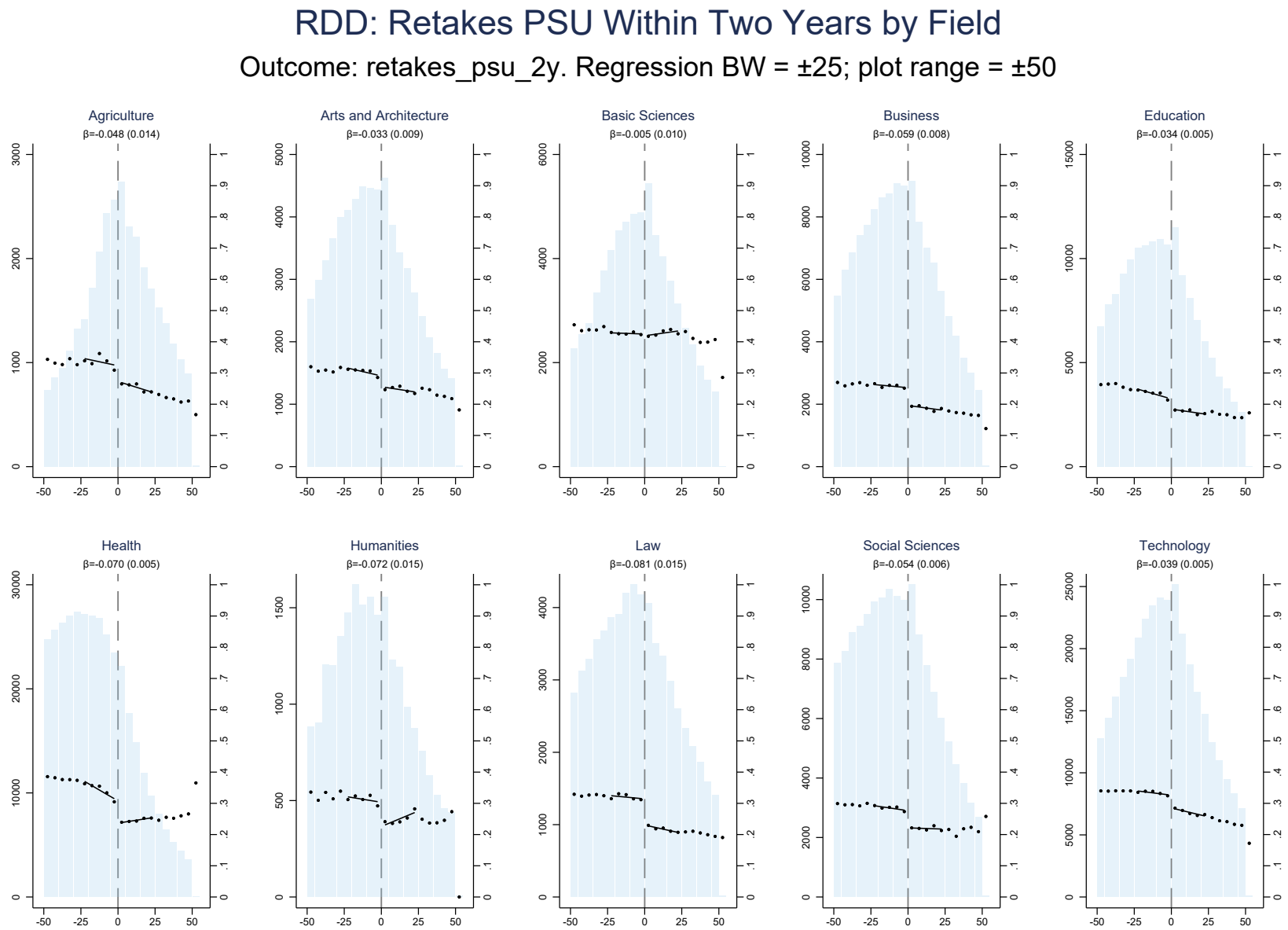
Non-harmonized field observations

Institution	Program	Years	Code	Obs.
UV	Bachillerato ingreso común	2007–2008	1900	558
USACH	Tecnólogo en alimentos (vesp)	2007–2010	1690	208
UV	Administración pública – diurno	2007	1997	26
UV	Actuación teatral	2008	1928	9
Total		2007–2010		801

These observations are excluded from the field-enriched sample because their program field could not be reliably harmonized. They represent 0.095 % of the baseline RDD sample within the main bandwidth.

← Back

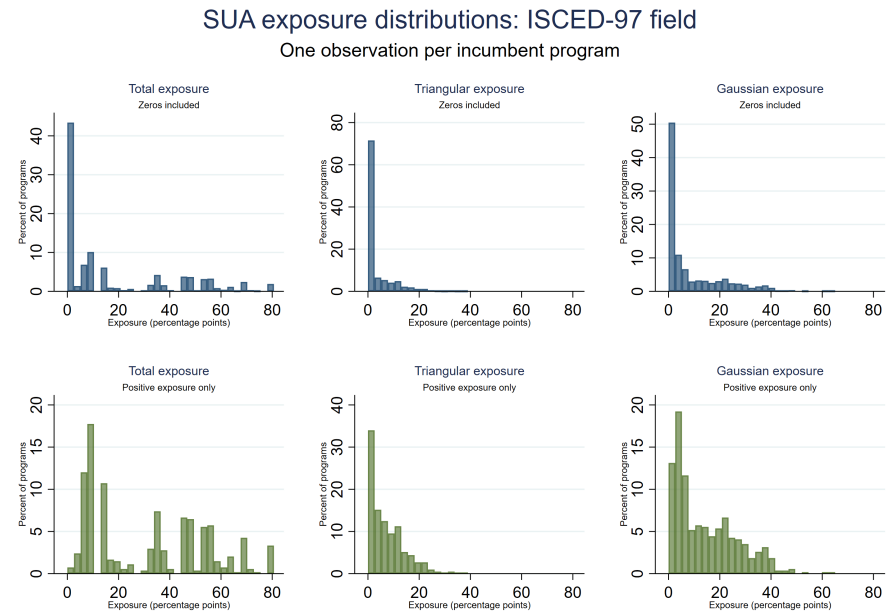
Appendix note 2: PSU retakes by field



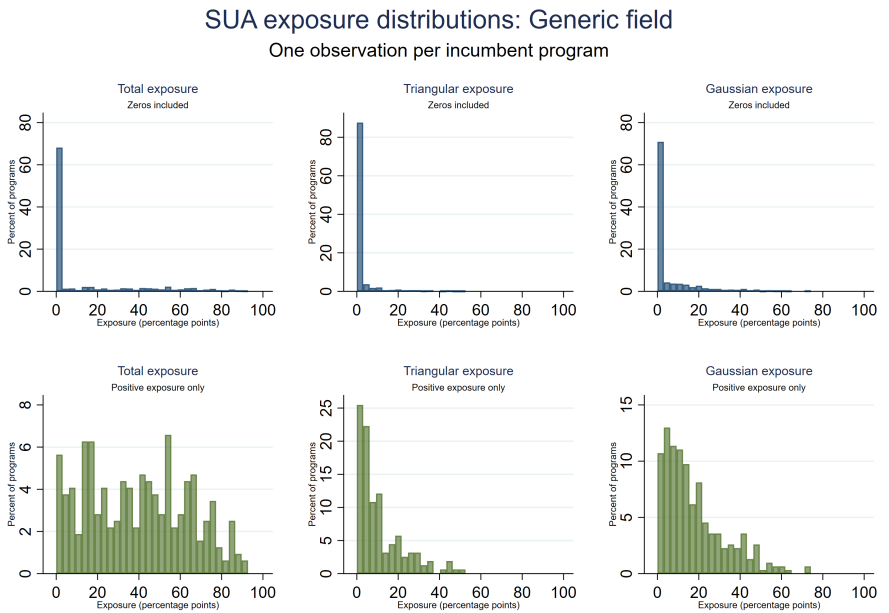
The reduction in PSU retaking is generally negative across fields, although magnitudes vary by academic area. ← Back

SUA exposure: alternative field definitions

← Broad field



Top row includes zeros; bottom row conditions on positive exposure.



Top row includes zeros; bottom row conditions on positive exposure.

Narrower field definitions generate more markets and a larger share of zero-exposure programs, particularly under the Generic-field definition.

First stage: regions with entrant universities only

← Main ES →

$$N_{pt}^{\text{first}} = \pi \left(E_p^{k,m} \times Post_t \right) + \mu_p + \alpha_{f_m(p),t} + \gamma_{r(p),t} + \nu_{pt}.$$

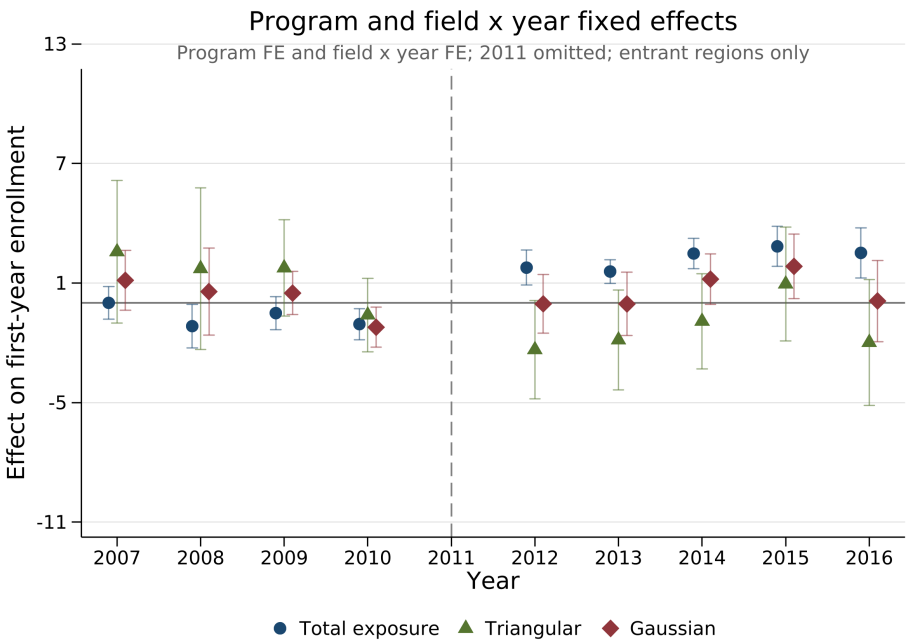
	Field × year FE			+ Region × year FE		
Exposure measure	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
Total	2.807***	2.609***	2.378***	-0.492	0.652	0.151
	(0.457)	(0.415)	(0.381)	(0.940)	(0.573)	(0.504)
Wald *F*	37.81	39.49	38.90	0.27	1.30	0.09
Programs with zero exposure	10	60	256	10	60	256
Triangular	-2.133	-0.830	0.711	-6.390**	-4.179***	-1.575
	(1.482)	(1.253)	(1.071)	(2.434)	(1.581)	(1.033)
Wald *F*	2.07	0.44	0.44	6.89	6.99	2.33
Programs with zero exposure	96	192	416	96	192	416
Gaussian	0.545	1.118	1.652**	-4.168**	-2.416**	-1.240
	(0.892)	(0.764)	(0.725)	(1.863)	(1.177)	(0.773)
Wald *F*	0.37	2.14	5.19	5.01	4.21	2.58
Programs with zero exposure	10	60	267	10	60	267
Observations	5,090	5,080	4,781	5,090	5,080	4,781
Programs	601	601	571	601	601	571
Markets	40	80	303	40	80	303
Regions	4	4	4	4	4	4
Program FE	Yes	Yes	Yes	Yes	Yes	Yes
Field × year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region × year FE	No	No	No	Yes	Yes	Yes

Notes: Sample restricted to incumbent programs in the four pre-treatment regions with entrant programs in 2009–2011 (Valparaíso, Biobío, La Araucanía, Metropolitana); outside them exposure is zero by construction. Same specification as the main first stage. Exposure in 10 percentage-point units; SE clustered by pre-treatment market. $^{***}p < 0{,}01$, $^{**}p < 0{,}05$, $^{*}p < 0{,}10$.

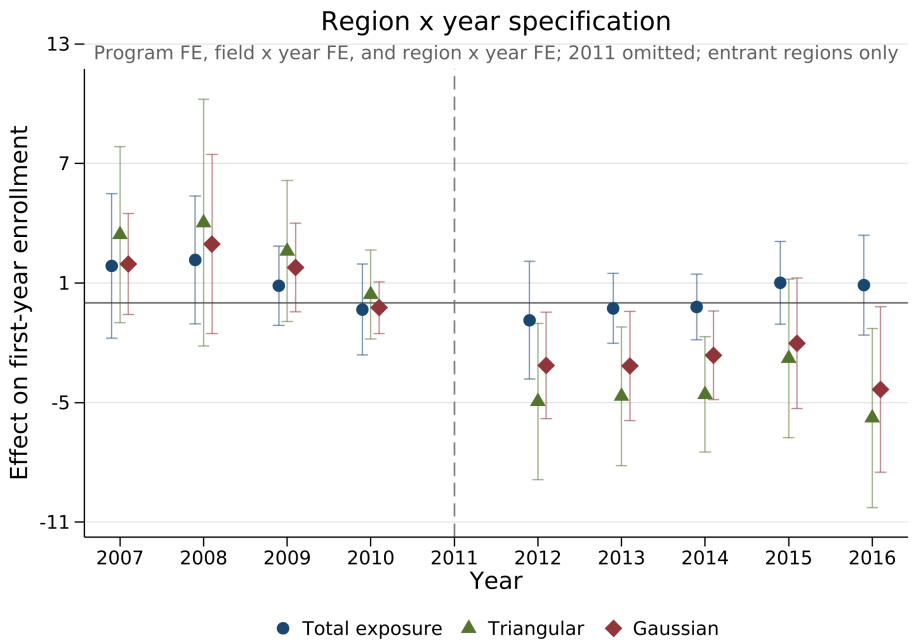
Event studies: regions with entrant universities only

← First stage

Field × year FE



+ Region × year FE



Notes: Sample restricted to incumbent programs in the four pre-treatment regions with entrant programs in 2009–2011 (Valparaíso, Biobío, La Araucanía, Metropolitana); outside them exposure is zero by construction. Broad-field markets; separate event studies for Total, Triangular and Gaussian exposure. Coefficients measure the enrollment difference associated with a 10 percentage-point increase in exposure in each year relative to 2011. 95 % confidence intervals; SE clustered by pre-treatment market. Joint pretrend p-values (2007–2010) for Total, Triangular and Gaussian are 0.046, 0.068 and 0.010 in the left panel, and 0.151, 0.302 and 0.209 in the right panel.

First stage: positive exposure only

← Main ES →

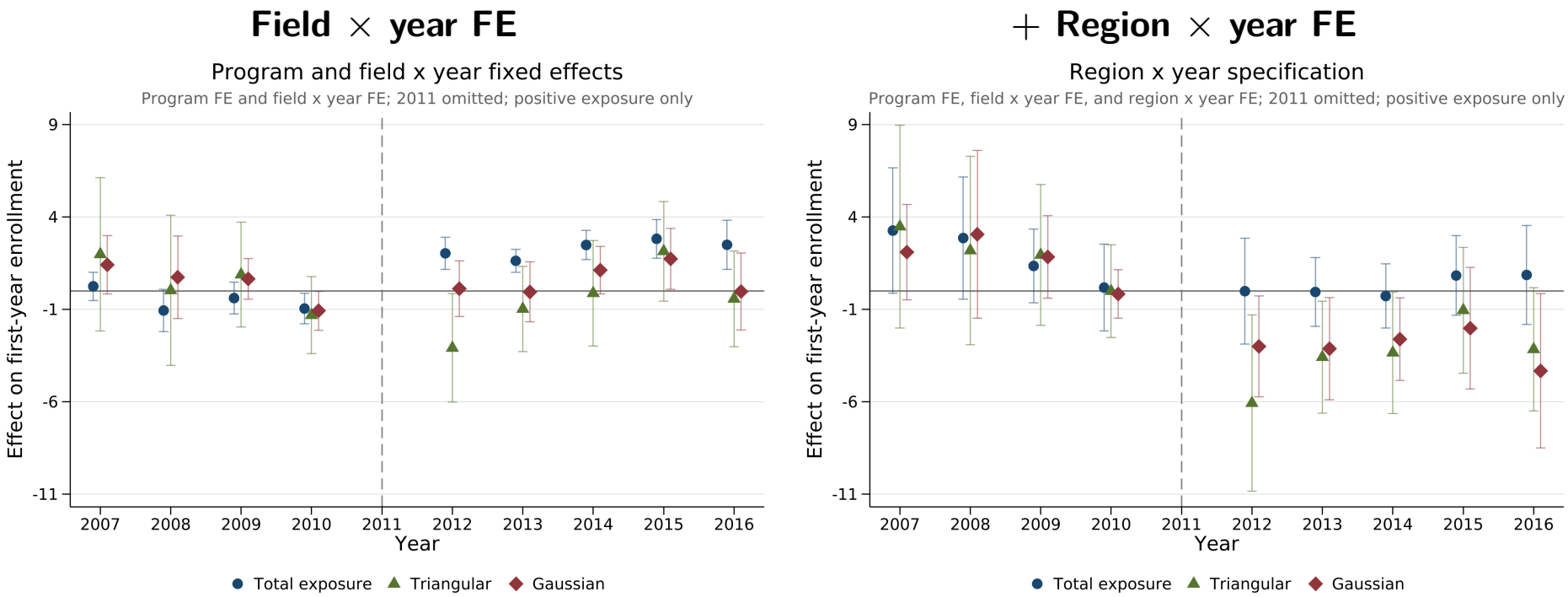
$$N_{pt}^{\text{first}} = \pi \left(E_p^{k,m} \times Post_t \right) + \mu_p + \alpha_{f_m(p),t} + \gamma_{r(p),t} + \nu_{pt}.$$

	Field × year FE			+ Region × year FE		
Exposure measure	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
Total	2.753***	2.423***	2.579***	-0.863	1.086	-1.266
	(0.474)	(0.443)	(0.575)	(0.952)	(0.911)	(1.144)
Wald *F*	33.68	29.84	20.11	0.82	1.42	1.23
Programs	591	540	304	591	540	304
Triangular	-0.668	0.555	0.385	-4.698**	-2.671	-1.705
	(1.454)	(1.264)	(1.496)	(2.119)	(1.678)	(1.624)
Wald *F*	0.21	0.19	0.07	4.91	2.54	1.10
Programs	505	407	142	505	407	142
Gaussian	0.378	0.805	0.489	-4.196**	-2.576*	-2.133**
	(0.908)	(0.798)	(0.874)	(1.865)	(1.289)	(0.897)
Wald *F*	0.17	1.02	0.31	5.06	4.00	5.65
Programs	591	540	292	591	540	292
Observations (Total)	5,009	4,571	2,516	5,009	4,571	2,516
Programs (Total)	591	540	304	591	540	304
Markets (Total)	36	59	131	36	59	131
Regions	4	4	4	4	4	4
Program FE	Yes	Yes	Yes	Yes	Yes	Yes
Field × year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region × year FE	No	No	No	Yes	Yes	Yes

Notes: Sample restricted to programs with positive exposure, defined separately for each measure (Triangular exposure can be zero where Total and Gaussian are positive). All such programs lie in the four regions with entrant programs, so the only difference with the entrant-region sample is that zero-exposure programs within those regions are dropped. Same specification as the main first stage. Exposure in 10 percentage-point units; SE clustered by pre-treatment market. $^{***}p < 0{,}01$, $^{**}p < 0{,}05$, $^{*}p < 0{,}10$.

Event studies: positive exposure only

← First stage



Notes: Sample restricted to programs with positive exposure, defined separately for each measure (Triangular exposure can be zero where Total and Gaussian are positive). All such programs lie in the four regions with entrant programs, so the only difference with the entrant-region sample is that zero-exposure programs within those regions are dropped. Broad-field markets; separate event studies for Total, Triangular and Gaussian exposure. Coefficients measure the enrollment difference associated with a 10 percentage-point increase in exposure in each year relative to 2011. 95 % confidence intervals; SE clustered by pre-treatment market. Joint pretrend *p*-values (2007–2010) for Total, Triangular and Gaussian are 0.072, 0.055 and 0.012 in the left panel, and 0.132, 0.348 and 0.207 in the right panel.

Log first stage: regions with entrant universities only

← Main ES →

$$\log N_{pt}^{\text{first}} = \beta_k \left[\widetilde{\log E_p^k} \times Post_t\right] + \theta_k \left[D_p^k \times Post_t\right] + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \varepsilon_{pt}.$$

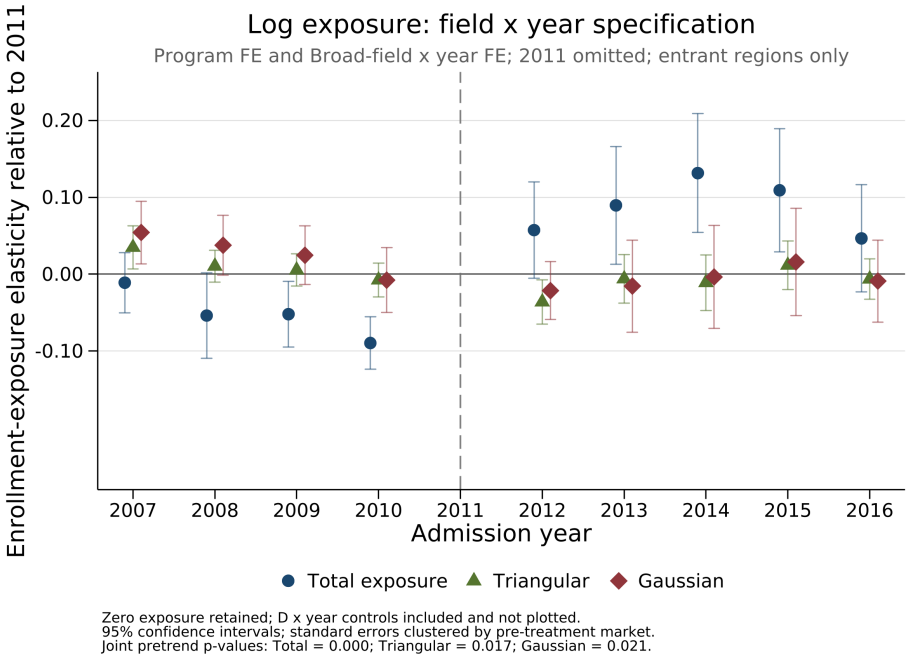
	Field × year FE			+ Region × year FE		
Exposure measure	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
Total	0.133***	0.109***	0.059*	0.025	0.012	0.006
	(0.039)	(0.035)	(0.031)	(0.086)	(0.041)	(0.031)
Wald *F*	11.66	9.63	3.51	0.09	0.09	0.04
Programs with zero exposure	10	60	256	10	60	256
Triangular	-0.015	0.006	0.029	-0.052***	-0.023	0.005
	(0.017)	(0.018)	(0.022)	(0.018)	(0.017)	(0.018)
Wald *F*	0.77	0.10	1.70	8.22	1.84	0.09
Programs with zero exposure	96	192	416	96	192	416
Gaussian	-0.024	-0.009	0.019	-0.117***	-0.075**	-0.021
	(0.036)	(0.034)	(0.019)	(0.034)	(0.031)	(0.017)
Wald *F*	0.44	0.06	0.96	11.57	5.83	1.46
Programs with zero exposure	10	60	267	10	60	267
Observations	5,090	5,080	4,781	5,090	5,080	4,781
Programs	601	601	571	601	601	571
Markets	40	80	303	40	80	303
Regions	4	4	4	4	4	4
Program FE	Yes	Yes	Yes	Yes	Yes	Yes
Field × year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region × year FE	No	No	No	Yes	Yes	Yes

Notes: Sample restricted to incumbent programs in the four pre-treatment regions with entrant programs in 2009–2011 (Valparaíso, Biobío, La Araucanía, Metropolitana); outside them exposure is zero by construction. Same specification as the main log first stage: $\widetilde{\log E_p^k} = \log E_p^k$ if $E_p^k > 0$ and 0 otherwise, $D_p^k = \mathbf{1}\{E_p^k > 0\}$. Observations with zero first-year enrollment are excluded. SE clustered by pre-treatment market. $^{***}p < 0{,}01$, $^{**}p < 0{,}05$, $^{*}p < 0{,}10$.

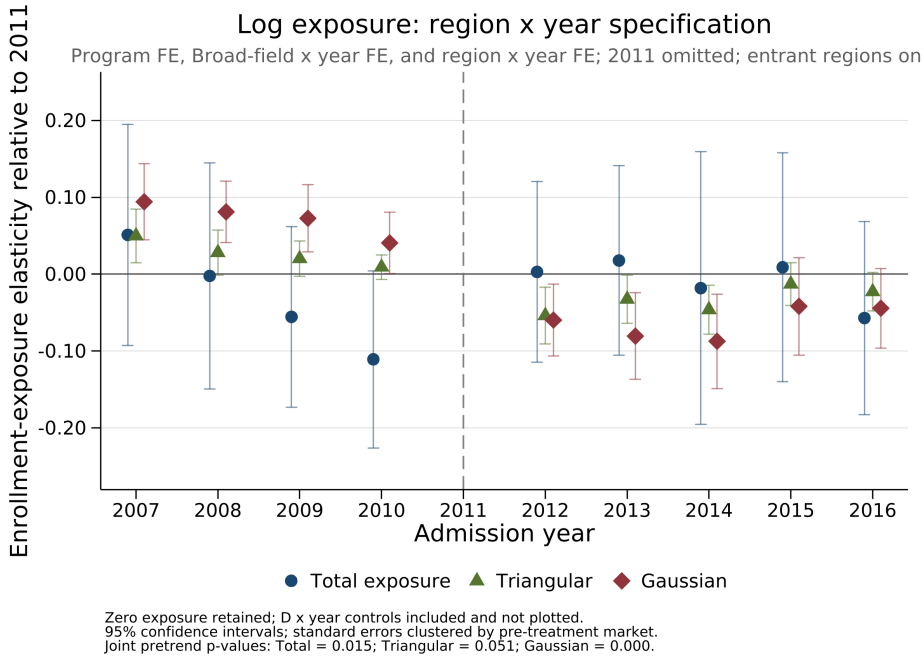
Log event studies: regions with entrant universities only

← First stage

Field × year FE



+ Region × year FE



Notes: Sample restricted to incumbent programs in the four pre-treatment regions with entrant programs in 2009–2011 (Valparaíso, Biobío, La Araucanía, Metropolitana); outside them exposure is zero by construction. Broad-field markets; separate event studies for Total, Triangular and Gaussian exposure. Plotted coefficients are those on the log-exposure × year terms, relative to 2011. 95 % confidence intervals; SE clustered by pre-treatment market. Joint pretrend *p*-values (2007–2010) for Total, Triangular and Gaussian are 0.000, 0.018 and 0.021 in the left panel, and 0.015, 0.051 and 0.000 in the right panel.

Log first stage: positive exposure only

← Main ES →

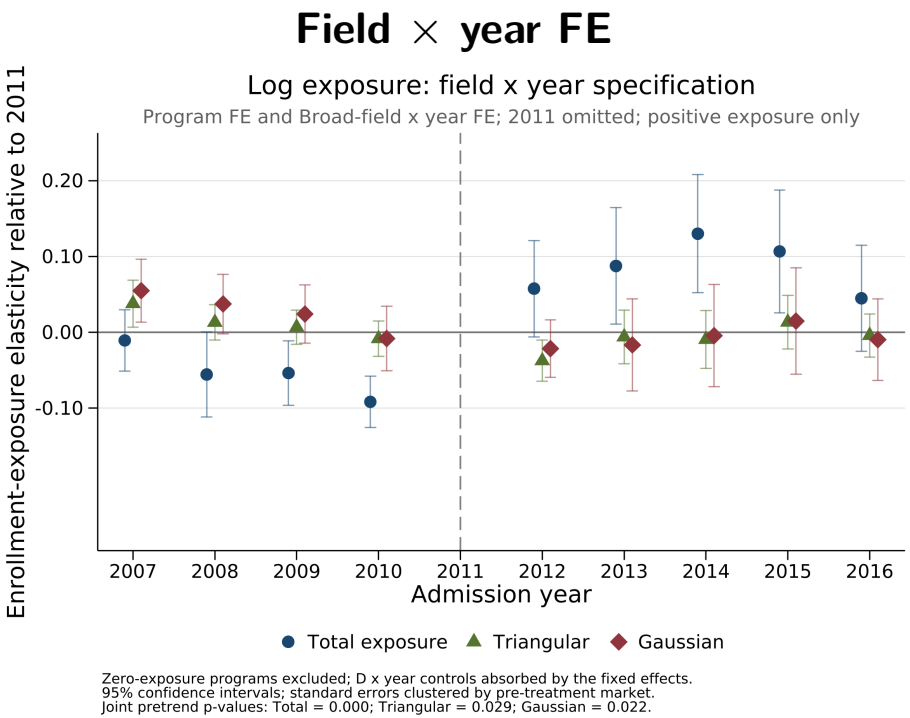
$$\log N_{pt}^{\text{first}} = \beta_k \left[\log E_p^k \times Post_t\right] + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \varepsilon_{pt}, \qquad E_p^k > 0.$$

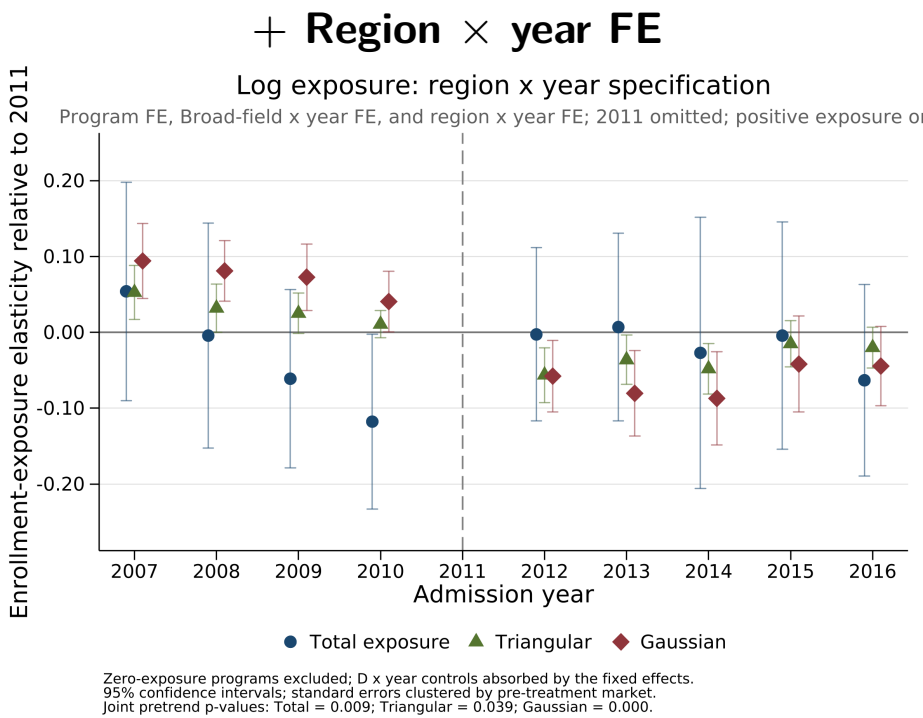
	Field × year FE			+ Region × year FE		
Exposure measure	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
Total	0.132***	0.126***	0.134***	0.019	0.046	0.100***
	(0.039)	(0.036)	(0.021)	(0.086)	(0.061)	(0.024)
Wald *F*	11.39	12.01	41.28	0.05	0.58	17.13
Programs	591	540	304	591	540	304
Triangular	-0.015	0.004	0.021	-0.056***	-0.028	0.004
	(0.019)	(0.022)	(0.027)	(0.020)	(0.021)	(0.023)
Wald *F*	0.61	0.03	0.56	8.06	1.84	0.03
Programs	505	407	142	505	407	142
Gaussian	-0.025	-0.004	0.018	-0.117***	-0.078**	-0.026
	(0.037)	(0.036)	(0.022)	(0.034)	(0.035)	(0.020)
Wald *F*	0.46	0.01	0.64	11.55	4.97	1.79
Programs	591	540	292	591	540	292
Observations (Total)	5,009	4,571	2,516	5,009	4,571	2,516
Programs (Total)	591	540	304	591	540	304
Markets (Total)	36	59	131	36	59	131
Regions	4	4	4	4	4	4
Program FE	Yes	Yes	Yes	Yes	Yes	Yes
Field × year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region × year FE	No	No	No	Yes	Yes	Yes

Notes: Sample restricted to programs with positive exposure, defined separately for each measure (Triangular exposure can be zero where Total and Gaussian are positive). All such programs lie in the four regions with entrant programs, so the only difference with the entrant-region sample is that zero-exposure programs within those regions are dropped. Log first stage without zero-exposure programs, so $D_p^k \times Post_t$ equals $Post_t$ and is absorbed by the fixed effects. Observations with zero first-year enrollment are excluded. SE clustered by pre-treatment market. $^{***}p < 0{,}01$, $^{**}p < 0{,}05$, $^{*}p < 0{,}10$.

Log event studies: positive exposure only

← First stage



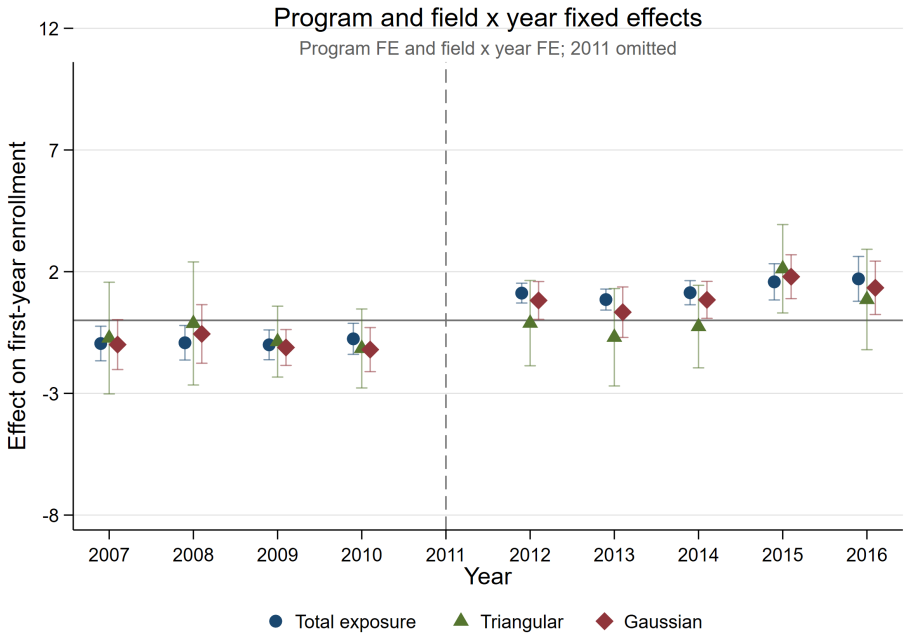


Notes: Sample restricted to programs with positive exposure, defined separately for each measure (Triangular exposure can be zero where Total and Gaussian are positive). All such programs lie in the four regions with entrant programs, so the only difference with the entrant-region sample is that zero-exposure programs within those regions are dropped. Broad-field markets; separate event studies for Total, Triangular and Gaussian exposure. Plotted coefficients are those on the log-exposure × year terms, relative to 2011. 95 % confidence intervals; SE clustered by pre-treatment market. Joint pretrend *p*-values (2007–2010) for Total, Triangular and Gaussian are 0.000, 0.029 and 0.022 in the left panel, and 0.009, 0.039 and 0.000 in the right panel.

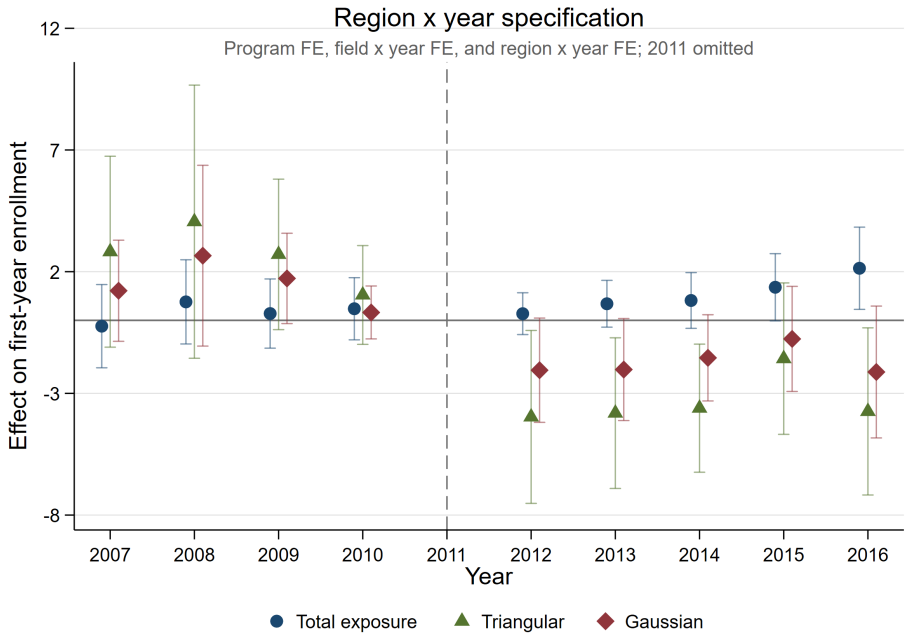
SUA exposure: event-study estimates

← First stage

Field × year FE



+ Region × year FE

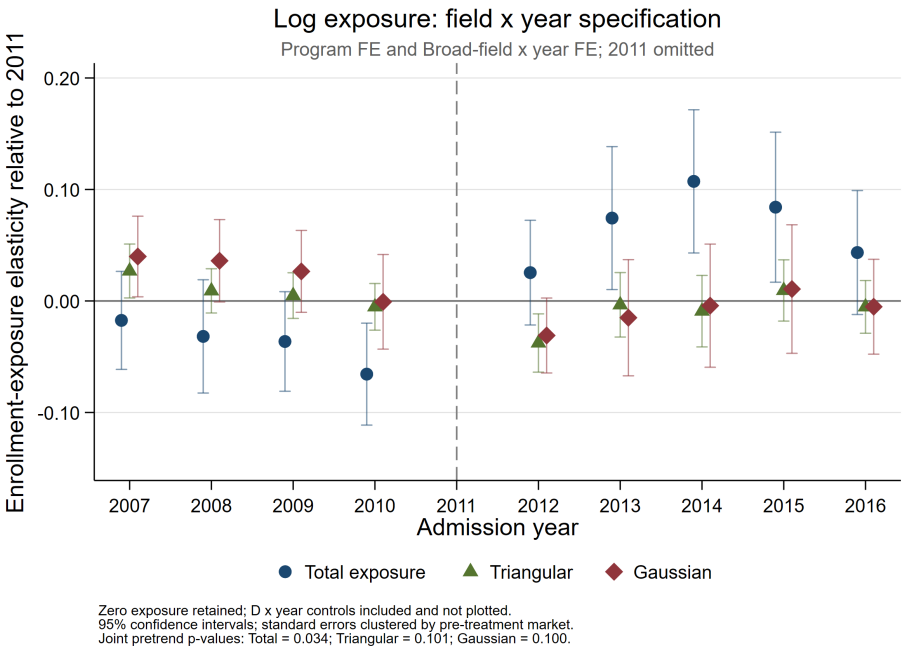


Notes: The figures use the Broad-field market definition and report separate event studies for Total, Triangular and Gaussian exposure. The left panel absorbs program and Broad-field × year fixed effects. The right panel additionally absorbs pre-treatment region × year fixed effects. Coefficients measure the enrollment difference associated with a 10 percentage-point increase in exposure in each year relative to 2011, the omitted year. Vertical bars report 95 % confidence intervals and standard errors are clustered by the pre-treatment market. Joint pretrend p-values for Total, Triangular and Gaussian exposure are 0.024, 0.568 and 0.020 in the left panel, and 0.285, 0.385 and 0.368 in the right panel.

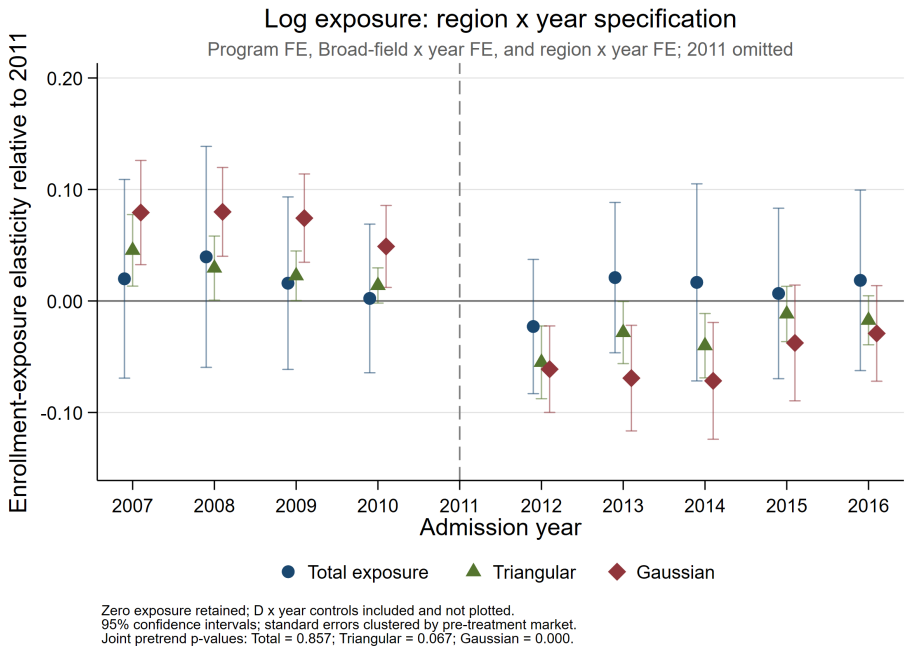
SUA exposure: log event-study estimates

← Log first stage

Field × year FE



+ Region × year FE

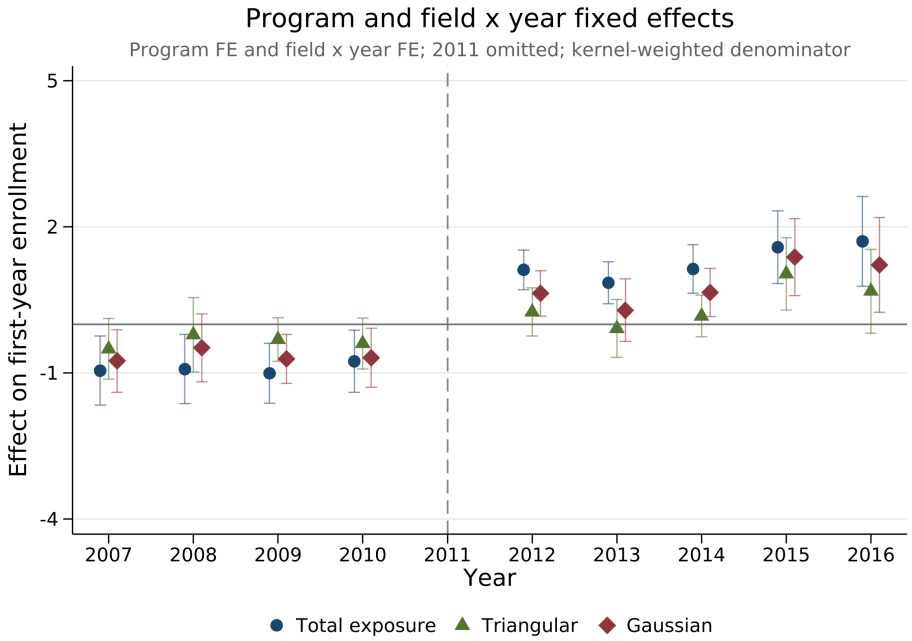


Notes: The figures use the Broad-field definition and report separate event studies for Total, Triangular, and Gaussian exposure. The left panel absorbs program and Broad-field × year fixed effects; the right panel additionally absorbs pre-treatment region × year fixed effects. For each exposure measure, zero-exposure programs are retained by setting $\widetilde{\log} E_p^k = 0$ when $E_p^k = 0$ and including $\mathbf{1}\{E_p^k > 0\} \times \mathbf{1}\{t = s\}$ separately for every event year. The plotted coefficients correspond to $\widetilde{\log} E_p^k \times \mathbf{1}\{t = s\}$ and are interpreted relative to 2011, the omitted year. Observations with zero first-year enrollment are excluded because the dependent variable is in logs. Vertical bars report 95 % confidence intervals; standard errors are clustered by the pre-treatment market. Joint tests of the 2007–2010 log-exposure coefficients yield pretrend p-values of 0.034, 0.101, and 0.100 for Total, Triangular, and Gaussian exposure in the left panel, and 0.857, 0.067, and 0.0001, respectively, in the right panel.

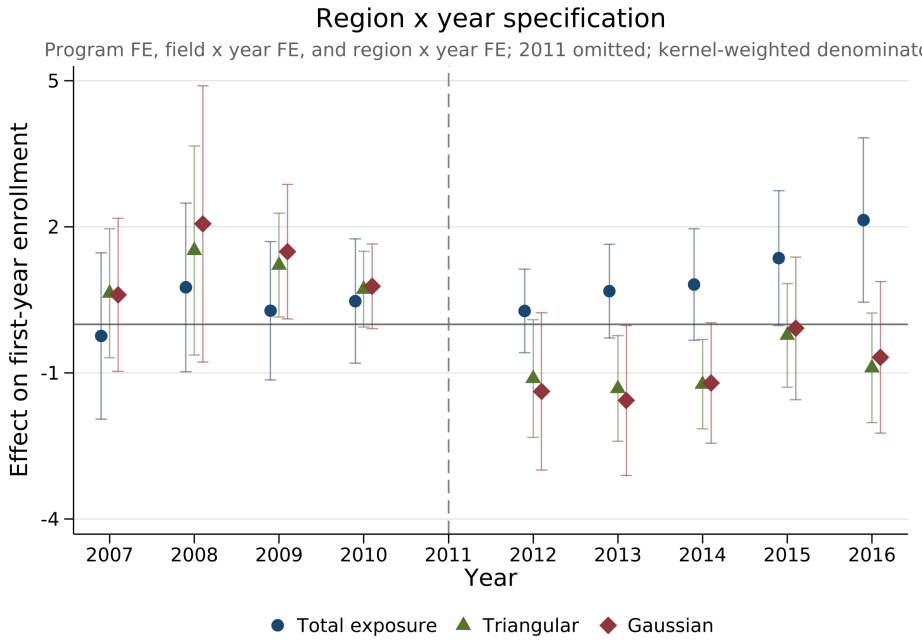
Event studies: kernel-weighted denominator

← First stage

Field × year FE



+ Region × year FE

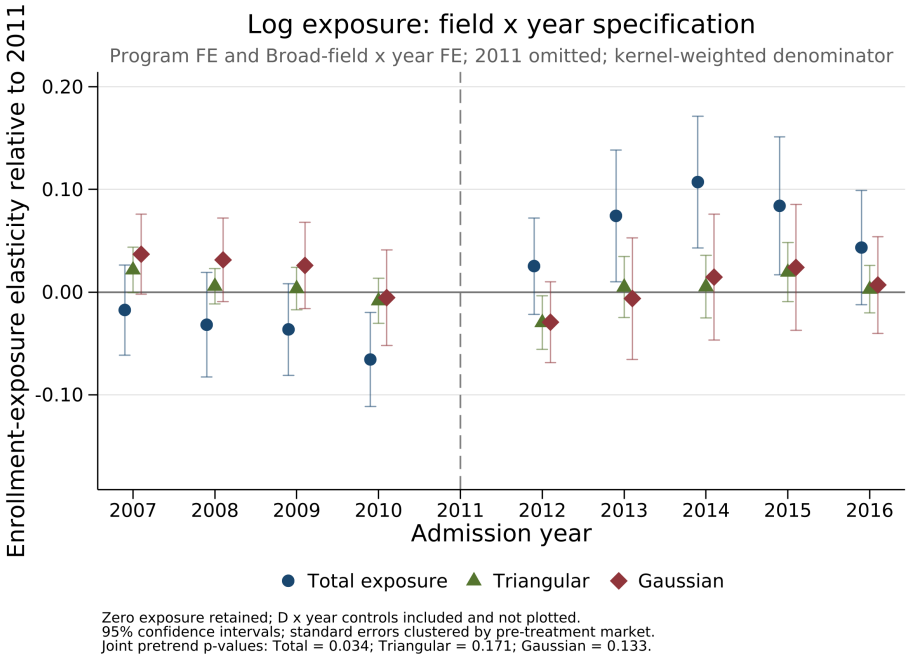


Notes: Triangular and Gaussian exposure use the kernel-weighted denominator; Total exposure is unchanged and shown for reference. Broad-field markets; separate event studies for each measure. Coefficients measure the enrollment difference associated with a 10 percentage-point increase in exposure in each year relative to 2011. 95 % confidence intervals; SE clustered by pre-treatment market. Joint pretrend *p*-values (2007–2010) for Total, Triangular KD and Gaussian KD are 0.024, 0.170 and 0.022 in the left panel, and 0.285, 0.146 and 0.177 in the right panel.

Log event studies: kernel-weighted denominator

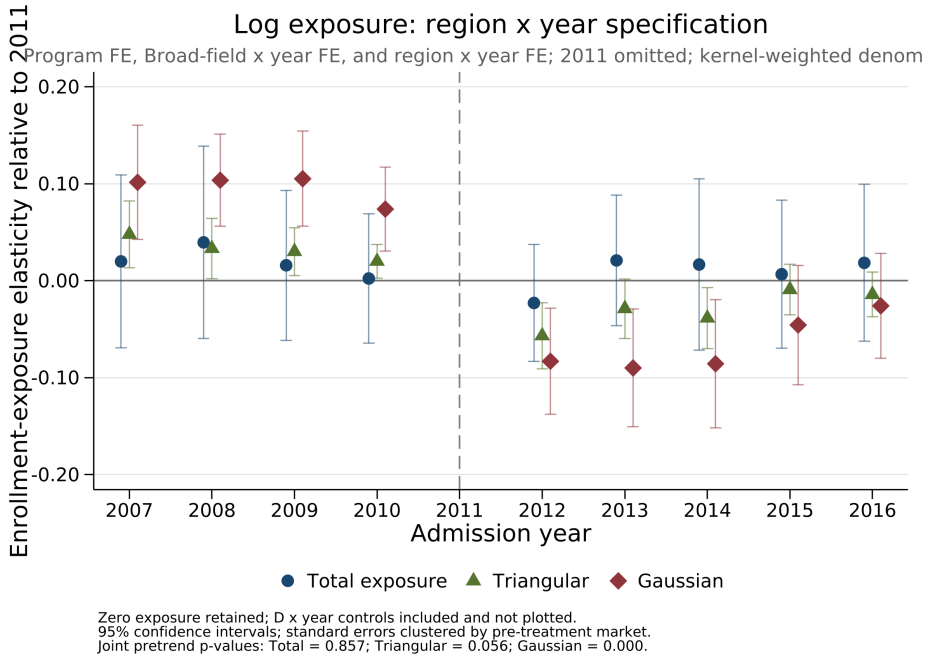
← Log first stage

Field × year FE



Zero exposure retained; D x year controls included and not plotted.
95% confidence intervals; standard errors clustered by pre-treatment market.
Joint pretrend p-values: Total = 0.034; Triangular = 0.171; Gaussian = 0.133.

+ Region × year FE



Zero exposure retained; D x year controls included and not plotted.
95% confidence intervals; standard errors clustered by pre-treatment market.
Joint pretrend p-values: Total = 0.857; Triangular = 0.056; Gaussian = 0.000.

Notes: Triangular and Gaussian exposure use the kernel-weighted denominator; Total exposure is unchanged and shown for reference. Broad-field markets; separate event studies for each measure. Plotted coefficients are those on the log-exposure × year terms, relative to 2011; $\mathbf{1}\{E > 0\}\times$ year controls included. 95 % confidence intervals; SE clustered by pre-treatment market. Joint pretrend p-values (2007–2010) for Total, Triangular KD and Gaussian KD are 0.034, 0.171 and 0.133 in the left panel, and 0.857, 0.056 and 0.0001 in the right panel.

Decomposing weighted exposure

$$\underbrace{E_p^k = \frac{\sum_{q \in \mathcal{E}_{m(p)}} k(|S_p - S_q|)\, N_q}{\sum_{\ell \in m(p)} N_\ell}}_{\text{Direct construction of weighted exposure}} \equiv E_p^k = \left[\frac{\sum_{q \in \mathcal{E}_{m(p)}} N_q}{\sum_{\ell \in m(p)} N_\ell}\right]\left[\frac{\sum_{q \in \mathcal{E}_{m(p)}} k(|S_p - S_q|)\, N_q}{\sum_{q \in \mathcal{E}_{m(p)}} N_q}\right]$$

$$E_p^k = \underbrace{M_{m(p)}}_{\substack{\text{entrant} \\ \text{market share}}} \times \underbrace{Q_p^k}_{\substack{\text{similarity} \\ \text{with entrants}}}$$

Entrant presence

$$M_m = \frac{\sum_{q \in \mathcal{E}_m} N_q}{\sum_{\ell \in m} N_\ell}.$$

M_m is the entrant share of total market enrollment. It captures the **size of entrant presence** and is common to all incumbent programs in market m.

Conditional similarity

$$Q_p^k = \frac{\sum_{q \in \mathcal{E}_m} k\left(|S_p - S_q|\right) N_q}{\sum_{q \in \mathcal{E}_m} N_q}.$$

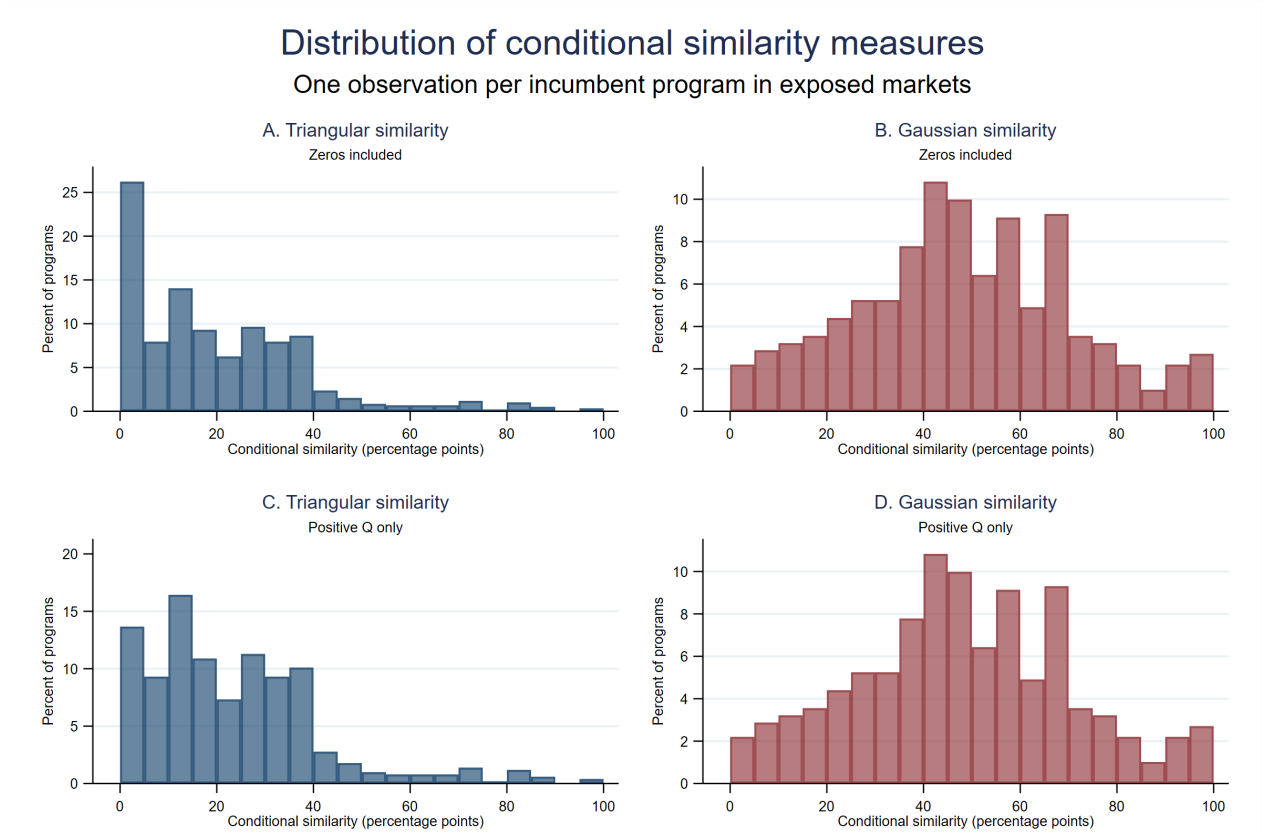
Q_p^k is the entrant-enrollment-weighted PSU similarity between incumbent p and the entrant programs in its market. It is higher when larger entrants have PSU profiles closer to that of p. Conditional on entrant presence, it captures **competitive proximity** rather than the overall size of entry.

Using similarity as the independent variable

$$N_{pt}^{\text{first}} = \beta_k \left[Q_p^k \times Post_t\right] + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \varepsilon_{pt}, \qquad M_{m(p)} > 0.$$

The sample is restricted to markets with positive entrant presence. Program fixed effects absorb time-invariant differences across incumbents, while Broad-field $\times$ year and pre-treatment region $\times$ year fixed effects absorb common field-specific and regional shocks over time. The model therefore asks whether, among incumbents in exposed markets, programs whose pre-treatment PSU profiles are more similar to the entrants experience different post-2012 enrollment changes. Unlike E_p^k, which combines entrant market share and similarity, Q_p^k isolates the **similarity component** of exposure. The coefficient β_k measures the differential post-2012 enrollment change associated with a 0.10 increase in entrant-enrollment-weighted PSU similarity.

Distribution of conditional similarity measures



- Each incumbent program appears once. The horizontal axis reports conditional similarity Q in percentage points and the vertical axis the share of programs.
- The upper row includes all exposed programs, while the lower row excludes zeros and shows the distribution conditional on positive Q.
- Triangular similarity is concentrated at relatively low values and contains a relevant mass near zero. Gaussian similarity is more dispersed and shifted toward higher values, indicating a less restrictive measure of similarity across exposed markets.

Similarity within exposed markets

Event studies →

Common fixed-effects specification

$$N_{pt}^{\text{first}} = \beta_k \left[Q_p^k \times Post_t \right] + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \varepsilon_{pt}, \qquad \log N_{pt}^{\text{first}} = \delta_k \left[\widetilde{\log Q_p^k} \times Post_t \right] + \theta_k \left[D_p^k \times Post_t \right] + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \varepsilon_{pt},$$

$$\widetilde{\log Q_p^k} = \begin{cases} \log(Q_p^k), & Q_p^k > 0, \\ 0, & Q_p^k = 0, \end{cases} \qquad D_p^k = \mathbf{1}\{Q_p^k > 0\}, \qquad M_m > 0.$$

	Levels		Logs	
	Triangular	Gaussian	Triangular	Gaussian
Similarity term × Post	−2,381***	−2,241***	−0,067***	−0,142***
	(0,491)	(0,487)	(0,020)	(0,035)
Wald F	23.50	21.18	11.37	16.29
Observations	5,009	5,009	5,009	5,009
Programs	591	591	591	591
Markets	36	36	36	36

Notes: In levels, a 0.10 increase in conditional similarity is associated with approximately 2.2–2.4 fewer first-year students after 2012. In logs, a 10 % increase in similarity is associated with approximately 0.67 % lower enrollment under the triangular measure and 1.42 % lower enrollment under the Gaussian measure. The sample is restricted to markets with positive entrant presence, $M_m > 0$. Levels retain economically meaningful observations with $Q_p^k = 0$. In logs, zero-similarity observations are also retained by setting $\widetilde{\log Q_p^k} = 0$ when $Q_p^k = 0$ and including $D_p^k \times Post_t$ separately. The coefficient on this indicator is not reported. Gaussian similarity is strictly positive in the exposed-market sample, so the corresponding indicator is absorbed by the fixed effects. All regressions absorb program, Broad-field × year, and pre-treatment region × year fixed effects. Standard errors clustered by pre-treatment market are in parentheses. Event studies reject the absence of differential pre-treatment trends, so estimates are interpreted as descriptive rather than causal. $^{***}p < 0{,}01$.

Conditional similarity: heterogeneity by incumbent selectivity

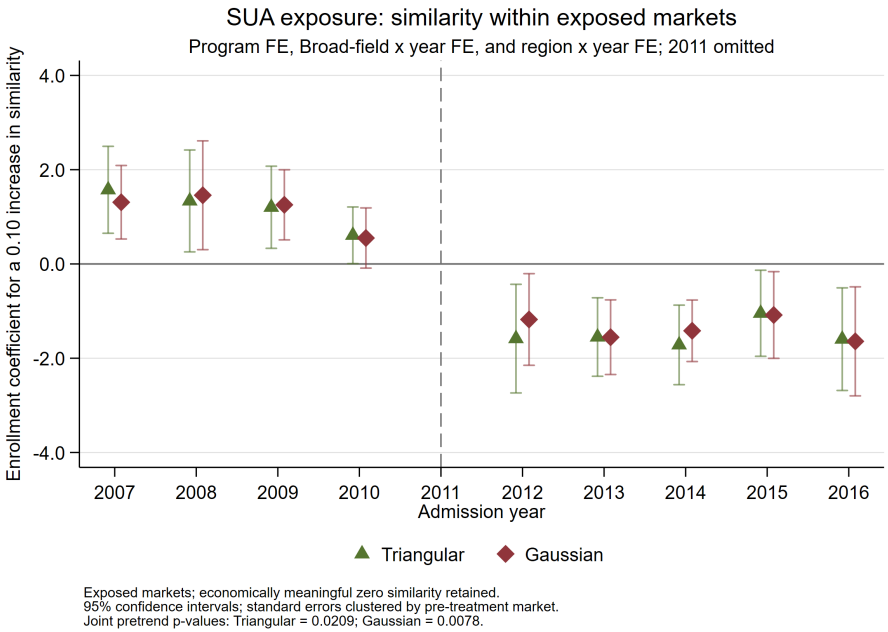
	PSU $<$ 550		PSU 550–649		PSU $\geq$ 650	
	Tri.	Gau.	Tri.	Gau.	Tri.	Gau.
Conditional similarity Q						
Coefficient	$-1{,}95^{*}$	$-3{,}26$	$-2{,}20^{***}$	$-2{,}17^{***}$	$-5{,}70$	$-3{,}48$
	(0,93)	(1,96)	(0,48)	(0,37)	(4,24)	(3,15)
F-statistic	4.40	2.77	21.27	34.14	1.80	1.22
Observations	836		2,932		1,038	
Programs	104		339		128	
Clusters	15		28		16	

Notes: Standard errors clustered by pre-treatment market in parentheses. All regressions include program and market $\times$ year FE and are restricted to exposed markets. Coefficients correspond to a 0.10 increase in conditional similarity. The same split for SUA exposure is in the main text. $^{*}p < 0{,}10$, $^{**}p < 0{,}05$, $^{***}p < 0{,}01$.

Similarity within exposed markets: event studies

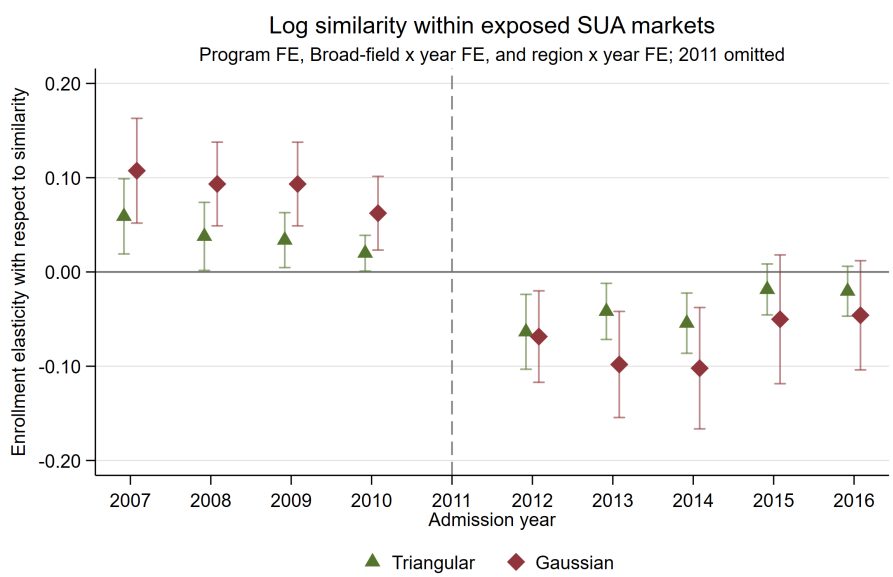
← Back to table

Levels



Exposed markets; economically meaningful zero similarity retained.
95% confidence intervals; standard errors clustered by pre-treatment market.
Joint pretrend p-values: Triangular = 0.0209; Gaussian = 0.0078.

Logs

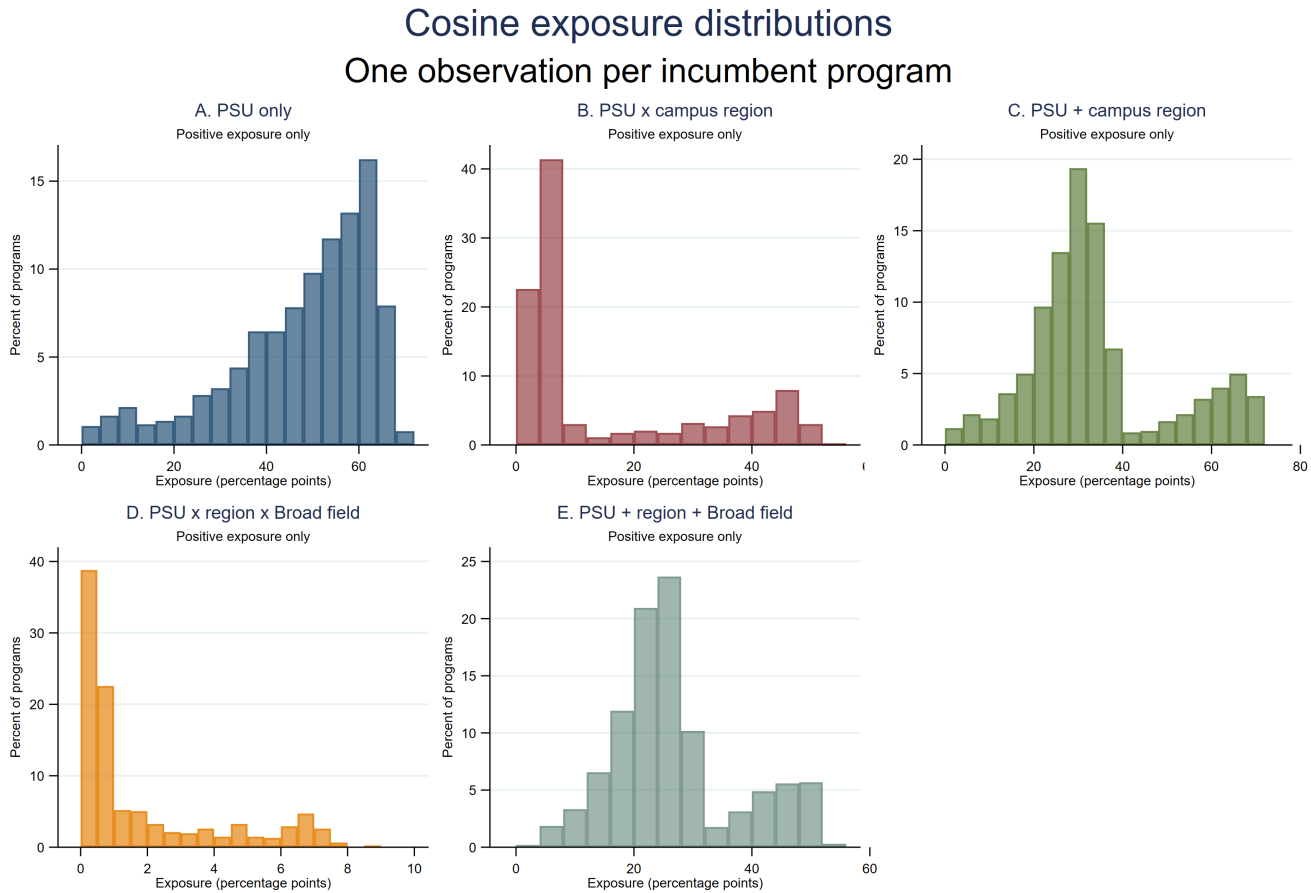


Exposed markets; Q = 0 retained; D x year controls included and not plotted.
95% confidence intervals; standard errors clustered by pre-treatment market.
Joint pretrend p-values: Triangular = 0.0348; Gaussian = 0.0001.

Notes: Coefficients are relative to 2011. All models absorb program, Broad-field × year, and pre-treatment region × year fixed effects; standard errors are clustered by pre-treatment market. The sample is restricted to markets with positive entrant presence, $M_m > 0$. Levels retain economically meaningful observations with $Q_p^k = 0$. In logs, zero-similarity observations are also retained by setting $\widetilde{\log Q_p^k} = 0$ when $Q_p^k = 0$ and including $\mathbf{1}\{Q_p^k > 0\} \times \mathbf{1}\{t = s\}$ separately in each event year; these indicator coefficients are not plotted. Observations with zero first-year enrollment are excluded from the log specification. Joint pretrend p-values are 0.0209 and 0.0078 for the triangular and Gaussian measures in levels, and 0.0348 and 0.0001 in logs. All four tests reject the absence of differential pre-treatment trends at the 5 % level; estimates are therefore interpreted as descriptive associations rather than causal effects.

Cosine exposure distributions conditional on positive exposure

← Back

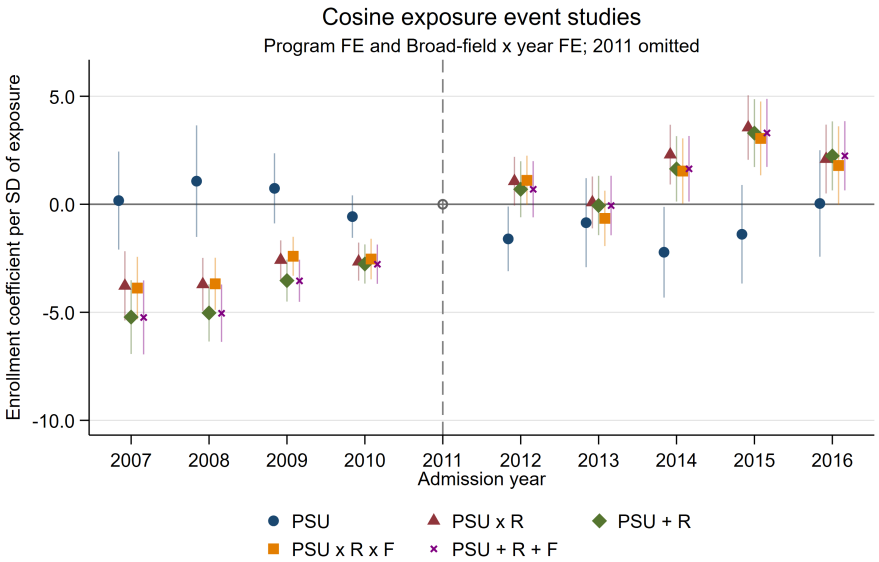


Positive exposure only. Exposure measured in percentage points; each panel uses its own x-axis scale.

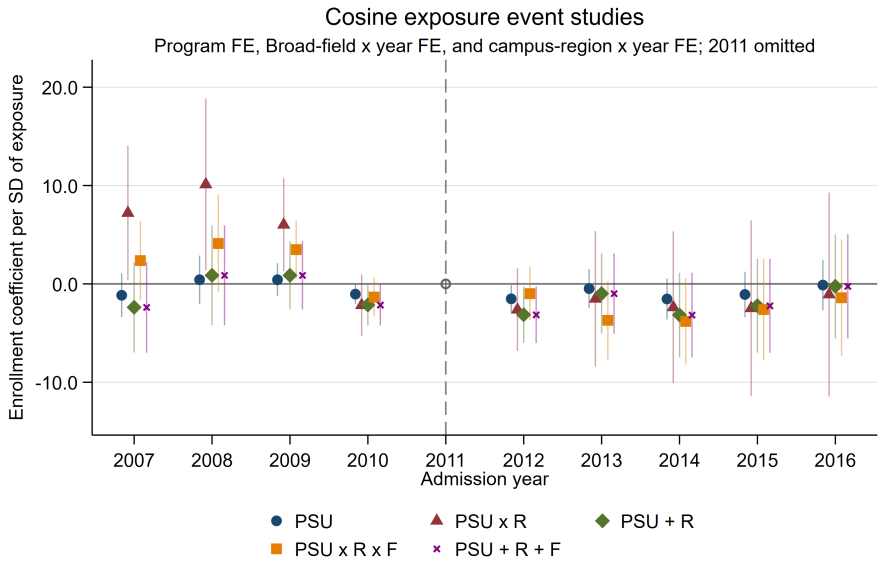
- Programs with zero exposure are excluded, allowing the shape of the positive part of each distribution to be seen more clearly.
- The multiplicative PSU $\times$ campus region $\times$ Broad field measure remains concentrated at relatively low exposure values even after conditioning on positive exposure.

Cosine exposure event studies

← First-stage results



R denotes campus region; F denotes Broad field. 95% CIs; SEs clustered by program.
Joint pretrend p-values: PSU = 0.3950; PSU x R = 0.0000; PSU + R = 0.0000.
PSU x R x F = 0.0000; PSU + R + F = 0.0000.



R denotes campus region; F denotes Broad field. 95% CIs; SEs clustered by program.
Joint pretrend p-values: PSU = 0.0990; PSU x R = 0.0023; PSU + R = 0.0990.
PSU x R x F = 0.0105; PSU + R + F = 0.0990.

- Without campus-region × year FE, PSU only does not reject joint pretrends ($p = 0{,}395$), but all four measures that incorporate campus region do ($p < 0{,}001$).
- Adding campus-region × year FE improves the diagnostic for PSU and the additive measures (all $p = 0{,}099$), but the multiplicative measures still reject: PSU × region ($p = 0{,}0023$) and PSU × region × field ($p = 0{,}0105$).
- Post-2012 estimates become mostly negative and imprecise once campus-region × year FE are included. The min-10 sample is not shown and delivers virtually the same conclusions.
- Overall, the cosine results are best interpreted as descriptive rather than causal.